

Renovascular Hypertension and Ischemic Nephropathy

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CHAPTER OUTLINE

INTRODUCTION, 1580

PATHOPHYSIOLOGY OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY, 1583

PHASES OF DEVELOPMENT OF RENOVASCULAR HYPERTENSION, 1587

EPIDEMIOLOGY OF RENAL ARTERY STENOSIS AND RENOVASCULAR HYPERTENSION, 1593

CLINICAL FEATURES, 1595

CHANGING POPULATION DEMOGRAPHICS, 1599

DIAGNOSTIC TESTING FOR RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY, 1600

MANAGEMENT OF RENAL ARTERY STENOSIS AND ISCHEMIC NEPHROPATHY, 1606

PREDICTORS OF LIKELY BENEFIT REGARDING RENAL REVASCULARIZATION, 1619

SUMMARY, 1620

INTRODUCTION

Few areas within nephrology have undergone more dramatic paradigm shifts with competing treatment options than occlusive renovascular disease (RVD). Aging population demographics, coupled with advances in imaging technology, effective medical therapy, and negative clinical trials, must be reconciled with an established record of clinical success and improving techniques of renal revascularization. Although prospective trial data have failed to provide compelling evidence in favor of stenting for many patients with atherosclerotic disease, experienced clinicians recognize that renal revascularization in these disorders sometimes should be undertaken to improve hypertension and protect renal function. Selecting patients and determining optimal timing for vascular intervention at reasonable risk are rarely straightforward.

The evaluation and treatment of RVD overlaps numerous medical disciplines and subspecialties, including nephrology, internal medicine, cardiovascular diseases, interventional radiology, and vascular surgery. These subspecialty groups often deal with different patient subgroups and clinical issues that shape different points of view. Cardiologists, for example, usually manage more patients with refractory congestive heart failure (CHF) at risk for flash pulmonary edema than internists, who may deal with hypertensive patients with progressive hypertension or a rise in serum creatinine levels (Fig. 47.1). Nephrologists regularly encounter declining kidney function with high-grade stenosis to a solitary functioning kidney. All these conditions can represent clinical

manifestations of RVD but present different comorbid risk and management issues. Not surprisingly, perceptions related to renovascular hypertension and ischemic nephropathy differ, even among informed clinicians. Initial results from prospective randomized trials, such as the Angioplasty and Stenting for Renal Artery Lesions (ASTRAL) in the United Kingdom and Cardiovascular Outcomes in Renal Artery Lesions (CORAL) in the United States comparing optimal medical therapy, with or without endovascular stent procedures, continue to provoke controversy. Some individual authors of these negative trials have separately reported high-risk subsets not enrolled in those studies that have major clinical and mortality benefits from restoring renal blood flow.¹⁻³

These diverse observational studies underscore the ambiguity that clinicians encounter in practice. Ultimately, RVD and impaired blood flow not only affect blood pressure and cardiovascular risk but also threaten the viability of the kidney. These can lead to irreversible loss of kidney function, sometimes designated as “ischemic nephropathy” or “azotemic RVD.”⁴ Restoring blood flow and perfusion by relieving vascular occlusion intuitively offers a means to halt or reverse this process. It must be recognized, however, that renal revascularization is a double-edged sword. The benefits include the potential to improve systemic arterial blood pressures and preserve or salvage renal function. The potential risks of renal intervention are all too familiar to nephrologists. Endovascular procedures themselves may threaten the affected kidney through vascular thrombosis, dissection, restenosis, or atheroemboli. These events sometimes precipitate the need for renal replacement therapy, including dialysis or transplantation. It is therefore important that nephrologists

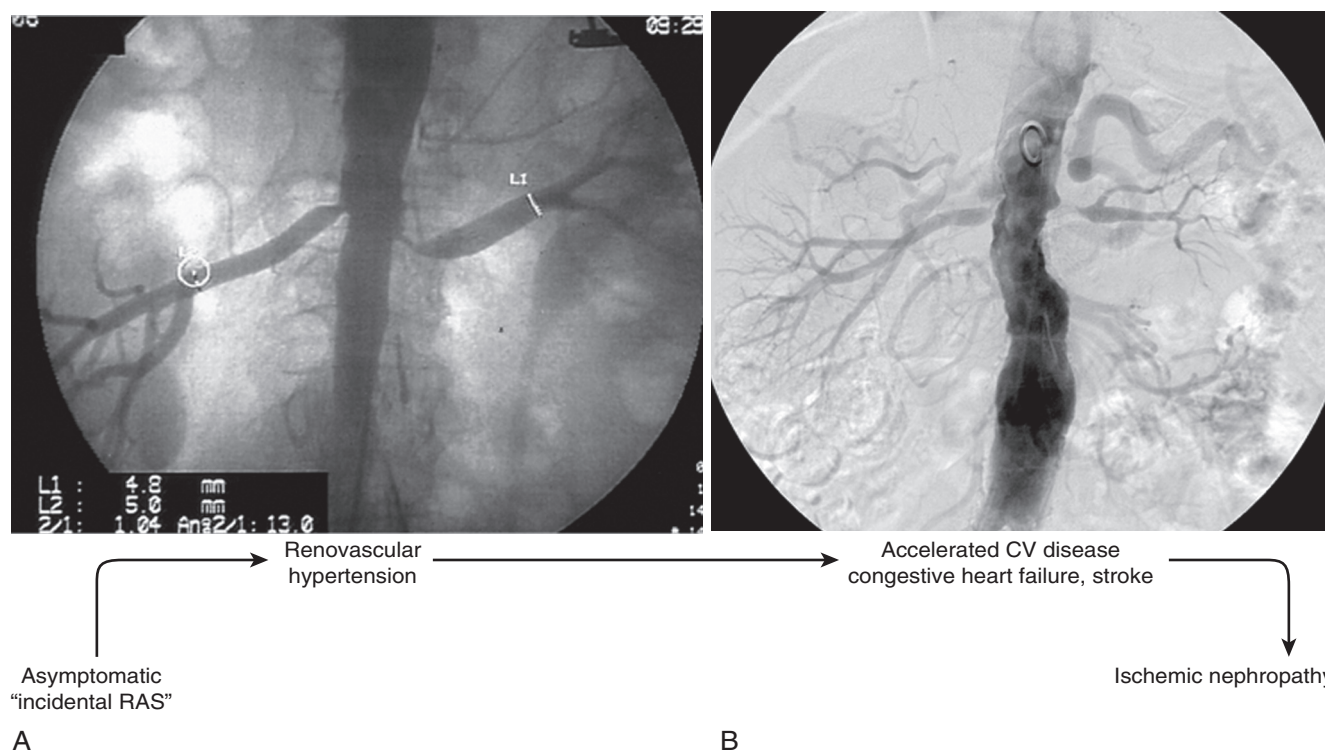


Fig. 47.1 (A and B) Spectrum of atherosclerotic renovascular disease (ARVD) manifestations. *Left panel*, Aortogram obtained during coronary angiography demonstrating moderate incidental stenosis of both renal arteries in a 67-year-old man with symptomatic coronary disease. *Right panel*, More severe occlusive disease observed in a 68-year-old woman presenting with severe hypertension and episodes of flash pulmonary edema. RVD commonly develops in the setting of atherosclerotic disease elsewhere and may be associated with any of many clinical syndromes, ranging from renovascular hypertension to accelerated cardiovascular decompensation and ischemic nephropathy. Recent trends favor initial management with effective medical therapy, often including agents that block the renin-angiotensin system. Clinicians face the challenge of recognizing the role of RVD in more advanced disease and balancing the risks and benefits of renal revascularization in high-risk subsets. RAS, renal artery stenosis. (Modified from Herrmann SM, Saad A, Textor SC. Management of atherosclerotic renovascular disease after Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL). *Nephrol Dial Transplant*. 2015;30(3):366–375.)

have a solid foundation related to the implications of reduced renal perfusion and the risks and benefits of medical management and restoration of renal artery patency.

HISTORICAL PERSPECTIVE

RVD is among the most extensively studied forms of secondary hypertension. Early observations regarding blood pressure regulation have revealed important connections among fluid volume, renal arterial pressures, and vascular resistance. The sequence of these observations related to identification of the renin-angiotensin-aldosterone system has been reviewed by Basso and Terragno⁵ and is summarized here. In 1898, Tigerstedt and Bergman established that extracts of the kidney had pressor effects in the whole animal and are credited with the identification of “renin.” Identification of each component of the renin-angiotensin system represents a remarkable series of research ventures, spanning more than a half-century and investigators in many countries.

Goldblatt and others carried out seminal experiments with the development of an animal model in which reduced renal perfusion regularly produced hypertension, published between 1932 and 1934. Numerous investigators thereafter identified the peptide nature of angiotensin, the role of renin substrate or angiotensinogen, the role of nephrectomy

in sensitizing the animal to the pressor effects of angiotensin, and the sequential phases of renovascular hypertension. Hence, the renin-angiotensin-aldosterone system owes its initial discovery and nomenclature primarily to early studies related to the regulation of blood pressure by the kidney. Only recently have many additional actions of angiotensin become evident regarding vascular remodeling, modulation of inflammatory pathways, and interaction with fibrogenic mechanisms. Understanding that reduced renal blood flow produces sustained elevations in arterial pressure has led to broad study of the mechanisms underlying many forms of hypertension. Experimental models of two-kidney and one-kidney “Goldblatt” hypertension—animals undergo clipping of one renal artery to reduce blood flow, with or without nephrectomy of the contralateral kidney (i.e., two-kidney, one-clip and one-kidney, one-clip models)—represent some of the most widely applied models of blood pressure and cardiovascular regulation. Extension of these studies into clinical medicine followed soon thereafter. Some forms of hypertension were designated as “malignant” in character during the late 1930s and 1940s based on poor survival if patients were untreated.

Few antihypertensive agents were known until the 1950s; intervention consisted mainly of lumbar sympathectomy and/or extremely low sodium intake diets. Some of the major

Time line of milestones related to renovascular hypertension

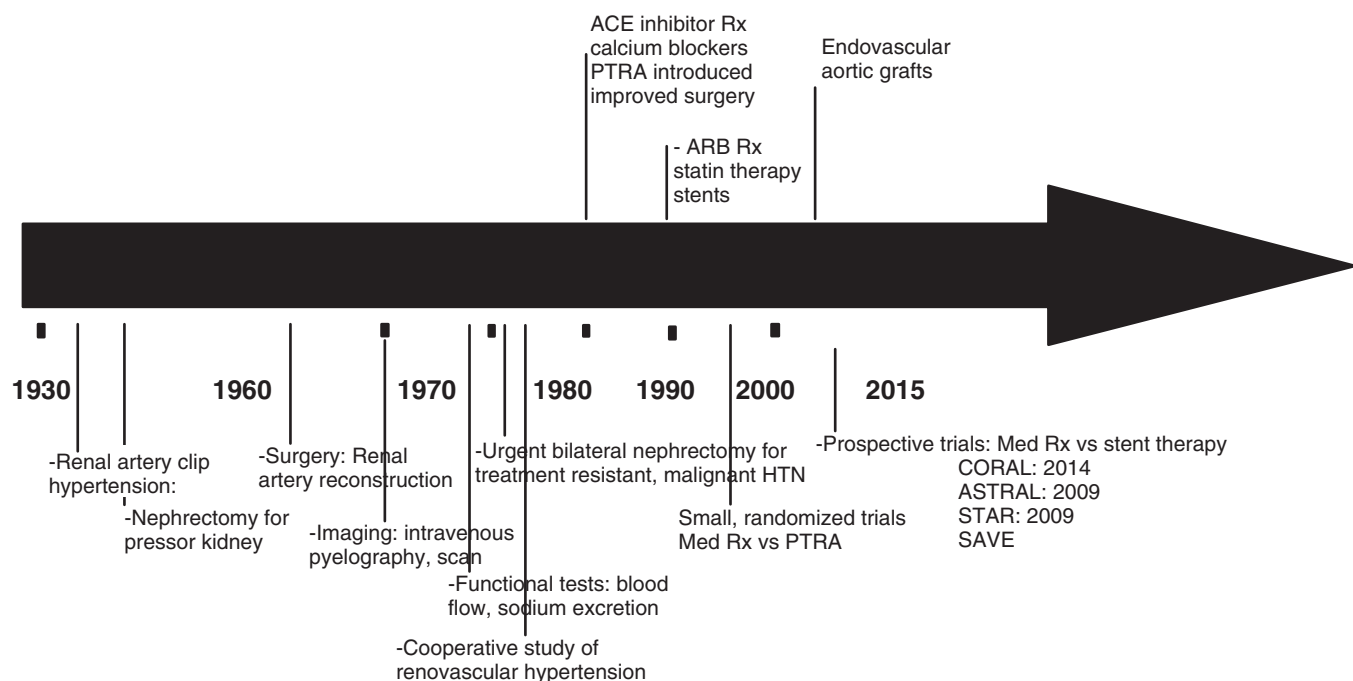


Fig. 47.2 Historical time line of major milestones in renovascular disease (RVD). Although pressor substances derived from the kidney were identified more than a century ago, recognition that renal perfusion regulates arterial blood pressure occurred only in the 1930s. Technical advances allowing surgical reconstruction, more effective arterial imaging, and endovascular revascularization gradually evolved from 1960 to the present era. Important advances in antihypertensive drug therapy, particularly with agents that block the renin-angiotensin system, and other advances in managing atherosclerotic disease such as statins, continue to define optimal medical management of patients with RVD. The introduction of endovascular aortic stent grafts represents an iatrogenic form of RVD that benefits from protection of the renal artery patency. Results from several prospective, randomized clinical trials indicate that renal revascularization often fails to add substantial benefits to effective medical therapy in the near term for patients with moderate atherosclerotic RVD, although high-risk subsets are an exception. ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; PTRAs, percutaneous transluminal renal angioplasty.

events in the identification and treatment of renovascular hypertension are summarized in Fig. 47.2.⁶ Recognition that some forms of severe hypertension were secondary to occlusive vascular disease in the kidney led surgeons to undertake unilateral nephrectomy for small kidneys in 1937.⁶ The fact that some of these were indeed pressor kidneys, and blood pressure fell to normal levels, provided proof of concept and led to more widespread use of nephrectomy. Unfortunately, achieving cure of hypertension after nephrectomy was rare, and Smith reviewed the poor results overall in a 1956 paper discouraging this practice.⁷

The 1960s marked the introduction of methods of vascular surgery to restore renal blood flow. These carried substantial morbidity but offered an opportunity to improve the renal circulation and potentially to reverse renovascular hypertension. One result of this development was a series of studies that characterized the functional role of each vascular lesion in producing hypertension, thereby allowing prediction of the outcomes of vascular surgery.⁶ A large cooperative study of renovascular hypertension included major vascular centers and reported on the results of more than 500 surgical procedures.⁸ These results provided limited support for vascular repair but identified relatively high associated morbidity and mortality, particularly for patients with atherosclerotic disease.

In the 1980s and 1990s, further developments led to improved medications and the introduction of endovascular procedures, including percutaneous angioplasty and stents.⁹ These both broadened the options for treating patients with vascular disease and raised new issues regarding timing and overall goals of intervention. Recent developments have highlighted the need for intensive cardiovascular risk factor reduction and more stringent standards of blood pressure control. Antihypertensive medications have improved dramatically, both in regard to efficacy and tolerability. As emphasized later, broad application of angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor antagonists for reasons other than hypertension alone changed the clinical presentation of disorders associated with renal artery stenosis (RAS). Recent prospective randomized trials have emphasized the limited additional benefits from restoring renal blood flow for many patients with moderate vascular disease.¹

Uncontrolled hypertension is now less commonly the reason to intervene in RVD than in the past. Often, the main objective is long-term preservation of renal function. In recent years, endovascular techniques have made renal revascularization possible with relatively low morbidity in many patients previously considered unacceptable surgical candidates. The challenge for clinicians is how and when to apply these tools most effectively in the management of individual patients.¹⁰

PATHOPHYSIOLOGY OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

RENAL ARTERY STENOSIS VERSUS RENOVASCULAR HYPERTENSION

As with most vascular lesions, the presence of a renovascular abnormality alone does not translate directly into functional importance. Some degree of RAS can be identified in as many as 20% to 45% of patients undergoing vascular imaging for other reasons, such as coronary angiography or lower extremity peripheral vascular disease.^{11,12} Most of these incidentally detected stenoses are of minor hemodynamic and/or clinical significance. Failure to limit treatment trials to patients with clinically important lesions has been a serious barrier to understanding the role for renal revascularization (see later). The term *renovascular hypertension* refers to a rise in arterial pressure definitely induced by reduced renal perfusion. A variety of lesions can lead to the syndrome of renovascular hypertension, some of which are listed in [Box 47.1](#). Studies of vascular obstruction using latex rubber casts have indicated that from 70% to 80% of lumen obstruction must occur before measurable changes in blood flow or pressure across the lesion can be detected.¹³ Measurements of pressure gradients during renal angiography confirm that a pressure gradient of at least 10 to 20 mm Hg between the aorta and poststenotic renal artery is required before measurable renin release occurs.^{14,15} When advanced stenosis is present, the pressure and flow drop steeply as illustrated in [Fig. 47.3](#). When lesions have reached this degree of hemodynamic significance, they are deemed “critical” stenoses.

When renal artery lesions reach critical dimensions, a series of events leads to a rise in systemic arterial pressure and restoration of renal perfusion pressure, as illustrated in

[Fig. 47.4](#). Hence, one can view the development of rising pressures in this context as an integrated renal response to maintain renal perfusion of the stenotic kidney. It is important to distinguish between experimental models of clip stenosis, at which time a sudden change in renal perfusion is induced, and the more common clinical situation of

Box 47.1 Some Vascular Lesions Producing Renal Hypoperfusion and Renovascular Hypertension Syndrome

Unilateral disease (analogous to one-clip, two-kidney hypertension)
 Unilateral atherosclerotic renal artery stenosis
 Unilateral fibromuscular dysplasia (FMD)
 Medial fibroplasia
 Perimedial fibroplasia
 Intimal fibroplasia

Medial Hyperplasia

Renal artery aneurysm
 Arterial embolus
 Arteriovenous fistula (congenital, traumatic)
 Segmental arterial occlusion, dissection (posttraumatic)
 Extrinsic compression of renal artery (e.g., pheochromocytoma)
 Renal compression (e.g., metastatic tumor)
 Bilateral disease or solitary functioning kidney (analogous to one-clip, one-kidney model)
 Stenosis to a solitary functioning kidney
 Bilateral renal arterial stenosis
 Aortic coarctation
 Systemic vasculitis (e.g., Takayasu arteritis, polyarteritis)
 Atheroembolic disease
 Vascular occlusion due to endovascular aortic stent graft

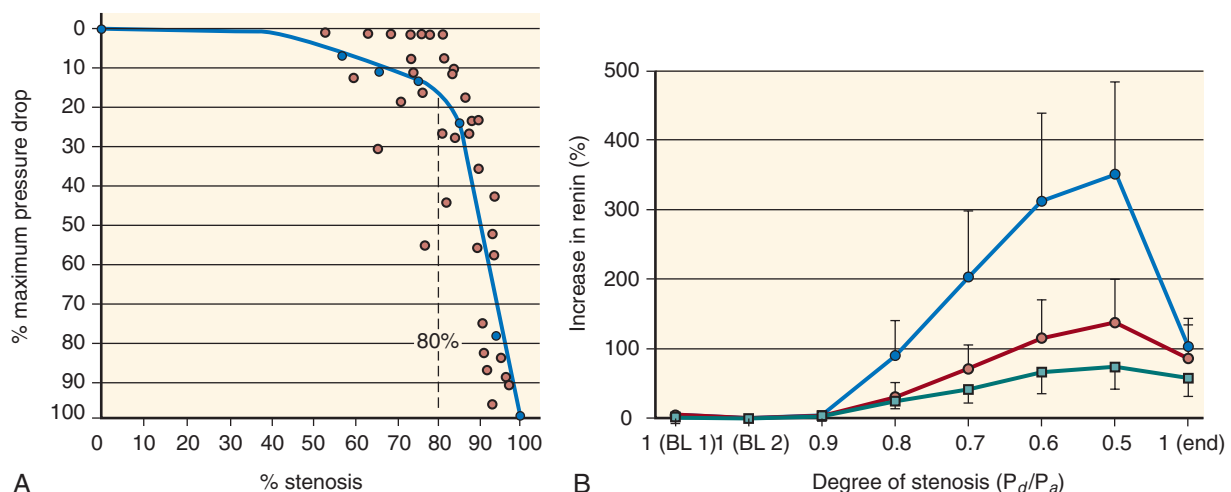


Fig. 47.3 (A) Measured fall in arterial pressure and blood flow across stenotic lesions induced in experimental animals. The degree of stenosis was determined using latex casts after completion of the experiment. These data indicate that “critical” lesions require 70% to 80% luminal obstruction before hemodynamic effects can be detected. (B) Studies from human subjects with translational pressure gradients indicate that an aortic-renal pressure gradient of 10% to 20% is necessary to detect renin release. P_a , aortic pressure; P_d , distal pressure. (A from May AG, Van de Berg L, DeWeese JA, et al. Critical arterial stenosis. *Surgery*. 1963;54:250–259; B from De Bruyne B, Manoharan G, Pijls NHJ, et al. Assessment of renal artery stenosis severity by pressure gradient measurements. *J Am Coll Cardiol*. 2006;48:1851–1855.)

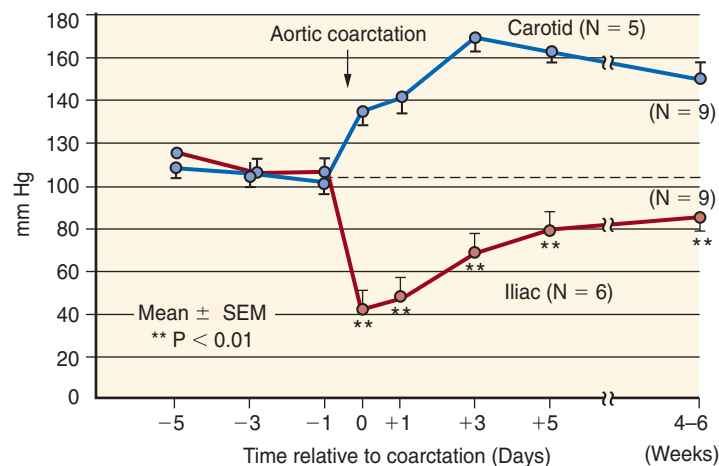


Fig. 47.4 Systemic arterial pressure (carotid) and poststenotic renal perfusion pressures (iliac) in an aortic coarctation model with a clip placed between the right and left renal arteries. Measurements were obtained in conscious animals during the development of renovascular hypertension. They illustrate the fact that despite a persistent gradient across the stenosis, renal perfusion pressure (inferred from Iliac pressure) returns to near-normal levels as a result of systemic hypertension. A corollary of this pressure gradient is that reduction of systemic pressures would reduce poststenotic perfusion pressures, sometimes below the range of autoregulation. SEM, Standard error of the mean. From Textor SC, Smith-Powell L. Post-stenotic arterial pressures, renal haemodynamics and sodium excretion during graded pressure reduction in conscious rats with one- and two-kidney coarctation hypertension. *J Hypertens.* 1988;6(4):311–319.)

gradually progressive lumen obstruction. In the latter case, hemodynamic characteristics change slowly and are likely to produce hypertension over a prolonged time interval. The rise in systemic pressure restores normal renal perfusion, often with normal-sized kidneys and no discernible hemodynamic compromise. If the renal artery lesion progresses further (or is experimentally advanced), the cycle of reduced perfusion and rising arterial pressures repeats until malignant-phase hypertension develops. Experimental models in swine have used gradually progressing vascular lesions to mimic human RVD.¹⁶

A corollary to critical arterial stenosis is that treatment aimed at the normalization of elevated systemic pressures in renovascular hypertension reduces renal perfusion pressure beyond the stenotic lesion. Poststenotic pressures may fall below the levels of autoregulation that maintain blood flow. This underperfusion of the kidney activates counterregulatory pathways and leads to a sequence of events, again directed toward restoring kidney perfusion.

ROLE OF THE RENIN-ANGIOTENSIN SYSTEM IN ONE-KIDNEY AND TWO-KIDNEY RENOVASCULAR HYPERTENSION

Reduction in renal perfusion pressures activates the release of renin from juxtaglomerular cells in the affected kidneys. Experimental studies have indicated that hypertension in two-kidney, one-clip models can be delayed indefinitely as long as agents that block this system are administered. Animals genetically modified to lack the angiotensin 1 (AT1) receptor fail to develop two-kidney, one-clip hypertension, as illustrated in Fig. 47.5.¹⁷ Experiments using kidney transplantation from AT1 receptor knockout mice have indicated that both systemic and renal angiotensin receptors participate in blood pressure regulation in additive fashion.¹⁸

Demonstration of the role of the renin-angiotensin axis in renovascular hypertension depends in part on whether

or not a contralateral nonstenotic kidney is present. Most human renovascular hypertension is considered analogous to two-kidney, one-clip experimental (Goldblatt) hypertension. The contralateral nonstenotic kidney is subjected to elevated systemic perfusion pressures. Effects of rising perfusion pressure force natriuresis from the nonstenotic kidney and suppress renin release. Hence, the nonstenotic kidney tends to prevent the rise in systemic pressures, thereby perpetuating reduced perfusion to the stenotic side and promoting continued renin release from the stenotic kidney. Blood pressure in these models is demonstrably angiotensin-dependent and is associated with elevated circulating levels of plasma renin activity, as illustrated in Fig. 47.6. The two-kidney, one-clip model of renovascular hypertension provided the basis for many of the early functional studies of surgically curable hypertension in which side-to-side function was compared regarding factors such as glomerular filtration and sodium excretion. This paradigm forms the basis for comparing kidneys side to side using radionuclide studies, such as captopril renography and renal vein renin determinations. Unilateral renal ischemia is a classic model for the study of angiotensin-dependent hypertension and target organ injury.

When no such contralateral kidney is present or able to respond to pressure natriuresis, the mechanisms sustaining hypertension differ. This corresponds to the one-kidney, one-clip hypertensive model. Although renin release occurs initially, elevated systemic pressures develop with sodium and volume retention because there is limited sodium excretion by the contralateral kidney. Rising pressures eventually restore renin levels to normal. Hypertension in this model is not demonstrably dependent on angiotensin II (Ang II) unless prior sodium depletion is achieved. Clinical examples of this situation include bilateral RAS or stenosis in a solitary functioning kidney, in which the entire renal mass is affected. In such cases, diagnostic comparison of side-to-side renin release is not possible or has little meaning.

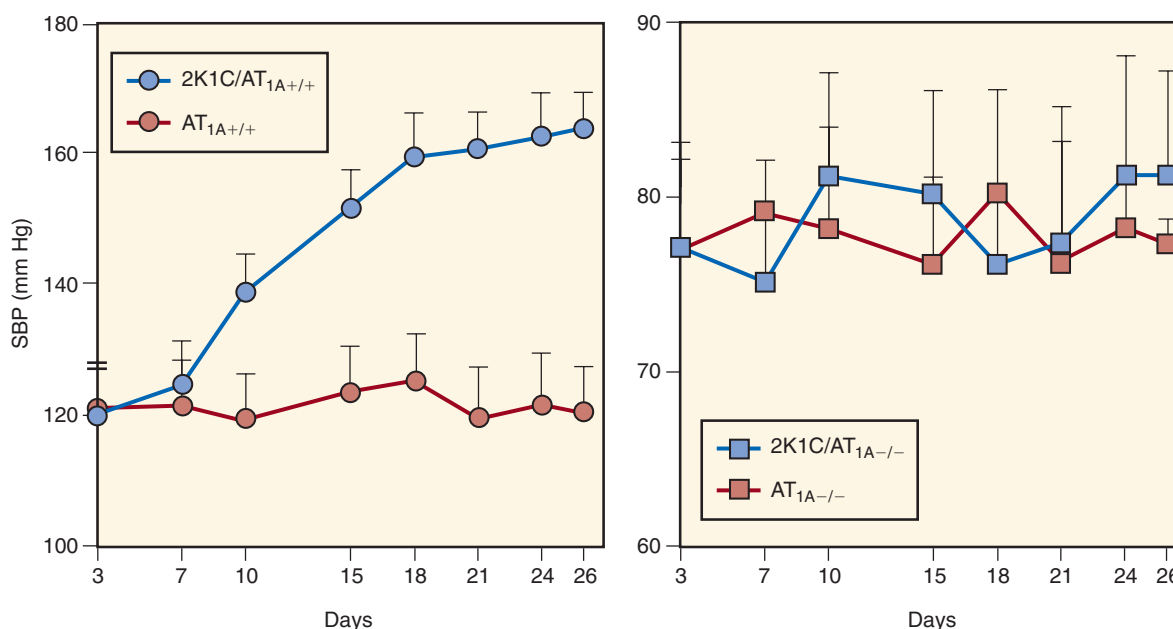


Fig. 47.5 Systolic blood pressures in mice before and after placement of a renal artery clip in experimental two-kidney, one-clip renovascular hypertension (2K1C). The rise in systolic blood pressure (SBP) after clip placement develops rapidly only in mice with an intact angiotensin 1A receptor (AT_{1A}^{+/+}; solid circles, left panel). This rise is blocked by administration of an angiotensin receptor blocker (open circles, left panel). A genetic knockout mouse strain with no AT1A receptor (AT_{1A}^{-/-}) has a lower SBP and no change after renal artery clipping (solid circles, right panel). No additional effect is noted with an angiotensin receptor blocker (open squares, right panel). These data reinforce the essential role of the renin-angiotensin system and an intact angiotensin 1 receptor for the development of renovascular hypertension. (Modified from Cervenka L, Horacek V, Vaneckova I, et al. Essential role of AT1-A receptor in the development of 2K1C hypertension. *Hypertension*. 2002;40:735–741.)

MECHANISMS SUSTAINING RENOVASCULAR HYPERTENSION

For more than a century, the kidney has been recognized has a source of multiple pressor substances. Recruitment of numerous pathways that raise arterial pressure increases the complexity of managing hypertension in this setting. Identification of components of the renin-angiotensin system provided a crucial link to understanding several of these systems. Circulating renin is derived primarily from the kidney in response to a reduction in renal perfusion pressure, detected by loss of afferent arteriolar stretch.¹⁹ Renin itself has biologic activity directed mainly at the enzymatic release of angiotensin I (Ang I) from its circulating substrate, angiotensinogen, in plasma and possibly other sites.⁵ Two more peptides are cleaved through the action of ACE to produce Ang II. Generation of Ang II in plasma occurs mainly during passage through the lung. Hence, the signal of reduced kidney pressures is amplified and transmitted through a major systemic vasopressor system and is one mechanism whereby renovascular hypertension develops.

RENIN-ANGIOTENSIN SYSTEM

Following its discovery, the renin-angiotensin system has been found to have widespread effects beyond its vasoconstrictor action in renovascular hypertension.^{20,21} Some actions of Ang II are illustrated in Fig. 47.7. Activation of this system increases vascular resistance, sodium retention, and aldosterone stimulation. Further studies have indicated that complex interactions between Ang II and tissue and cellular systems occur, leading to vascular remodeling, left ventricular

hypertrophy, and activation of inflammatory and fibrogenic mechanisms.

Hypertension and peripheral vasoconstriction reflect further complex interactions between angiotensin and other vasoactive systems. RVD leads to disturbances in sympathetic nerve signaling, which may differ between one-kidney and two-kidney models. Muscle sympathetic nerve activity is increased in humans with renovascular hypertension, and blood pressure responses to adrenergic inhibition are amplified.²²

A major transition occurs with recruitment of altered oxidative stress within the systemic vasculature, leading to increased oxygen free radicals.²³ Experimental animals with two-kidney, one-clip hypertension develop an increase in oxidative stress (reflected by F2 isoprostanes) that can be reversed in part with angiotensin blockade and/or antioxidants.²⁴ Vascular injury itself produces disturbances in endothelium-derived mechanisms, such as endothelin (ET) production and vasodilator systems, including prostacyclin.²⁵

The roles of endothelial dysfunction and increased oxidative stress in human RVD have been supported by clinical studies in patients with both atherosclerotic and fibromuscular RAS showing reversal after revascularization.^{22,26} Studies in a swine model of RVD have demonstrated an important interaction between atherosclerosis induced by cholesterol feeding and RVD.^{27,28} Endothelial dysfunction in the kidney and tissue fibrosis are augmented by the atherosclerotic process. Some of these changes can be abrogated experimentally by intensive therapy with statins and antioxidants. These data are supported by observations of a rise in nitric oxide (NO) levels and reduction in malondialdehyde within 24 hours of endovascular revascularization in atherosclerotic RVD.²⁹

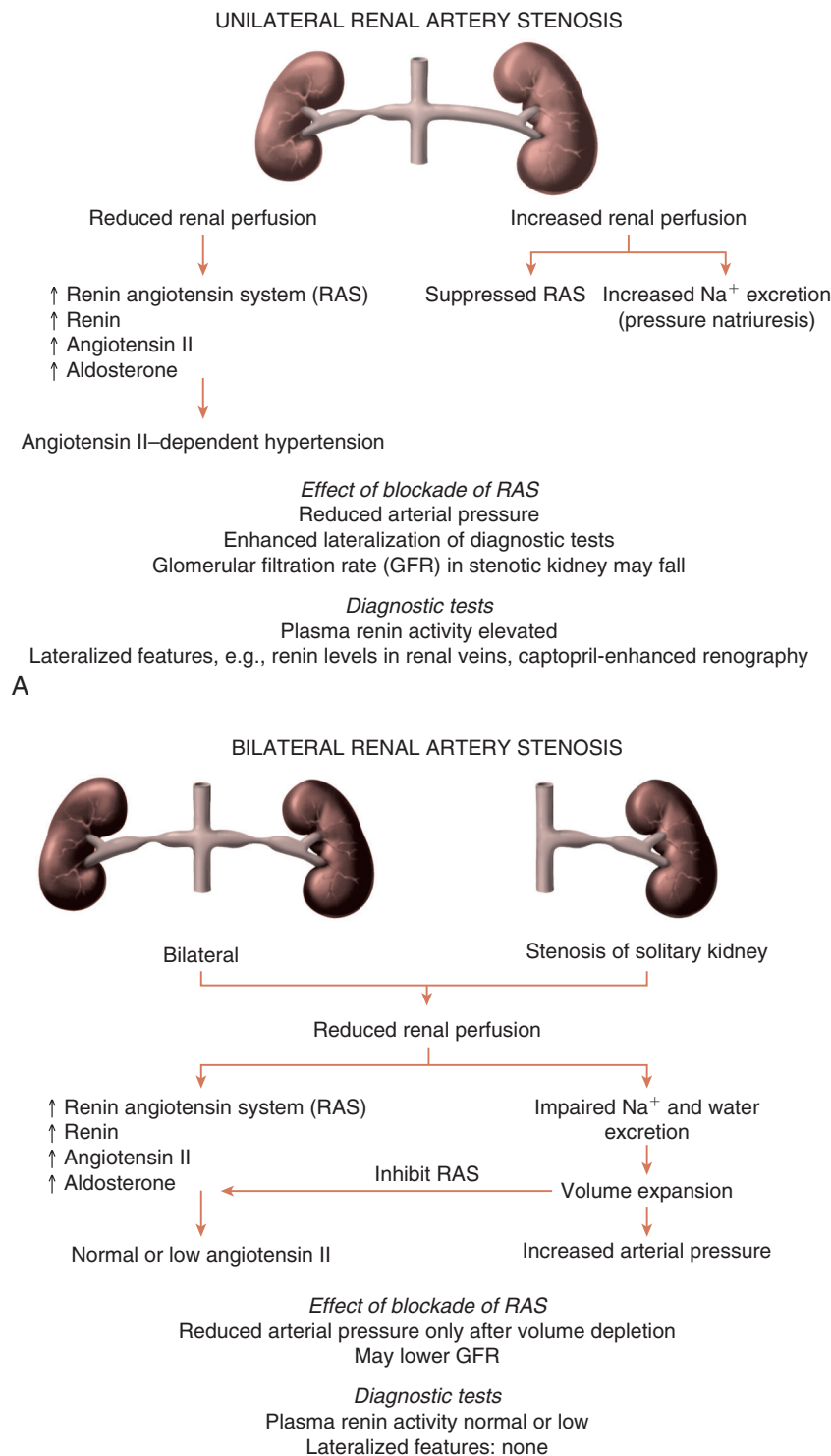


Fig. 47.6 Schematic view of two-kidney (A) and one-kidney (B) renovascular hypertension. These models differ by the presence of a contralateral kidney exposed to elevated perfusion pressures in two-kidney hypertension. The nonstenotic kidney tends to allow pressure natriuresis to ensue and produces ongoing stimulation of renin release from the stenotic kidney. The one-kidney model eventually produces sodium retention and a fall in renin, with minimal evidence of angiotensin dependence unless sodium depletion is achieved. *GFR*, Glomerular filtration rate; *RAS*, renin angiotensin system.

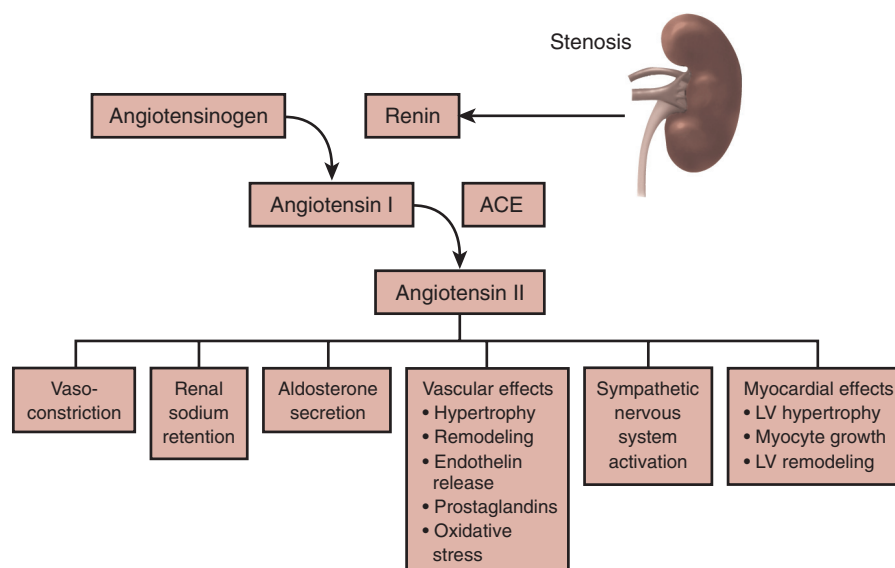


Fig. 47.7 Schematic view of activation of the renin-angiotensin system beyond a renal artery stenotic lesion. The generation of circulating and local angiotensin II leads to widespread effects, including sodium retention, efferent arteriolar vasoconstriction, and elevated systemic vascular resistance. Additional studies implicate angiotensin II in many other pathways of vascular and cardiac smooth muscle remodeling, activation of inflammatory and fibrogenic cytokines, coagulation factors, and induction of other vasoactive systems. *ACE*, Angiotensin-converting enzyme; *LV*, left ventricular.

Studies have suggested that oxidative stress can be reversed by infusion of antioxidants and successful revascularization.²⁰ Experimental infusion of agents that protect mitochondria from reactive oxygen species by inhibiting opening of mitochondrial transition pores has been associated with improved recovery of microvascular structures and renal function after revascularization.³⁰

PHASES OF DEVELOPMENT OF RENOVASCULAR HYPERTENSION

Experimental models of renovascular hypertension have indicated that mechanisms sustaining hypertension change over time (Fig. 47.8).³¹ Even before occlusive lesions are evident, accumulation of inflammatory monocytes in the renal vascular wall and elevation of circulating cytokines are evident.³² An early pressor phase is characterized by elevated circulating renin activity and hypertension, both of which return to normal after removing the vascular lesion (e.g., the clip). A second phase is described with a return of circulating renin activity to normal or low levels, during which hypertension persists and blood pressure can still respond to clip removal. A third phase occurs, during which removal of the clip no longer leads to a reduction in arterial pressure. These observations highlight the transition between differing mechanisms of vascular resistance, some of which become independent of renal perfusion. Some authors have suggested that microvascular injury in the contralateral kidney sustains hypertension in the third phase.³¹ Studies in a swine model have indicated that the fall in renin activity follows the transition to mechanisms related to oxidative stress, with persistent elevation of oxidative metabolites such as isoprostanes.²³ Whether these phases translate directly to human RVD is not well known.

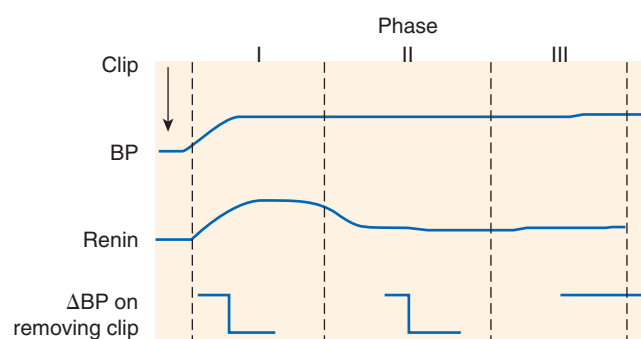


Fig. 47.8 Schematic depiction of phases observed in experimental renovascular hypertension. Initially, high levels of renin activity fall in the chronic phase, although removal of the renal artery clip corrects hypertension. These observations support the concept of renal artery stenosis leading to the recruitment of additional structural and pressor mechanisms after initial activation of the renin-angiotensin system. The degree to which human renovascular hypertension follows these patterns is not well known. *BP*, blood pressure. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

MECHANISMS OF ISCHEMIC NEPHROPATHY

Reduced renal perfusion beyond critical stenosis ultimately leads to a loss of viable kidney function, as illustrated in Fig. 47.9.³³ Patients with stenosis affecting the entire renal mass develop reduced blood flow and glomerular filtration when poststenotic pressures fall below the range of autoregulation. This process can be reversible if pressure is restored and/or the vascular lesion is removed. The mechanisms whereby this occurs differ from those that govern the development

**BLOOD PRESSURE, RENAL PLASMA FLOW, AND GFR IN PATIENTS WITH BILATERAL RAS
DURING NITROPRUSSIDE INFUSION BEFORE AND AFTER REVASCULARIZATION**

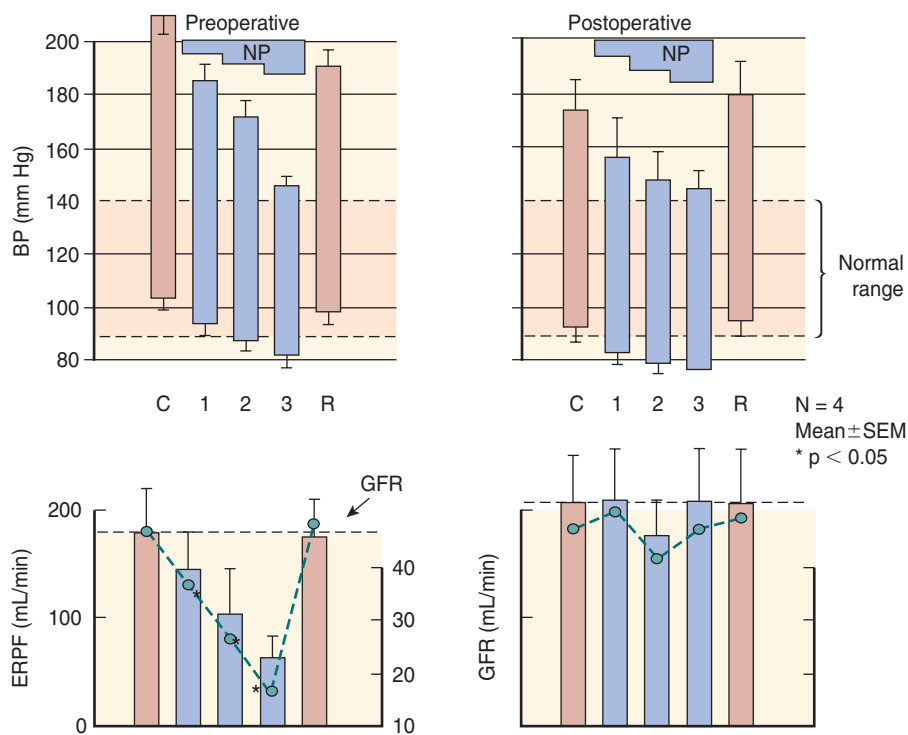


Fig. 47.9 Effective renal plasma flow (ERPF) and glomerular filtration rate (GFR) in patients with critical bilateral renal artery stenosis during pressure reduction with sodium nitroprusside (NP). Reducing systemic blood pressure to normal levels produced a reversible fall in both plasma flow and GFR. Repeat studies in the same patients (*right-hand panel*) after unilateral surgical revascularization demonstrates that the sensitivity of blood flow and GFR to pressure reduction can be reversed. SEM, Standard error of the mean. (From Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al., eds. *Brenner and Rector's: The kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.)

of hypertension. The term *ischemic nephropathy* may itself be a misnomer.^{15,34}

ADAPTIVE MECHANISMS TO REDUCED RENAL PERFUSION

Unlike brain or cardiac tissue, the kidney is highly perfused with oxygenated blood, consistent with its function as a filtering organ. Measurements of renal vein oxygen saturation and erythropoietin in patients with high-grade renovascular lesions indicate that whole-organ ischemia is rarely present.³⁵ It has long been postulated that local areas of reduced oxygen delivery in the kidney predispose to injury. The kidney maintains autoregulation of blood flow in the face of reduced arterial diameters of up to 75%. Under basal conditions, renal blood flow is among the highest of all organs reflecting its filtration function. Less than 10% of delivered oxygen is sufficient to maintain overall renal metabolic needs. Imaging using blood oxygen level–dependent (BOLD) magnetic resonance imaging (MRI) has demonstrated markedly reduced oxygenation in deeper medullary regions, despite preserved cortical oxygenation in normal kidneys and in patients with RAS.³⁴

Much of the oxygen consumption in the kidney occurs during solute transport. Studies of patients with RAS sufficient to reduce blood flow, kidney volume, and glomerular filtration rate (GFR) have indicated remarkable preservation of tissue

oxygenation using BOLD MRI³⁶ (Fig. 47.10). This stability of oxygen gradients is due partly to the surplus of oxygenated blood perfusing the cortex and to reduced oxygen consumption as a result of reduced solute filtration leading to reduced reabsorptive work of the affected kidney.^{37,38} When severe, reduced blood flow leads to the accumulation of deoxygenated molecular hemoglobin (Fig. 47.11).^{39,40} Under conditions of impaired renal perfusion, oxygen delivery may be maintained by the development of collateral vessels associated with intrarenal redistribution of blood flow. The renal medulla normally functions at levels close to hypoxia and is therefore sensitive to acute changes in perfusion.³⁷ During chronic reduction of blood flow, the medulla is partially protected by adaptive maintenance of tissue perfusion at the expense of cortical blood flow, which parallels whole-kidney renal blood flow.⁴¹ Hence, gradual reduction of renal perfusion pressures allows recruitment of protective mechanisms that remain incompletely understood and that lead to different functional and morphologic changes from those observed after acute ischemic injury.

More severe vascular occlusion produces cortical hypoxia as these adaptive measures are overwhelmed, and tissue injury ensues.³⁸ Eventually, structural atrophy of the renal tubules occurs, partly due to necrosis and apoptosis.⁴² The latter is an active, programmed form of cellular death, which appears to be closely regulated and differs from tissue necrosis. Tubular

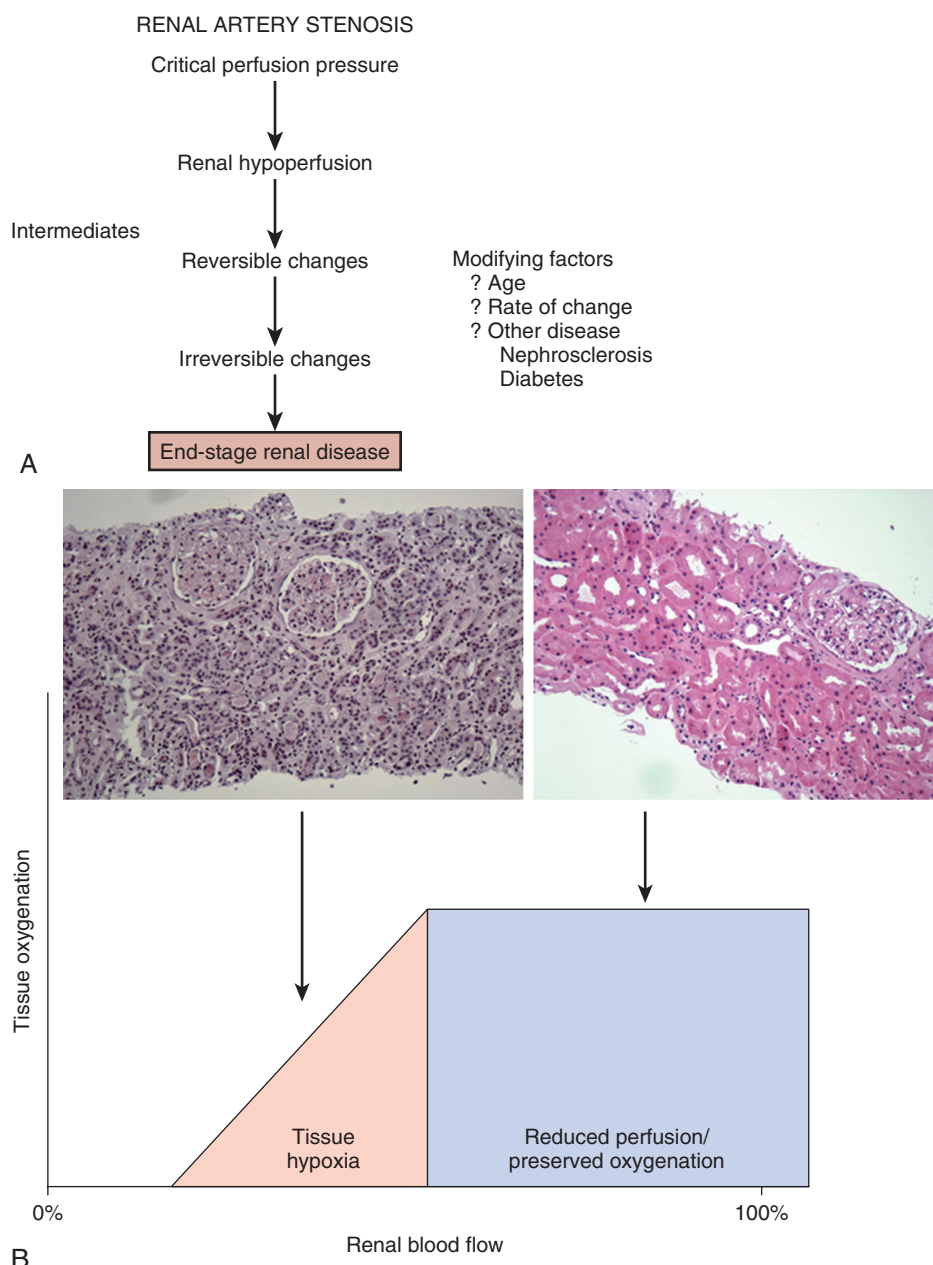


Fig. 47.10 (A) General clinical paradigm that a reduction in the glomerular filtration rate (GFR) related to hypoperfusion of the kidney can be reversible, but can undergo a transition to irreversible injury eventually that no longer responds to restoration of blood flow alone. (B) Schematic depiction of the adaptation of tissue oxygenation in the kidney to moderate reductions in blood flow. Lowering renal blood flow up to 30% to 40% in human subjects (*right side*) is associated with preservation of both oxygen gradients (*lower graph*) and tissue histology (*upper images*). The fact that such adaptation occurs partly explains the stability of kidney function during antihypertensive drug therapy that lowers systemic pressure and renal blood flow in patients with atherosclerotic renal artery stenosis. Eventually, more severe reductions of blood flow overwhelm “adaptation,” leading to overt tissue hypoxia (*left side*). Transjugular biopsy samples from patients with cortical hypoxia demonstrate loss of tubular structures and accumulation of inflammatory mmmT cells and macrophages with diffuse activation of transforming growth factor beta (TGF- β). (Magnification: low power: $\times 40$.) (Modified from Gloviczki ML, Keddis MT, Garovic VD, et al. TGF expression and macrophage accumulation in atherosclerotic renal artery stenosis. *Clin J Am Soc Nephrol*. 2013;8(4):546–553; and Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol*. 2015;26:2074–2080.)

atrophy is potentially reversible, and the kidney maintains the capacity for tubular cell regeneration under many conditions. These features support the concept that underperfused kidney tissue sometimes can achieve a “hibernating” state capable of restoring function if blood flow is restored.⁴³ In late stages, histologic examination demonstrates reduced glomerular volume, loss of tubular structures near under-

perfused glomeruli, and areas of local inflammatory reaction, as illustrated in Fig. 47.10B.⁴⁴

MECHANISMS OF TISSUE INJURY IN AZOTEMIC RENOVASCULAR DISEASE

Reduction of blood flow to the kidney activates numerous pathways of vascular and tissue injury, including increased

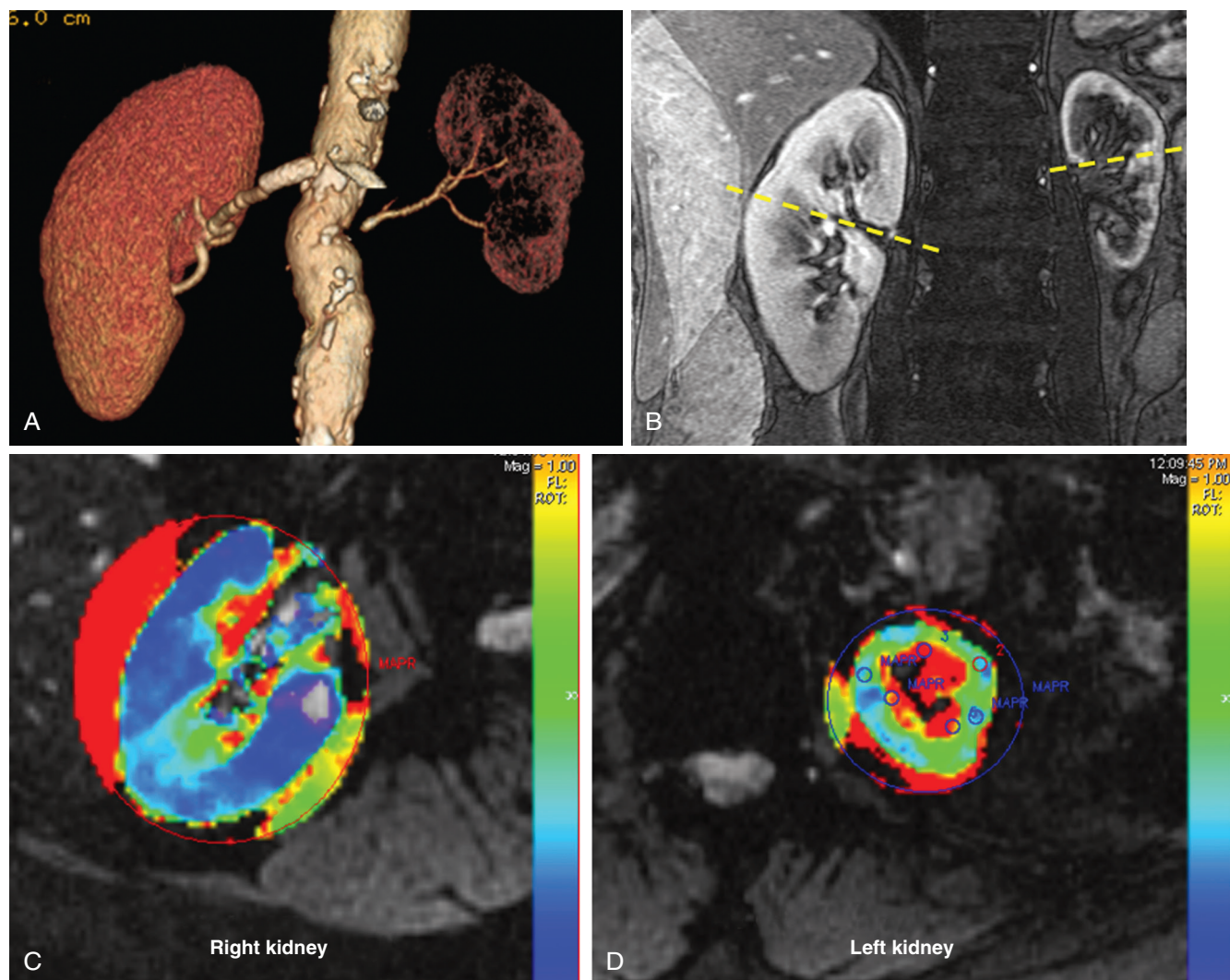


Fig. 47.11 (A) Computed tomography angiogram of high-grade vascular occlusion leading to left kidney atrophy while the right kidney is well perfused beyond an endovascular stent. (B) T2 signals from magnetic resonance imaging (MRI) illustrate a gradient of perfusion from the cortex to lower levels in the deeper medullary sections, especially in the left kidney (*dashed line* depicts the level of axial sections shown in panels [C] and [D]). (C and D) Mapping of the blood oxygen level–dependent (BOLD) magnetic resonance R2* levels (a function of deoxyhemoglobin) illustrates the normal gradient in the right kidney from the cortex (low R2*, *blue*) to deeper medullary segments that have progressively higher levels of R2* (*green to red*). The R2* map of the left kidney (D) illustrates overt cortical hypoxia evident beyond critical vascular occlusion associated, with a larger fraction of the slice having high R2* values (*red*) and fractional medullary hypoxia.

Ang II, ET release, and oxidative stress, as noted previously. Experimental studies have demonstrated complex interactions with renal microvessels, leading to vascular rarefaction (Fig. 47.12), stimulation of inflammatory signaling, and cellular infiltration in injured renal tubules.⁴⁴ Under the right conditions, these factors trigger the ongoing release of inflammatory cytokines, which eventually activates fibrogenic mechanisms and tissue fibrosis.

The role of Ang II during renal hypoperfusion is complex. The generation of Ang II acts to raise perfusion pressure and protect glomerular filtration through efferent arteriolar constriction.⁴⁵ Ang II induces hypertrophy and hyperplasia in several cell types, in addition to direct stimulation of local hormone production and ion transport.⁴⁶ Experimental infusion of Ang II leads to parenchymal renal injury, with focal and segmental glomerulosclerosis.⁴⁷ ACE inhibition and

Ang II receptor blockade in several experimental models diminish renal cell proliferation and suppress infiltration of mononuclear cells, which triggers expression of extracellular matrix proteins and progressive nephrosclerosis.⁴³ Ang II may also participate in vascular smooth muscle cell growth, platelet aggregation, generation of superoxide radicals, activation of adhesion molecules and macrophages, induction of gene transcription for protooncogenes, and oxidation of low-density lipoproteins (LDLs).⁴⁸ These observations demonstrate the multiple roles of Ang II in the adaptation and maintenance of kidney function and the modulation of many steps along the cascade toward progressive renal injury.

Experimental studies have suggested an independent effect of hypercholesterolemia in modifying parenchymal renal injury in ischemic nephropathy. Cholesterol feeding is used as a model of early atherosclerosis and itself alters renal

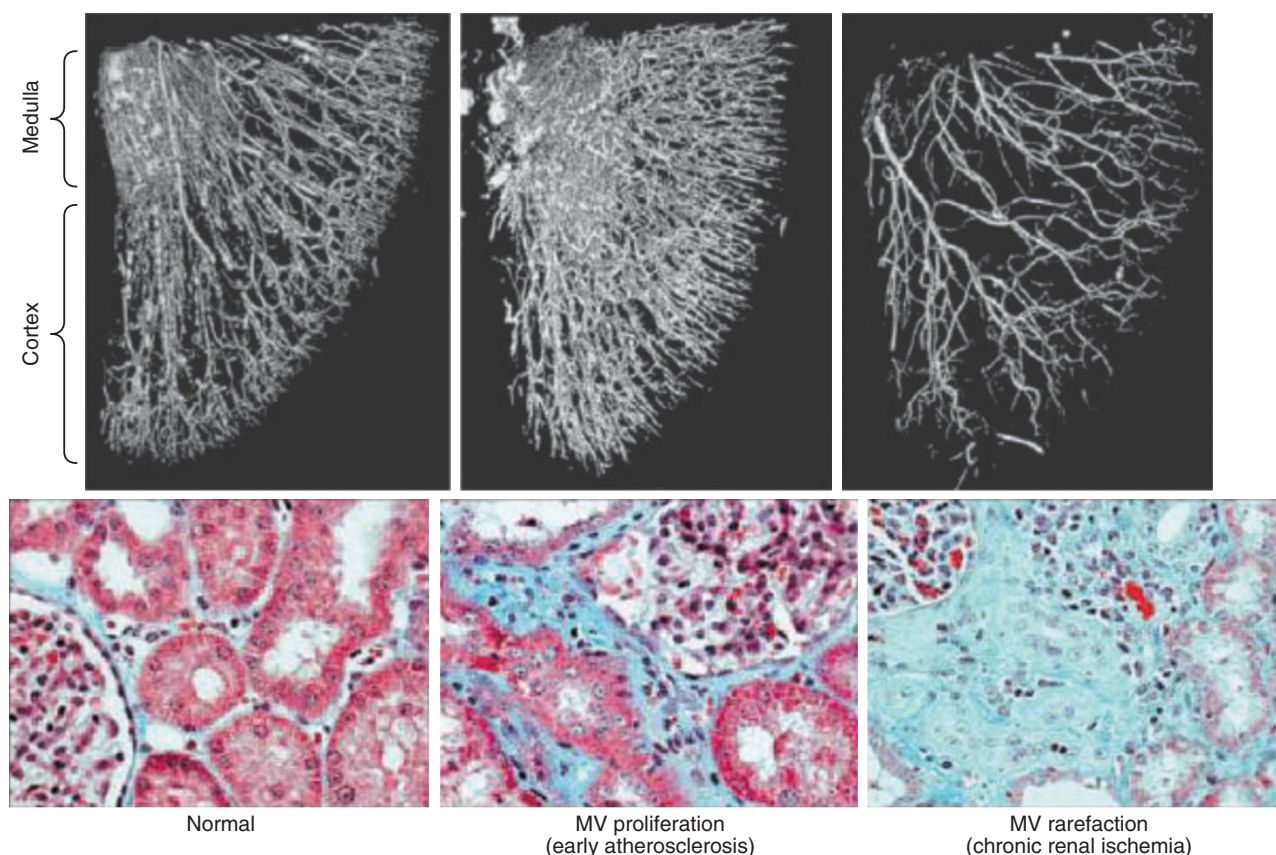


Fig. 47.12 Upper panels, Micro-computed tomography reconstructions of vessels in experimental atherosclerosis and superimposed large-vessel renovascular disease at an early stage (middle) and after chronic ischemia (right), compared with normal (left). Lower panels, Corresponding renal histology (trichrome stain $\times 400$). Complex microvascular dysfunction and rarefaction develop in poststenotic kidneys. These are accelerated and/or modified by angiogenic stimuli, oxidative stress pathways, and a variety of cytokines, leading eventually to interstitial fibrosis. Studies in experimental animals suggest that angiogenic stimuli such as intrarenal infusion of endothelial progenitor cells or mesenchymal stem cells may offer the potential to repair microvascular injury. MV, microvascular. (From Lerman LO, Chade AR. Angiogenesis in the kidney: a new therapeutic target? *Curr Opin Nephrol Hyper*. 2009;18:160–165.)

vascular reactivity to acetylcholine and changes renal tubular functional characteristics.⁴⁹ Levels of oxidized LDLs rise in this model, associated with markers of oxidative stress and activation of tissue nuclear factor kappa B (NF- κ B), transforming growth factor beta (TGF- β), and inducible nitric oxide synthase. The effect of these changes in producing kidney fibrosis is magnified in the presence of RAS. Many of these effects can be reduced experimentally by ET blockade, antioxidants, statins,⁵⁰ or mitochondrial protection.⁵¹

The vascular endothelium is a source of multiple vasoactive factors, the most widely recognized of which are NO and ET. Endothelial NO is synthesized from L-arginine by a family of NO synthases and participates in the regulation of kidney function by counteracting the vasoconstrictor effects of Ang II.⁵² In addition to its effects on blood flow and tubular reabsorption of sodium, NO inhibits the growth of vascular smooth muscle cells, mesangial cell hypertrophy and hyperplasia, and synthesis of extracellular matrix.⁵²

A reduction in renal perfusion leads to diminished shear stress distal to the stenosis.⁵³ This condition reduces the production of NO and accelerates the release of renin and generation of Ang II in the stenotic kidney. Hence, the effects of NO are diminished in the poststenotic kidney, allowing predominance of intrarenal vasoconstrictors, including Ang

II and vasoconstrictor prostaglandins, such as thromboxane.⁵⁴ The antithrombotic effects and the inhibition of responses to tissue injury of NO are also lost. A direct consequence of reduced perfusion in the poststenotic kidney is progressive rarefaction of small vessels in both the cortex and medulla, eventually producing tubular collapse⁵⁵ (see Fig. 47.12).

Glomerulosclerosis is a late event and usually reflects severe loss of the GFR. Experimental studies infusing autologous endothelial progenitor cells and/or vascular endothelial growth factors (VEGF) into the renal artery have demonstrated that functional recovery of GFR and restoration of vascular function can sometimes be achieved, suggesting that angiogenesis may be an achievable goal in this vascular bed.^{56,57}

The ET peptides are potent and long-lasting vasoconstrictor peptides produced and released from endothelial cells. ET itself is released from renal epithelial cells after stimulation with a variety of substances, such as thrombin and local cytokines, including TGF- β , interleukin-1, and tumor necrosis factor (TNF).⁵⁸ Renal ischemia is a potent stimulus for expression of ET-1 in the kidney. Sustained vascular effects of ET may contribute to hypoperfusion that lasts well beyond the vascular insult in postischemic kidneys.

The kidney is a rich site for the production of prostaglandins, which are cyclooxygenase derivatives of arachidonic

acid. These mediators are produced in arteries, arterioles, and glomeruli in the cortex, where they have important actions to maintain renal blood flow and filtration, particularly under conditions of elevated levels of Ang II.⁵⁹ Enhanced synthesis of prostacyclin (prostaglandin I₂ [PGI₂]) and prostaglandin E₂ (PGE₂) occurs during tissue hypoperfusion and ischemia, which may protect against some forms of hypoxic injury.⁶⁰ Conversely, thromboxane A₂ (TXA₂) is a vasoconstrictor prostaglandin that lowers the GFR by reducing renal plasma flow and can accelerate structural renal damage.⁶¹ It is stimulated by the production of Ang II and reactive oxygen species and may in turn modify the hemodynamic properties of Ang II. TXA₂ modulates ET in its actions on vascular permeability, which may contribute to interstitial matrix composition and target organ damage. Blockade of TXA₂ receptors in some models can reduce the severity of experimental tissue damage, including acute ischemic injury.

The term *oxidative stress* refers to an imbalance between tissue oxygen radical-generating systems and radical scavenging systems toward prooxidant species.⁶² This process increases the presence and toxicity of reactive oxygen species, which in turn can promote the formation of vasoactive mediators including ET-1, leukotrienes, and prostaglandin F₂ alpha and isoprostanes, which are products of lipid peroxidation. As noted previously, these mediators affect renal function and hemodynamics, both by inducing renal vasoconstriction and by changing glomerular capillary ultrafiltration characteristics.⁶³ Reactive oxygen species themselves can magnify ischemic renal injury by causing lipid peroxidation of cell and organelle membranes. These disrupt structural integrity and the capacity for cell transport and energy production, particularly in the proximal tubule. Other cytokine pathways, including activation of NF- κ B and growth factors, may play a role.⁶⁴

The role of TGF- β merits emphasis. It belongs to a family of polypeptides that regulate normal cell growth, development, and tissue remodeling after injury.⁶⁵ TGF- β is an important and ubiquitous fibrogenic factor, which modifies extracellular matrix synthesis by glomerular and extraglomerular mesenchymal cells.⁶⁵ These actions modify tissue healing and progression to advanced renal failure. TGF- β is essential for tissue repair after many forms of injury, including ischemic conditions during which it participates in restoring extracellular matrix in proximal tubular basement membranes. Activation of the AT₁ receptor stimulates the generation of TGF- β , which plays a major role in tissue fibrosis through increases in type IV collagen deposition. TGF- β acts synergistically with ET and has interactions with platelet-derived growth factor (PDGF), interleukin-1, and basic fibroblast growth factor in progressive interstitial fibrosis.⁶⁶ Investigators have proposed that some forms of renal scarring represent an overabundance of TGF- β activity due to failure to suppress its activity after repair of an original injury.⁶⁷ Activation of TGF- β develops in experimental models of RAS and is magnified by hypercholesterolemia. Biopsies from patients with atherosclerotic RAS demonstrate diffuse TGF- β staining and macrophage infiltration.⁶⁸ Conversely, animals lacking downstream effector pathways for TGF- β (e.g., Smad3 knockout mice) are protected from parenchymal injury in RVD models.⁶⁹

Although whole-kidney oxygen saturation and delivery remain preserved in the poststenotic kidney, it is inescapable that local areas in the kidney are exposed to at least

intermittent recurrent ischemia. The potential for repetitive acute renal injury to induce long-term irreversible fibrosis is evident from studies of acute heme protein exposure.⁷⁰ The hallmark of acute ischemia is a rapid decline in cellular adenosine triphosphate (ATP), which in turn allows the accumulation of intracellular calcium, activation of phospholipases, and generation of oxygen free radicals.⁷¹

Tissue ischemia appears to be a common denominator in many forms of progressive tubulointerstitial injury.⁷² Such injury is associated commonly with interstitial inflammatory reactions and activation of fibroblasts and heat shock proteins. Disruption of the tubular epithelium alters the antigenic profile of these cells, initiating a cell-mediated immune response, sometimes associated with B lymphocyte, T lymphocytes, and macrophage infiltrates.⁷³ As noted, sustained tubulointerstitial injury leads to increased TGF- β production, enhanced expression of plasminogen activator inhibitor 1 (PAI-1), tissue inhibitor of metalloprotease-1 (TIMP-1), alpha 1 (IV) collagen, and fibronectin-EDA and thus to increased synthesis of extracellular matrix.⁷⁴

Many of the mechanisms mentioned interact with each other. Taken together, the kidney is subject to a wide variety of vasoactive and inflammatory mediators, which can be disturbed by loss of blood flow and perfusion pressure. These disturbances appear to activate a variety of fibrogenic and local destructive mechanisms, which can lead to irreversible parenchymal damage in the kidney. For further discussion of vasoactive molecules in the kidney, see Chapter 11.

CONSEQUENCES OF RESTORING RENAL BLOOD FLOW

As illustrated in Fig. 47.9A, restoring renal perfusion can allow recovery of renal function if accomplished when these changes remain reversible. At some point, both inflammatory and fibrogenic mechanisms no longer to respond appear with recovery of renal function.

RENAL REPERFUSION INJURY

The course of recovery after restoration of blood flow in an underperfused kidney depends on the extent and duration of the reperfusion injury, in addition to the adequacy of reperfusion.⁷⁵ Paradoxically, some tissues subjected to ischemia undergo morphologic and functional changes, which worsen during the reperfusion phase.⁷⁶ This is thought to reflect vascular endothelial damage and activation of leukocytes, which may be primed to obstruct distal capillaries after restoration of perfusion pressure contributing to a so-called no-reflow phenomenon. Under experimental conditions, reperfusion injury appears to require major degrees of prooxidant stress with excess PGF₂ alpha isoprostanes and free oxygen radicals, particularly with a deficit of NO.⁷⁷ Hence, antioxidants and reactive oxygen metabolite scavengers improve outcomes following experimental reperfusion. In the kidney, ischemia-reperfusion injury is most pronounced in the proximal tubules, with local necrosis and tubular obstruction, as observed in acute tubular necrosis (ATN). Studies in experimental RVD in swine have suggested that pretreatment with a mitochondrial transition pore-inhibiting agent allows improved recovery of blood flow, preservation of microvascular integrity, and improved function, consistent with prevention of ischemia and reperfusion injury.⁵¹

Ang II may participate in some of these changes because the activation of AT1 receptors impairs glomerular filtration in the postischemic kidney.⁷⁸ Local imbalance of NO production in the kidney and the dual actions of NO have the potential drawback of accelerating reoxygenation injury and initiating lipid peroxidation.⁷⁶ However, systemic treatment with NO donors improves renal function and blunts local inflammation before reperfusion in some conditions.⁷⁹

EPIDEMIOLOGY OF RENAL ARTERY STENOSIS AND RENOVASCULAR HYPERTENSION

The syndrome of renovascular hypertension can be produced by a wide variety of lesions affecting renal blood flow. In unselected mild to moderate hypertensive populations, the frequency appears to be between 0.6% to 3%, whereas in a referral clinic of treatment-resistant hypertensive patients, the prevalence may exceed 20%.⁸⁰ Some specific example lesions producing renal ischemia are listed in [Box 47.1](#). A rapidly developing form of this disorder can be seen after spontaneous or traumatic renal artery dissection or iatrogenic occlusion, sometimes caused by endovascular aortic stent grafts.⁹ Most stenotic lesions are fibromuscular diseases or atherosclerotic RAS. The reported prevalence depends heavily on differences between patient groups studied. As noted later, the prevalence of anatomic RAS far exceeds that of renovascular hypertension.

Clinical Relevance

Renal artery stenosis is much more common than renovascular hypertension. Not all stenoses are associated with hypertension.

Fibromuscular disease (FMD) commonly refers to one of several noninflammatory conditions affecting the intima or fibrous layers of the vessel wall. In some cases, multiple layers of the vessel wall may be affected. Reports from arteriograms obtained in “normal” renal organ donors have indicated that 3% to 5% of individuals may have one of these lesions, many of which are present at an early age and do not affect renal blood flow or arterial pressure.⁸¹ Such lesions can lead to renovascular hypertension, sometimes associated with dissection or progression. Smoking is a risk factor for disease progression. Medial fibroplasia is the most common subtype, often associated with a string-of-beads appearance, as illustrated in [Fig. 47.13A](#).⁸² These lesions consist primarily of intravascular webs, each of which may have only moderate hemodynamic effect. The combination of multiple webs in series, however, can impede blood flow characteristics and activate responses in the kidney to reduce perfusion. FMD appears in the renal arteries in 65% to 70% of cases and in cerebral arteries in 25%.⁸¹ Both renal and cerebral vessels may be abnormal in 10% to 25% of patients. The preponderance of hypertensive cases requiring vascular intervention occur in women, with a bias toward the right renal artery.⁸³



Fig. 47.13 (A and B) Angiographic appearance of medial fibroplasia with serial intravascular webs and small aneurysmal dilations between them. These lesions appear in the midportion of the vessel, have a predilection for the right renal artery, and are most commonly found in women. As shown in B, these lesions can often be improved substantially by balloon angioplasty.

FMD lesions are typically located away from the origin of the renal artery, often in the midportion of the vessel or at the first arterial bifurcation.⁸¹ Some of these expand to develop small vascular aneurysms. Although less common, other dysplastic lesions, particularly intimal hyperplasia, can progress and lead to renal ischemia and atrophy. Although loss of renal function is unusual with FMD, quantitative imaging of cortical and medullary kidney volumes indicate that parenchymal thinning occurs both in the stenotic and contralateral kidneys beyond FMD.⁸⁴ Interventional studies have suggested that among patients referred for renal revascularization for hypertension, FMD accounts for 16% or less.⁸⁵

Atherosclerosis affecting the renal arteries is the most common renovascular lesion in the United States. Atherosclerotic renal artery stenosis (ARAS) can be identified commonly in patients with disease affecting other vascular beds and may be magnified by inflammatory vascular injury.³² Population-based surveys have identified incidental RAS (>60% occlusion by Doppler criteria) in 6.8% of individuals older than 65 years in the United States.⁸⁶ A systematic review of imaging studies for other vascular conditions has confirmed that the prevalence of lesions with more than 50% luminal occlusion rises progressively with the extent of overall atherosclerotic burden.¹² Hence, patients undergoing coronary angiography have identifiable ARAS in 14% to 20% of cases.^{11,87,88} Studies of patients with peripheral vascular disease have indicated that 30% to 50% of such patients have renal artery lesions of some degree.¹² Table 47.1 summarizes multiple reports related to the coexistence of atherosclerotic lesions in various vascular territories. The prevalence of ARAS increases with age and with the presence of atherosclerotic risk factors such as elevated cholesterol levels, smoking, and hypertension. The probability of identifying high-grade renal arterial stenosis in hypertensive patients with azotemia rises from 3.2% in the sixth decade to above 25% in the eighth decade.⁸⁹ Additional population-based studies in the United States have confirmed that more than 70% of older patients

with RAS above 60% occlusion have clinical manifestations of cardiovascular disease.⁹⁰ The prevalence of ARAS may be rising as a result of more people surviving to older ages. These figures confirm previous postmortem observations indicating that many patients dying of cardiovascular disease have renal artery lesions at autopsy. They underscore the fact that some renal artery lesions remain undetected on clinical grounds for many years (see later).

The location of atherosclerotic disease is usually near the origin of the artery (Fig. 47.14A), although it can be observed anywhere. Many such lesions represent a direct extension of an

Table 47.1 Prevalence Rates of Atherosclerotic Renal Artery Stenosis ^a	
Vascular Disease	Prevalence Rate (%)
Suspected renovascular hypertension	14.1
Coronary angiography	10.5
With hypertension	17.8
Peripheral vascular disease	25.3
AAA	33.1
ESKD	40.8 (?) ^b
Congestive heart failure	54.1 (?) ^b

^a40 studies, 15,879 patients. Shown are prevalence rates of patients with vascular disease affecting other regional beds identified by angiography.

^b(?) marks refer to limited data suggesting high prevalence in unexplained advanced chronic kidney disease (CKD) in older subjects and patients with congestive heart failure.

AAA, Abdominal aortic aneurysm; ESKD, end-stage kidney disease.

Modified from de Mast Q, Beutler JJ. The prevalence of atherosclerotic renal artery stenosis in risk groups: a systematic literature review. *J. Hypertens.* 2009;27: 1333–1240.

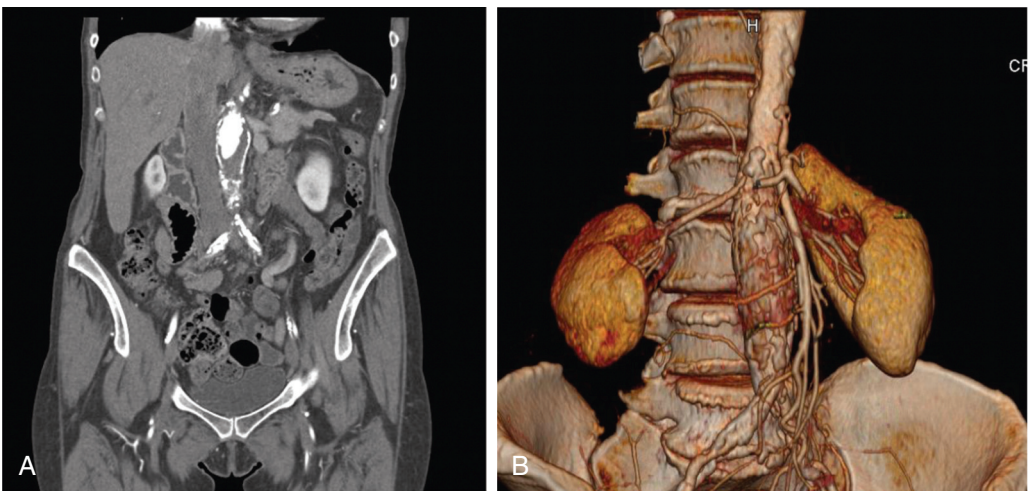


Fig. 47.14 (A and B) Atherosclerotic disease commonly affects the renal arteries and abdominal aorta. Shown is a computed tomography angiogram with high-grade stenosis to a small right kidney arising from diffuse aortic disease. (A) The coronal section demonstrates extensive thrombotic debris along the calcified aortic wall. (B) A reconstructed view of the aorta demonstrates diffuse calcification and early aortic aneurysm formation. The mean age of series presenting for renal revascularization for atherosclerotic disease has risen to more than 70 years. The decision about endovascular or surgical intervention in such cases must balance the hazards of aortic manipulation against the potential benefits regarding blood pressure control and/or renal function.

aortic plaque into the renal arterial segment. ARAS is strongly associated with preexisting hypertension, cardiovascular lipid risk, diabetes, smoking, and abnormal renal function.^{11,87}

CLINICAL FEATURES

RENOVASCULAR HYPERTENSION: FIBROMUSCULAR DISEASE VERSUS ATHEROSCLEROSIS

As noted earlier, renovascular hypertension may develop as a result of many lesions (see Box 47.1). The two most common are FMD and atherosclerosis. FMD represents several types of noninflammatory intimal or medial disorders of the vessel wall, commonly affecting midportions of the renal artery detected in individuals between 15 and 55 years of age.⁸² These lesions rarely lead to major renal functional loss, although some progression may be seen, particularly in smokers. FMD lesions most often appear clinically as hypertension of early onset (in those between 30 and 50 years of age) and unusual severity. Occasionally, it presents as hypertension during pregnancy. Many lesions respond well to percutaneous angioplasty (see Fig. 47.13B).⁹¹

By contrast, atherosclerotic lesions usually arise near the origin of the renal artery and amplify the risks of systemic atherosclerosis elsewhere. ARAS is commonly associated with a reduced glomerular filtration rates.⁹² Clinical manifestations fall across a wide spectrum (see Fig. 47.1) and are not related simply to the anatomic severity of the lesion.⁹³ Ambulatory blood pressure recordings indicate exaggerated systolic pressure variability and frequent loss of the circadian pressure rhythm,⁹⁴ commonly associated with left ventricular hypertrophy.⁹⁵ Sympathetic nerve traffic recordings indicate heightened adrenergic outflow.⁹⁶ A population-based study of 870 subjects older than 65 years has indicated that those with RAS had a two- to threefold increased risk for adverse cardiovascular events during the subsequent 2 years.⁹⁰ These data are supported by a review of Medicare claims data between 1999 and 2001 for a random sample of Medicare recipients older than 67 years.⁹⁷ The authors indicated that the incidence of newly identified atherosclerotic RVD was 3.7/1000 patient-years and was associated with preexisting peripheral and coronary disease. After detection, subsequent development of claims for heart disease, transient ischemic attack, renal replacement therapy, and CHF are 3- to 20-fold higher in such patients as compared with contemporaries without RVD.⁹⁷ Adverse cardiovascular events developed more than tenfold more commonly than the need for renal replacement therapy. As a result, a persistent controversy in cardiovascular disease has been how to manage clinically significant RAS as a modifying factor for cardiovascular outcomes. The major US trial of renal revascularization (CORAL) specifically targeted overall combined cardiovascular outcomes in treating this disorder (see later).⁹⁸ This controversy is compounded by changes produced by evolving medical therapy and changing population characteristics.

RENAL ARTERY STENOSIS

Manifestations of renal artery disease vary widely across a spectrum, as illustrated in Fig. 47.1, Box 47.2, and Table 47.2.

Box 47.2 Syndromes Associated With Renovascular Hypertension^a

1. Early- or late-onset hypertension (<30 years or >50 years)
2. Acceleration of treated essential hypertension
3. Deterioration of renal function in treated essential hypertension
4. Acute renal failure during treatment of hypertension
5. Flash pulmonary edema
6. Progressive renal failure
7. Refractory congestive cardiac failure

^aThese "syndromes" should alert the clinician to the possible contribution of renovascular disease in a given patient. Syndromes 5 to 7 are most common in patients with bilateral disease, many of whom are treated as "essential hypertension" until these characteristics appear.

Table 47.2 Clinical Features of Patients With Renovascular Hypertension^a

Clinical Feature	Essential HTN (%)	Renovascular HTN (%)
Duration <1 year	12	24
Age of onset >50 years	9	15
Family history of HTN	71	46
Grade 3 or 4 fundi	7	15
Abdominal bruit	9	46
Blood urea nitrogen > 20 mg/dL	8	16
Potassium <3.4 mEq/L	8	16
Urinary casts	9	20
Proteinuria	32	46

^aClinical features that differed ($P < .05$) between closely matched groups of 131 patients with essential versus renovascular hypertension taken from the Cooperative Study of Renovascular Hypertension in the 1960s. (From Simon N, Franklin SS, Bleifer KH, Maxwell MH. Clinical characteristics of renovascular hypertension. *JAMA*. 1972;220:1209–1218.) These observations underscore the potential severity of hypertension in candidates for surgery, but none of these features allows clinical discrimination with confidence. HTN, Hypertension.

This spectrum ranges from an incidental finding noted during imaging for other indications to advancing renal failure leading to the need for dialysis support. As lesions progress to more severe levels of occlusion, hypertension develops, and multiple manifestations may appear. As described previously, a number of mechanisms raise systemic arterial pressure and tend to restore renal perfusion pressures to levels close to baseline. Clinical features of patients with essential hypertension have been compared with those in patients subjected to revascularization for renovascular hypertension in the Cooperative Study in the 1960s are summarized in Table 47.2.⁹⁹ Many features, including short duration of hypertension, early age of onset, funduscopic findings, and hypokalemia, were more common in those with renovascular hypertension, but they have limited discriminating or predictive value.⁹⁹

As renal artery lesions progress to critical stenosis, they can produce a rapidly developing form of hypertension, which

may be severe and associated with polydipsia, hyponatremia, and central nervous system findings.¹⁰⁰ Such cases are usually seen with acute renovascular events, such as the sudden occlusion of a renal artery or branch vessel.

More commonly, RAS presents as a progressive worsening of preexisting hypertension, often with a modest rise in serum creatinine level. Because the prevalence of hypertension and atherosclerosis increases with age, this disorder must be considered particularly in older subjects with progressive hypertension. Some of the most striking examples of renovascular hypertension are older individuals whose previously well-controlled hypertension has deteriorated, with an accelerated rise in systolic blood pressure and target injury, such as stroke. Studies from hypertension referral centers in the Netherlands are typical in this regard. Of 477 patients undergoing detailed evaluation for RAS because of treatment resistance, 107 (22.4%) were identified with RVD (>50% stenosis by angiography).⁸⁵ Clinical features predictive of RAS included older age, recent progression, other vascular disease (e.g., claudication), an abdominal bruit, and elevated serum creatinine levels. The authors derived a multivariate regression equation of predictive features for the presence of angiographic RAS. They presented a clinical scoring system to determine the pretest probability of identifying renal artery disease (Fig. 47.15).⁸⁵ The strongest predictors include age and serum creatinine level. Clinical features alone could provide pretest predictive value nearly as accurate as radio-nuclide scans.⁸⁵

Declining renal function during antihypertensive therapy is a common manifestation of progressive renal arterial disease. Not surprisingly, blood flow and perfusion pressures to the kidney fall distally to a critical RAS. This can be worsened by reduction in systemic arterial pressure by any antihypertensive regimen. A reduced GFR during antihypertensive therapy has become particularly common since the introduction of ACE inhibitors and angiotensin receptor blockers (ARBs). A precipitous rise in serum creatinine levels soon after starting these agents may occur due to a loss of transcapillary filtration pressure produced by removing the efferent arteriolar vasoconstriction from Ang II. This particular “functional” loss of GFR is reversible if detected promptly and should lead the clinician to consider large-vessel RVD when it occurs.^{101,102} Clinically important changes in serum creatinine levels become apparent mainly when the entire renal mass is affected, such as with bilateral RAS or stenosis to a solitary functioning kidney. Most such patients tolerate the reintroduction of renin-angiotensin blockade when challenged after successful revascularization.¹⁰³

Other syndromes heralding occult RAS have become more commonly recognized. Among the most important are rapidly developing episodes of circulatory congestion (so-called “flash” pulmonary edema).¹⁰⁴ These usually arise in patients with hypertension and with left ventricular systolic function, which may be well preserved. Underlying arterial compromise may favor volume retention and resistance to diuretics in such cases. A sudden rise in arterial pressure impairs cardiac

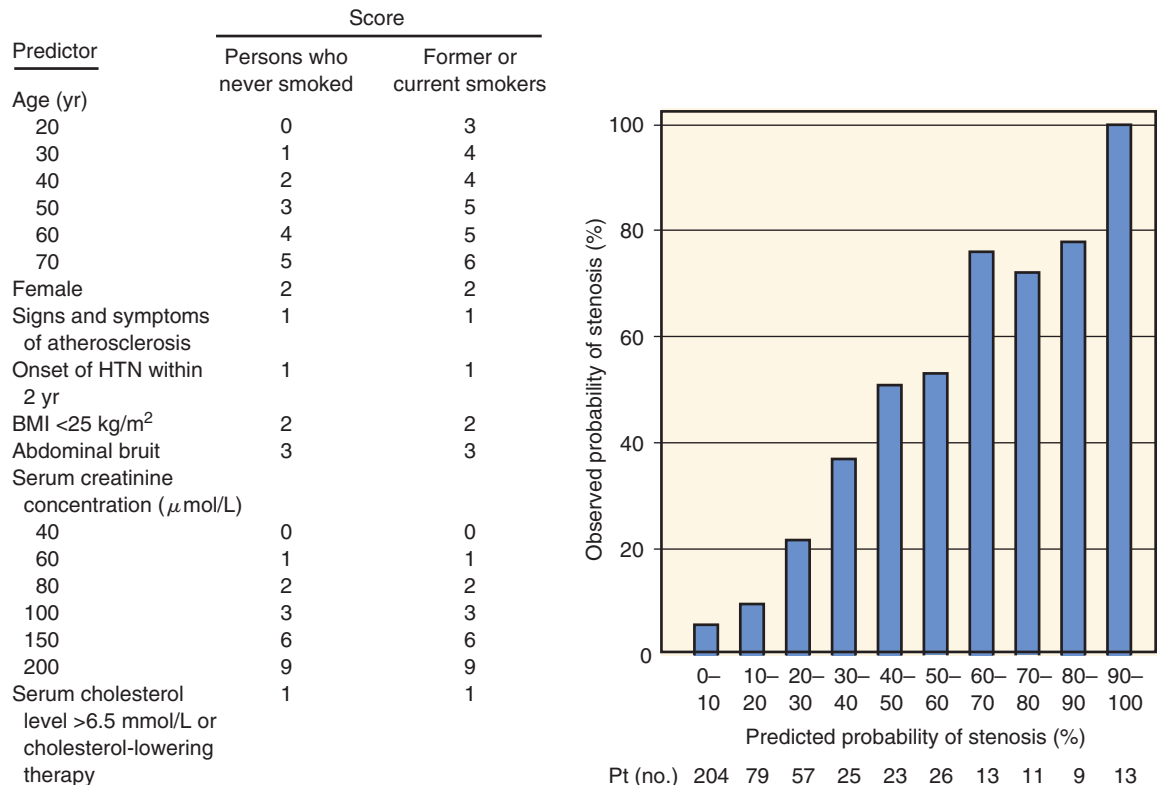


Fig. 47.15 Probability of identifying renal artery stenosis based on clinical features. These data were derived from 477 patients in referral centers for treatment-resistant hypertension (HTN) in the Netherlands. Overall prevalence was 22.4%, illustrating that even in “enriched” patient populations, renovascular disease is not present in the majority. Clinical features allowed selection of patients for testing with a relatively high pretest probability of disease, which affects the validity of testing schemes. (From Krijnen P, van Jaarsveld BC, Steyerberg EW, et al. A clinical prediction rule for renal artery stenosis. *Ann Intern Med.* 1998;129:705–711.)

function due to rapidly developing diastolic dysfunction. These episodes tend to be rapid both in onset and in resolution. Patients with treatment-resistant CHF, often with reduced arterial pressures, may also harbor unsuspected RVD. Restoration of renal blood flow in such patients can improve volume control and sensitivity to diuretics, with a lower risk of azotemia during therapy.¹⁰⁵ A similar sequence of events may produce symptoms of “crescendo angina” from otherwise stable coronary disease.¹⁰⁶ Registry data have indicated that patients with episodic pulmonary edema with RAS have substantially increased hospitalization and mortality rates, which can be reduced with successful revascularization.^{107,108}

Another clinical presentation of RAS is advanced renal failure, occasionally at end stage, requiring renal replacement therapy. This manifestation has been controversial partly because it raises the possibility of an undetected, potentially reversible form of chronic kidney disease. As discussed earlier, this is designated by some as ischemic nephropathy or azotemic RVD¹⁰⁹ and is defined as a loss of renal function distal to an arterial stenosis due to impaired renal blood flow. Studies in patients with bilateral RAS have indicated that reduction of systemic pressures to normal levels using sodium nitroprusside can abruptly reduce renal plasma flow and the GFR, indicating that poststenotic pressures are at critical levels beyond autoregulation (see Fig. 47.9).¹¹⁰ Some estimates have suggested that as many as 12% to 14% of patients reaching end-stage kidney disease (ESKD), with no other identifiable primary renal disease, may have occult, bilateral RAS.^{111,112} A survey using spiral CT angiography has examined 49 of 80 patients starting dialysis therapy identified as having ARAS (estimated as >50% lumen occlusion); in 20 of them (41%) and in 8 (16%) it was bilateral. Assuming that these lesions somehow were the primary cause of ESKD, the authors have proposed that up to 27% of new ESKD patients may have lost kidney function on this basis.¹¹³ A more conservative review of US Renal Data System (USRDS) data for patients older than 67 years in the United States starting dialysis has suggested that identified RVD may be present in between 7.1% and 11.1% of patients, although clinicians caring for such patients attributed their renal failure to RAS in only 5.0%.¹¹⁴ Multivariate analysis has indicated that male gender and advancing age correlated positively with this disorder, whereas a African-American, Asian, or Native American background correlated negatively.¹¹¹ Importantly, patients with rapidly progressive dysfunction and accelerated hypertension have the potential to recover kidney function and a mortality benefit from revascularization.^{2,108,115} However, such individuals have rarely been included in prospective, randomized trials (see later).

The causal role of vascular impairment in producing renal dysfunction is established most firmly when renal revascularization leads to recovery of renal function. Unfortunately, this does not occur commonly.^{1,4} Studies over the last decade have produced several paradigm shifts related to hemodynamically induced renal dysfunction. As noted, the overall high perfusion of the kidneys protects against significant tissue hypoxia to some extent.³⁶ Hence, many subjects can be treated with antihypertensive drug therapy without evident further loss of GFR, sometimes for many years as demonstrated by prospective randomized controlled trials (RCTs) such as ASTRAL and CORAL.¹¹⁶ A subset of patients with more severe or long-standing vascular occlusion (usually defined as a peak

systolic velocity >385 cm/s on duplex ultrasound) manifests overt cortical hypoxia using BOLD MRI.¹¹⁷ Such patients demonstrate more advanced tissue histologic injury with loss of tubular structures on biopsy and interstitial cellular infiltrates, consistent with inflammatory injury^{44,117} (see Figs. 47.10B and 47.11). These observations support the gradual transition from a primarily hemodynamic reduction in kidney function (potentially improved by restoring blood flow) to inflammatory-mediated injury that no longer recovers predictably after the restoration of vessel patency. Patients with advanced renal dysfunction have high comorbidity associated with cardiovascular disease and commonly have interstitial renal injury on biopsy.^{118,119} Prospective treatment trials specifically directed at slowing the loss of kidney function by renal revascularization have failed to identify much benefit in the intermediate term.^{98,120,121} Those with declining renal function have a poor survival rate regardless of intervention, the strongest predictor of which is a low baseline GFR.¹²²

The potential benefit of revascularization regarding salvage, or at least stabilization, of renal function is greatest when the serum creatinine level is less than 3 mg/dL, so the diagnosis of ischemic nephropathy is best considered early in its course.¹²² Remarkably, RAS can be associated with proteinuria, occasionally to nephrotic levels.^{123–125} Proteinuria can diminish or resolve entirely following renal revascularization,¹²⁶ suggesting that intrarenal hemodynamic changes and/or stimulation of local hormonal or cytokine activity alter glomerular membrane permeability in a reversible fashion. Although other glomerular diseases can develop in patients with renal artery disease, including diabetic nephropathy and focal sclerosing glomerulonephritis (FSGS), the presence of proteinuria alone does not establish a second disorder.

Clinical manifestations and prognosis differ when RVD affects one of two kidneys or affects the entire functioning renal mass. Although poststenotic kidneys may lose GFR and tissue volume, the contralateral kidney without stenosis undergoes compensatory hypertrophy. These changes mask the damage to the affected kidney, making the overall GFR an unreliable marker of the severity and progression of atherosclerotic RAS.^{127,128} Although blood pressure levels may be similar, the fall in blood pressure after renal revascularization is greater in bilateral disease.¹²⁹ Most patients with episodic pulmonary edema have bilateral disease or a solitary kidney. Long-term mortality during follow-up is higher when bilateral disease is present, regardless of whether renal revascularization is undertaken.^{93,130,131} These data have suggested that the extent and severity of RVD reflect the overall atherosclerotic burden of the individual. Patients with incidental RAS (>70%) managed without revascularization have reduced survival with bilateral disease, despite reasonable blood pressure control, as illustrated in Fig. 47.16.⁹ The causes of death are mainly related to cardiovascular disease, including stroke and CHF.

PROGRESSIVE VASCULAR OCCLUSION

Atherosclerosis is a progressive disorder, although individual rates of progression vary widely. The clinical manifestations from RAS depend partly on the severity and extent of vascular occlusion. The most ominous of these manifestations, such as renal failure or pulmonary edema, are related to bilateral disease or stenosis to a solitary functioning kidney and pose

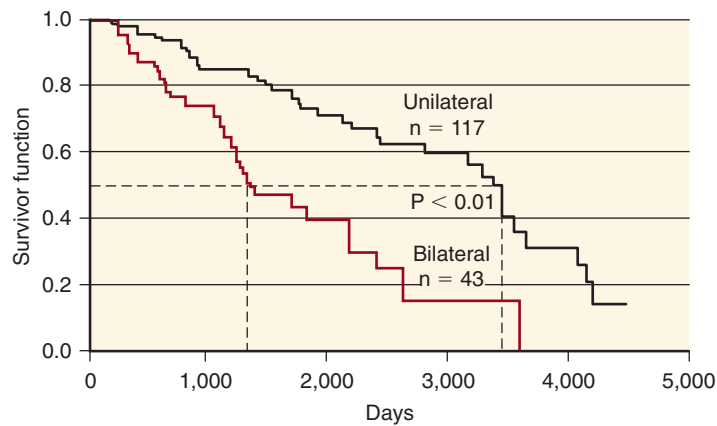


Fig. 47.16 Kaplan-Meier survival curve of 160 patients with more than 70% renal artery stenosis managed without revascularization. Those with bilateral disease had lower survival, primarily due to associated cardiovascular disease. The mean age of death was 79 years. These data underscore the close relationship between the extent of vascular disease and mortality. Less than 10% of these subjects developed advanced kidney disease during follow-up, although long-term survival, even in treated patients, is related to levels of kidney function at the time of intervention. Recent prospective randomized clinical trials indicate that between 16% and 22% of medically treated patients with atherosclerotic renal artery stenosis progress more advanced renal dysfunction over 3 to 5 years. (From Foster JH, Maxwell MH, Franklin SS, et al. Renovascular occlusive disease: results of operative treatment. *JAMA*. 1975;223:1043-1198; Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567-1609.)

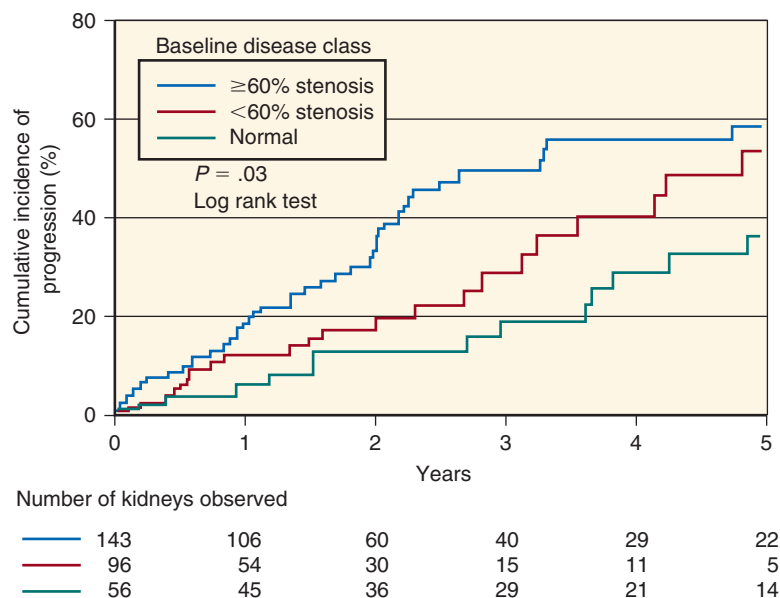


Fig. 47.17 Cumulative rates of anatomic disease progression in atherosclerotic renal artery stenosis, as measured by renal artery Doppler ultrasound. During a follow-up period of 5 years, overall progression was 31%, but those with the most severe baseline lesion progressed in 60% of cases. The progression of vascular disease was not closely related to changes in serum creatinine levels or renal atrophy. (Modified from Caps MT, Perissinotto C, Zierler RE, et al. Prospective study of atherosclerotic disease progression in the renal artery. *Circulation*. 1998;98:2866-2872.)

the greatest hazard when total arterial occlusion develops. Hence, the impetus to intervene in RAS depends heavily on predicting or establishing the natural history of vascular stenosis in an individual. Retrospective studies of serial angiograms obtained in the 1970s and early 1980s have indicated that atherosclerotic lesions progress to more severe levels in 44% to 63% of patients followed from 2 to 5 years.¹³² Up to 16% of renal arteries developed total occlusion. Subsequent prospective studies in patients undergoing cardiac catheterization or serial Doppler ultrasound measurements

have suggested that current rates of progression may be lower.¹³³ Zierler and colleagues have reported a 20% rate of disease progression overall, with 7% advancing to total occlusion over 3 years.¹³³ A later report from the same group using different Doppler velocity criteria has suggested higher rates of progressive stenosis.¹³⁴ Overall progression was detectable in 31%, but primarily those with the most severe baseline stenosis (>60%) and severe hypertension were more likely to progress (51%; Fig. 47.17). The occurrence of total occlusion was uncommon (9 of 295 [3%]). Data from medical

treatment trials have suggested that progressive occlusion can develop silently in up to 16% of treated subjects.¹³⁵ These observations are supported by the development of renal endpoints (defined as progressive renal deterioration) ranging from 16% to 22% in recent prospective RCTs (ASTRAL and CORAL).¹

Importantly, clinical events such as detectable changes in renal function or accelerating hypertension bear only a limited relationship to anatomic vascular progression. The occurrence of renal “atrophy” (loss in renal size by 1 cm or more by ultrasound) developed in 20.8% of the most severe lesions in the prospective series.¹⁰⁶ Most series of medically treated patients have indicated that despite evident progression of vascular disease, changes in kidney function are modest and uncommon.¹³⁶ Results reported during medical follow-up of 41 patients managed medically before the introduction of ACE inhibitors for an average of 36 months have identified a loss of renal length in 35%, whereas a significant rise in serum creatinine levels developed in 8 of 41 (19.5%).¹³⁶ Blood pressure control improved during the intervals followed in the 1980s, associated with the introduction of ACE inhibitors and angiotensin-blocking agents. Medical management was associated with increased requirements for antihypertensive agents. The percentage developing clinical progression with refractory hypertension or progressive renal insufficiency fell from 21% in the earliest period to less than 10% in the later cohort after the introduction of ACE inhibitors.¹³⁰ This conclusion is consistent with long-term studies from Europe, in which incidental renal artery lesions were rarely associated with progressive renal failure over more than 9 years of follow-up.¹³⁷ These data support the observation that renal artery lesions remain stable in some patients over many years, without adverse clinical effects or evident progression¹³⁸ as observed in treatment trials, including ASTRAL and CORAL.¹³⁹ It is likely that overall rates of atherosclerotic disease progression have been falling due to the more widespread use of statins, aspirin, diminishing tobacco use, and more intense antihypertensive therapy. Hence, worsening vascular occlusion is an important clinical risk but does not occur in all patients.

DEVELOPMENTS IN MEDICAL AND ANTIHYPERTENSIVE DRUG THERAPY

Before the 1980s, the literature of RVD was concerned primarily with the identification of functionally important lesions in patients with severe hypertension. Drug therapy was limited in scope and often produced intolerable side effects. Most importantly, the available drugs did not yet include agents capable of interrupting the renin-angiotensin system (see Fig. 47.2). As a result, patients commonly appeared with accelerated or malignant hypertension, a large fraction of which was related to RAS. Among 123 patients whose average age was 44 years presenting with accelerated hypertension, more than 30% of whites were identified as having renovascular hypertension.¹⁴⁰ Some patients could not be effectively controlled with available medications and were subjected to urgent bilateral nephrectomy as a lifesaving measure. Evaluation for RAS has centered on identifying patients whose blood pressures could be improved, perhaps cured, by renal revascularization. Since the 1980s, antihypertensive agents have expanded to include calcium channel blockers and, most importantly, drugs that functionally block the renin-

angiotensin system, such as ACE inhibitors, ARBs, and direct renin inhibitors (i.e., aliskiren). The impact of these agents has been significant.

Reviews of medical therapy for renovascular hypertension older than indicated that regimens using these agents increase the likelihood of achieving good blood pressure control from 46% to more than 90%.⁹² The concept of emergency bilateral nephrectomy for control of hypertension has effectively disappeared. It is likely that many patients with RVD and hypertension now go undetected because blood pressure and renal function are well controlled and stable.⁸⁸ Importantly, systematic efforts to reach goal blood pressure levels, optimize statin and aspirin use, and withdraw tobacco exposure have been part of treatment trials, including CORAL.⁹⁸ Reduction of cardiovascular risk may be even more effective with the expanded use of ACE inhibitors for CHF, proteinuric renal disease, and other constellations of cardiovascular risk factors, particularly since the publication of the HOPE trial¹⁴¹ and others. Whether the use of ACE inhibitors and/or ARBs actually delays the onset of renovascular hypertension in humans, as it does in experimental animals, cannot be established with the present data.

CHANGING POPULATION DEMOGRAPHICS

The last several decades have been characterized by longer lifespans in many western countries. These likely result from several factors, including major declines in mortality related to stroke and cardiovascular disease. Population groups older than 65 years are now among the most rapidly growing segments in the United States.⁹⁰ One consequence of lower mortality from coronary and cerebrovascular events has been the emergence of vascular disease affecting other areas, such as the aorta and kidneys. As a result, clinical manifestations of RAS are appearing in older individuals, often combined with other comorbid conditions.^{89,142}

These features change the clinical presentation and affect the risk-benefit considerations inherent in undertaking renal revascularization. Series with renal artery intervention now routinely include average age values between 68 and 75 years, whereas 2 decades ago the mean age was between 61 and 63 years.^{143,144} These mean values are more than 15 years higher than those from the 1960s and 1970s. As might be expected, the prevalence of advanced coronary disease, CHF, previous stroke or transient ischemic attack (TIA), and aortic disease, as well as impaired renal function, has been rising in patients with atherosclerotic renal artery disease.⁹⁰

ROLE OF CONCURRENT DISEASES

Atherosclerotic RAS rarely occurs in isolation. It is a manifestation of atherosclerotic disease, which often affects multiple other sites. Follow-up studies related to the survival of incidentally identified renal artery disease have suggested that the presence of RAS independently predicts mortality, particularly in the presence of elevated serum creatinine levels,¹⁴⁵ although the mortality risk of a patient with a serum creatinine level above 1.4 mg/dL (but <2.3 mg/dL), for any reason, is higher than that of those with normal serum creatinine levels.¹⁴¹ The major causes of death are cardiovascular events,

including CHF, stroke, and myocardial infarction.⁹⁷ It is essential to consider these competing risks in planning the management of patients with all forms of vascular disease, especially older adults.¹⁴⁶

Clinical Relevance

Competing Risks

Competing risks of death from cardiovascular disease must be considered when weighing the risks and benefits of revascularization.

These disorders often dominate the clinical outcomes of patients with renal arterial disease, independently of the level of renal function. It has been difficult to establish improved survival in prospective trials of patients treated with medical therapy or renal revascularization.

Although many patients experience better blood pressure control, and some recover renal function, current methods of revascularization are not free of risks.¹⁴⁷ Even after successful renovascular procedures, other comorbid events may obscure any long-term benefit, challenging the cost-effectiveness of renal revascularization. Reviews of Medicare claims in the United States for 2 years after identification of atherosclerotic RAS have indicated that the risk of cardiovascular events, many of which are fatal, are more than tenfold more likely than progression to advanced renal failure.⁹⁷ Conversely, others have argued that RAS accelerates these cardiovascular risks by increasing arterial pressures and activating adverse neurohumoral pathways, predisposing both to CHF and renal dysfunction.¹⁴⁸ These divergent views have formed the basis for several prospective RCTs related to renal revascularization over the past decade (see later) that partly address these issues.

DIAGNOSTIC TESTING FOR RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

GOALS OF EVALUATION

The literature related to the diagnosis and evaluation of renovascular hypertension is complex and inconsistent (Box 47.3). Some of the confusion likely reflects the widely different patient groups being considered for evaluation and divergent goals for intervention. The clinician must identify the objectives of initiating expensive and sometimes ambiguous investigations beforehand. As with all tests, the reliability and value of diagnostic studies depend heavily on the pretest probability of disease¹⁴⁹ (Table 47.3). Furthermore, it is essential to consider from the outset exactly what is to be achieved:

- Is the major goal to exclude high-grade renal artery disease?
- Is it to exclude bilateral (as opposed to unilateral) disease?
- Is it to identify stenosis and estimate the potential for clinical benefit from renal revascularization?
- Is it to evaluate the role of RVD in explaining deteriorating renal function?

The specific approach to diagnosis will differ, depending on which of these is the predominant clinical objective.

Box 47.3 Renovascular Disease and Ischemic Nephropathy

Goals of Diagnostic Evaluation

Establish presence of renal artery stenosis, location and type of lesion.
Establish whether unilateral or bilateral stenosis (or stenosis to a solitary kidney) is present.
Establish presence and function of stenotic and nonstenotic kidneys.
Establish hemodynamic severity of renal arterial disease.
Evaluate progression of vascular occlusion, renal atrophy.
Plan vascular intervention—degree and location of atherosclerotic disease.

Goals of Therapy

1. Improved blood pressure control
 - Prevent morbidity and mortality of high blood pressure.
 - Improve blood pressure control and reduce medication requirements.
2. Preservation of renal function
 - Reduce risk of renal adverse perfusion from use of anti-hypertensive agents.
 - Reduce episodes of circulatory congestion (flash pulmonary edema).
 - Reduce risk of progressive vascular occlusion causing loss of renal function—preservation of renal function.
 - Salvage renal function—that is, recover glomerular filtration rate.
3. Recovery of volume regulation and fluid excretion
 - Reduce circulatory overload and diuretic resistance.
 - Reduce episodes of flash pulmonary edema.

DIAGNOSTIC TESTS

Noninvasive diagnostic tests for renovascular hypertension and ischemic nephropathy remain imperfect. For the purposes of this discussion, diagnostic tests fall into general categories as follows (Table 47.3): (1) perfusion and imaging studies to identify the presence and degree of vascular stenosis; (2) physiologic and functional studies to evaluate the role of stenotic lesions particularly related to activation of the renin-angiotensin system; and (3) studies to predict the likelihood of benefit from invasive maneuvers, including renal revascularization.

IMAGING OF THE RENAL VASCULATURE

The diagnosis of RVD fundamentally requires identification of vascular abnormalities. Therefore imaging of the renal vasculature is essential for these disorders. Advances in Doppler ultrasound, radionuclide imaging, magnetic resonance arteriography (MRA), and computed tomography angiography (CTA) continue to introduce major changes in the field of renovascular imaging. The details of these methods are discussed further in Chapter 25. The following is a discussion of some of the specific merits and limitations of each modality as they apply to renovascular hypertension and ischemic nephropathy.¹⁴⁹

Current practice favors limiting invasive arteriography to carrying out endovascular intervention (e.g., stenting and/or angioplasty). Although angiography remains the gold standard for evaluation of the renal vasculature, its invasive

Table 47.3 Noninvasive Assessment of Renal Artery Stenosis

Study	Rationale	Strengths	Limitations
Vascular Studies to Evaluate the Renal Arteries			
Duplex ultrasonography	Shows the renal arteries and measures flow velocity as a means of assessing the severity of stenosis	Inexpensive, widely available; suitable for sequential measurement to follow disease progression and/or restenosis	Heavily dependent on operator's experience; less useful than invasive angiography for diagnosis of fibromuscular dysplasia and abnormalities in accessory renal arteries
Magnetic resonance angiography	Shows the renal arteries and perirenal aorta	Not nephrotoxic, but concerns for gadolinium toxicity exclude use in GFR <30 mL/min/1.73 m ² ; provides excellent images	Expensive; gadolinium excluded in renal failure, unable to visualize stented vessels
Computed tomography angiography	Shows the renal arteries and perirenal aorta	Provides excellent images; stents do not cause artifacts	Expensive, moderate volume of contrast required, potentially nephrotoxic
Perfusion Studies to Assess Differential Renal Blood Flow			
Captopril renography with technetium ^{99m} Tc-mercaptatide (^{99m} Tc-MAG3)	Captopril-mediated fall in filtration pressure amplifies differences in renal perfusion	Normal study excludes renovascular hypertension	Multiple limitations in patients with advanced atherosclerosis or creatinine >2.0 mg/dL (177 μmol/L)
Nuclear imaging with technetium mercaptatide or technetium-labeled pentetic acid (DTPA) to estimate fractional flow to each kidney	Estimates fractional flow to each kidney	Allows calculation of single-kidney GFR	Results may be influenced by other conditions (e.g., obstructive uropathy)
Physiologic Studies to Assess the Renin-Angiotensin System			
Measurement of peripheral plasma renin activity	Reflects the level of sodium excretion	Measures the level of activation of the renin-angiotensin system	Low predictive accuracy for renovascular hypertension; results influenced by medications and many other conditions
Measurement of captopril-stimulated renin activity	Produces a fall in pressure distal to the stenosis	Enhances the release of renin from the stenotic kidney	Low predictive accuracy for renovascular hypertension; results influenced by many other conditions
Measurement of renal vein renin activity	Compares renin release from the two kidneys	Lateralization predictive of improvement in blood pressure with revascularization	Nonlateralization has limited predictive power of the failure of blood pressure to improve after revascularization; results influenced by medications and many other conditions

DTPA, Diethylenetriamine pentaacetic acid; *GFR*, glomerular filtration rate; *MAG3*, ^{99m}Tc-mercaptoacetyltri-glycine.
 Modified from Safian RD, Textor SC. Medical progress: renal artery stenosis. *N Engl J Med*. 2001;344:431–442.

nature, potential hazards, and cost make it most suitable for those in whom intervention is planned, often during the same procedure. As a result, most clinicians favor preliminary noninvasive studies. When noninvasive studies are equivocal, arterial angiography may be warranted to establish the presence of transstenotic pressure gradients, as recommended in treatment trials.^{150,151}

Noninvasive Imaging

Doppler Ultrasound of the Renal Arteries. Duplex interrogation of the renal arteries provides measurements of localized velocities of blood flow and characteristics of renal tissue. In many institutions, this provides an inexpensive means for measuring vascular occlusive disease at sequential time points, both to establish the diagnosis of RAS and monitor its progres-

sion. After renal revascularization, Doppler studies are commonly used to monitor restenosis and target vessel patency^{152,153} (Fig. 47.18). The main drawbacks relate to the difficulties of performing adequate studies in obese patients, operator dependency, and time required. These factors vary considerably among institutions.

The primary criteria for renal artery studies are a peak systolic velocity above 180 cm/sec and/or a relative velocity above 3.5, as compared with the adjacent aortic flow.^{154,155} Using these criteria, sensitivity and specificity with angiographic estimates of lesions exceeding 60% can surpass 90% and 96%, respectively,¹⁵⁴ although not universally.¹⁵⁶ Increasing the threshold for peak systolic velocities reduces the rate of false-positive estimates of stenosis. When main vessel velocities cannot be determined reliably, segmental waveforms in the

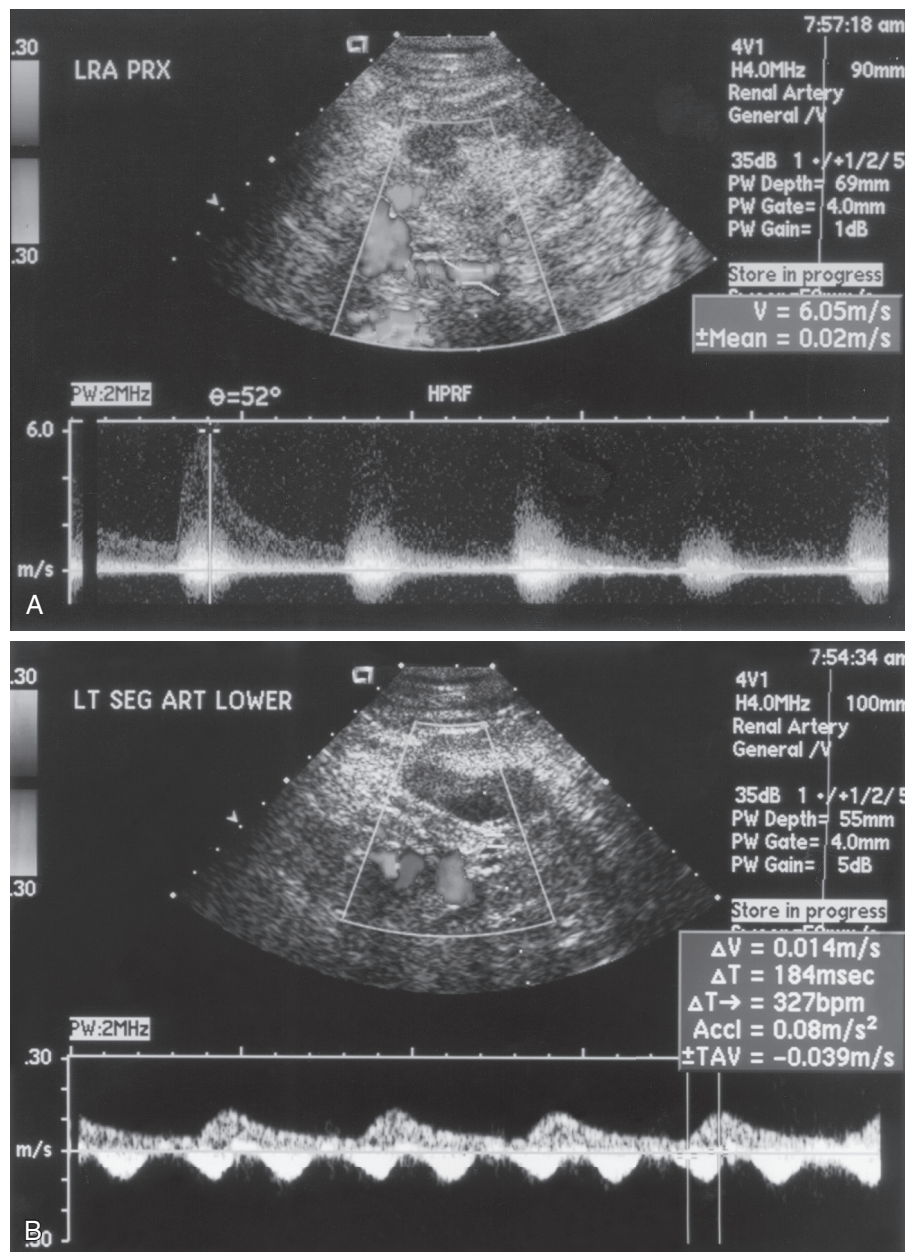


Fig. 47.18 (A) Duplex ultrasound velocity measurement in a patient with high-grade renal artery stenosis affecting the proximal left renal artery (LRA PRX). Peak systolic velocities reach 605 cm/s (6.05 m/s), well above the normal upper limit of 180 cm/s. (B) Segmental arterial branch ultrasound in the distal segmental renal arteries demonstrates “parvus” and “tardus” dampening of the signal characteristic of poststenotic waveforms. The utility of these measurements depends on the ability to obtain reliable identification of vessel segments and the skills of the operator. Once the location of a vascular lesion is known, subsequent studies can be performed more easily to track progression of vascular occlusion, restenosis, and/or the results of endovascular intervention.

arcuate vessels in the renal hilum can provide additional information. Damping of these waveforms, labeled as “parvus” and “tardus,” have been proposed as indirect signs of upstream vascular occlusive phenomena.¹⁵⁷ Some authors have challenged the use of angiographic estimates of stenosis as representing a gold standard altogether.¹⁵⁸ These authors argue that Doppler velocities correlate highly ($R = .97$) with a truer estimate of vascular occlusion, specifically luminal stenosis determined by intravascular ultrasound.

Doppler study of the renal arteries is highly reliable when adequate images of the renal arteries can be obtained. Positive

Doppler velocities in an artery clearly identified as the renal artery are rarely proven to be negative later. False-negative studies are more common. In subjects with accessible vessels, Doppler ultrasound provides the most practical means of following vessel characteristics sequentially over time. A drawback of renal artery Doppler studies includes a frequent failure to identify accessory vessels. Because the correlation between velocity and degree of stenosis is only approximate, clinical trials such as CORAL have raised the peak systolic velocity threshold to 300 cm/s.¹⁵⁴ This seems warranted, particularly when the risk of overdiagnosis of renal arterial

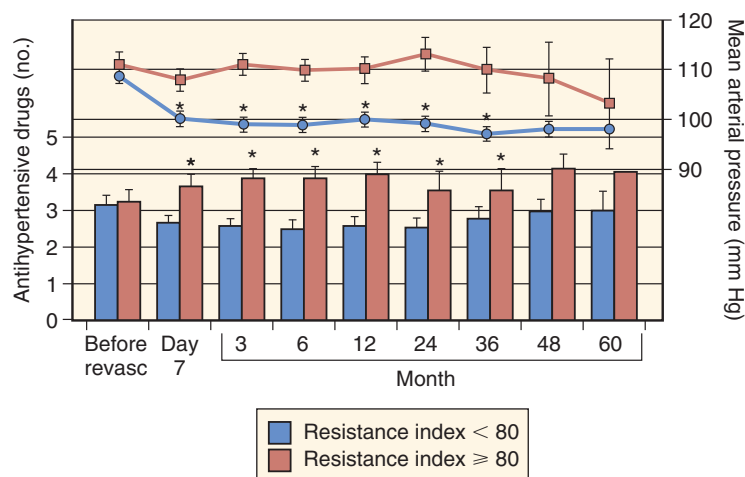


Fig. 47.19 Outcome of revascularization as measured by mean arterial blood pressure and number of antihypertensive agents in 138 patients with renal artery stenosis. These patients were divided into groups with ultrasound-determined resistive index above 80 and those lower than 80 in the most severely affected kidney. The authors indicate that a high resistive index reflects intrinsic parenchymal and small vessel disease in the kidney that does not improve after revascularization. Those with lower indices had both lower blood pressures during follow-up and lower antihypertensive medication requirements. (From Radermacher J, Chavan A, Bleck J, et al. Use of Doppler ultrasonography to predict the outcome of therapy for renal-artery stenosis. *N Engl J Med*. 2001;344:410–417.)

lesions is high, as in the STAR trial, in which 18 of 64 patients assigned to stenting were found not to have significant RVD at the time of angiography, despite noninvasive estimates to the contrary.¹²⁰

Additional studies have emphasized the potential for Doppler ultrasound to characterize small-vessel flow characteristics in the kidney.¹⁵⁹ The resistive index provides an estimate of the relative flow velocities in diastole and systole. In a study of 138 patients with RAS, a resistive index above 80 provided a predictive tool for the identification of parenchymal renal disease that did not respond to renal revascularization (Fig. 47.19). A sizeable portion of this group eventually progressed to renal failure. A resistive index less than 80 was associated with a more than 90% favorable blood pressure response and stable or improved renal function. The authors emphasized that accurate predictive power depend on using the highest resistive index observed, even when present in the nonstenotic kidney. A subsequent study of 215 subjects with mean preintervention serum creatinine levels of 1.51 mg/dL failed to confirm the predictive value of resistive index measurements.¹⁶⁰ Of 99 subjects with improved renal function after 1 year, 18% had a resistive index above 0.8 before intervention, whereas 15% of 92 subjects with no improvement had a resistive index above 0.8 (not significantly different [NS]). In this series, a preintervention level of serum creatinine itself was the strongest predictor of improved renal function. Most clinicians agree that detecting a low resistive index indicates a well-preserved vasculature in the kidney, with improved likelihood of recovering or stabilizing after vascular intervention.^{161,162}

Computed Tomography Angiography. CTA using helical and/or multiple detector scanners and intravenous contrast can provide excellent images of both kidneys and the vascular tree. Resolution and reconstruction techniques render this modality capable of identifying smaller vessels, vascular lesions, and parenchymal characteristics, including malignancy and

stones¹⁶³ (Fig. 47.20). When used for the detection of RAS, CTA agrees well with conventional arteriography (correlation, 95%), and sensitivity may reach 98% and specificity 94%.¹⁶⁴ Studies have indicated that CTA also provides excellent accuracy regarding the evaluation of in-stent restenosis,¹⁶⁵ and evolving quantitative, three-dimensional image analysis may improve on intraarterial methods.¹⁶⁴ Although this technique offers excellent noninvasive examination of the vascular tree, it does require iodinated contrast. As a result, it raises concerns for the evaluation of renovascular hypertension and/or ischemic nephropathy for older patients with impaired renal function and/or diabetes. Remarkably, patients with atherosclerotic RAS tolerate contrast from CTA imaging without evident toxicity, despite advanced age and a low GFR.¹⁶⁶ Some studies have suggested that release of cell cycle arrest markers tends to protect from acute injury in this setting.

Concerns regarding systemic toxicity associated with MRA contrast nonetheless have encouraged wider use of multidetector CT imaging for patients suspected of having RVD. Limitations include reduced visibility of vessel lumens in the presence of substantial calcium deposition. A single study comparing both CTA and MRA with intraarterial studies in 402 subjects has indicated lower CT performance for the detection of lesions with more than 50% stenosis.¹⁶⁷ In this particular study, CTA had a sensitivity of 64% and specificity of 92%, respectively, whereas MRA had a sensitivity of 62% and specificity of 84%, respectively. This was an unusual population, with only 20% of the screened population having stenotic lesions, nearly half of which were FMD. The results of such studies have reinforced the importance of careful patient selection for study and establishing exactly the purpose for which imaging is being undertaken in advance.¹⁶⁸

Magnetic Resonance Angiography. Gadolinium-enhanced images of the abdominal and renal vasculature had been a mainstay of evaluating RVD in many institutions.^{163,169}

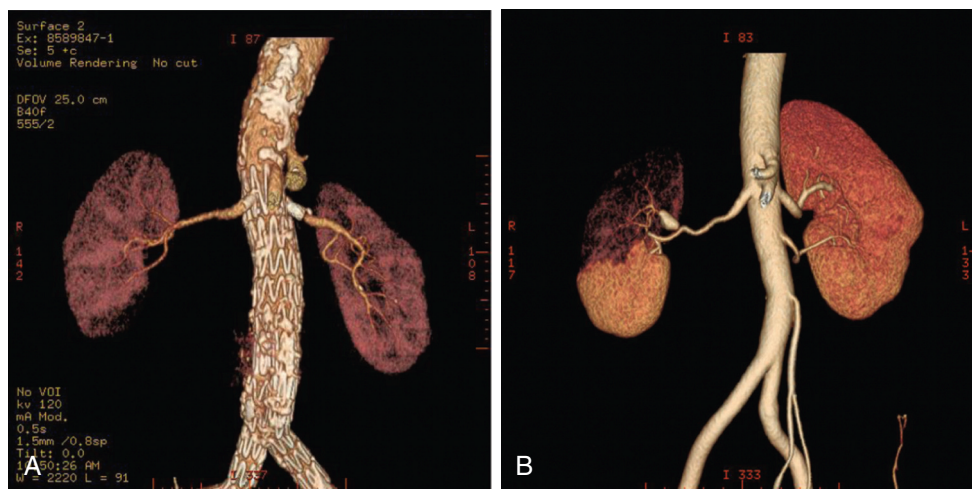


Fig. 47.20 Computed tomography (CT) angiograms illustrating reconstructed views of complex vascular disease. (A) Aortic endovascular stent extending beyond the origins of the renal arteries. Renal artery stents have been placed through the aortic graft to restore blood flow, although the nephrograms demonstrate patchy defects consistent with small-vessel occlusion and/or atheroembolic events. (B) CT angiogram with a small aneurysm of the right renal artery that has produced segmental infarction, leading to accelerated hypertension. Although CT angiography requires contrast, current multidetector CT studies allow excellent image resolution at rapid acquisition times and less contrast exposure than before.



Fig. 47.21 Magnetic resonance (MR) angiogram reconstructed without gadolinium contrast. After warnings about the association of gadolinium with nephrogenic systemic fibrosis, contrast MR is less commonly used for atherosclerotic renovascular imaging than it was used previously. Newer techniques can allow excellent vascular imaging without contrast, as shown here.

Comparative studies have indicated that sensitivity ranges from 83% to 100% and specificity from 92% to 97% in RAS.^{170,171} Meta-analyses of published literature including 998 subjects have found more than 97% sensitivity using gadolinium-enhanced imaging.¹⁷² The nephrogram obtained from gadolinium filtration provides an estimate of relative function and filtration, as well as parenchymal volume. The quantitative measurement of parenchymal volume determined by MRI appears to correlate closely with isotopically determined single-kidney GFR in some institutions.¹⁷³ Since 2006, however, concerns about the potential for gadolinium-based contrast to produce nephrogenic systemic fibrosis have effectively eliminated the use of contrast-enhanced MRI for patients with impaired kidney function in the United States.¹⁷⁴ This topic is further discussed in Chapter 25 because gadolinium may be used at lower renal function in other regions.

Technologic advances allow the use of high-resolution vascular MRI without contrast in many patients. An example of MRA without contrast is shown in Fig. 47.21. Drawbacks

include expense and the tendency to overestimate the severity of lesions, which appear as a signal void.^{163,175} The limits of resolution with current instrumentation, however, make the detection of small accessory vessels and quantitating fibromuscular lesions difficult. Both of these have been improving with use of the newer generations of scanners. High spatial resolution, three-dimensional, contrast-enhanced MR scanners provide up to 97% sensitivity and 92% specificity for renal artery stenotic lesions.¹⁷⁶ Signal degradation in the presence of metallic stents renders MRA unsuitable for follow-up studies after endovascular procedures in which stents are used.

Captopril Renography. Imaging the kidneys before and after the administration of an ACE inhibitor (e.g., captopril) can provide a functional assessment of the change in blood flow and GFR to the kidney, related both to changes in arterial pressure and removal of the efferent arteriolar effects of Ang II. Because hypertension often is not the primary motivation for evaluating RVD, these tests are less frequently carried out currently. The most commonly used radiopharmaceuticals are technetium-99 (^{99m}Tc) diethylenetriamine pentaacetic acid (DTPA) and ^{99m}Tc-mercaptoacetyltriglycine (MAG3). The latter agent has clearance characteristics similar to those of Hippuran and is often taken as reflecting renal plasma flow. Both can be used, although specific interpretive criteria differ.^{177,178} Both provide information regarding the size and filtration of both kidneys; the change in these characteristics after ACE inhibition allows inferences regarding the dependence of glomerular filtration on Ang II. Studies of patient groups with prevalence rates of RVD from 35% to 64% have indicated that sensitivity and specificity range from 65% to 96% and 62% to 100%, respectively.¹⁷⁸ Because of its high specificity, captopril renography can be used in populations with low pretest probability, with an expectation that a normal study will exclude significant renovascular hypertension in more than 96% of cases.¹⁷⁹ Some series have reported 100% accurate negative predictive values.¹⁸⁰

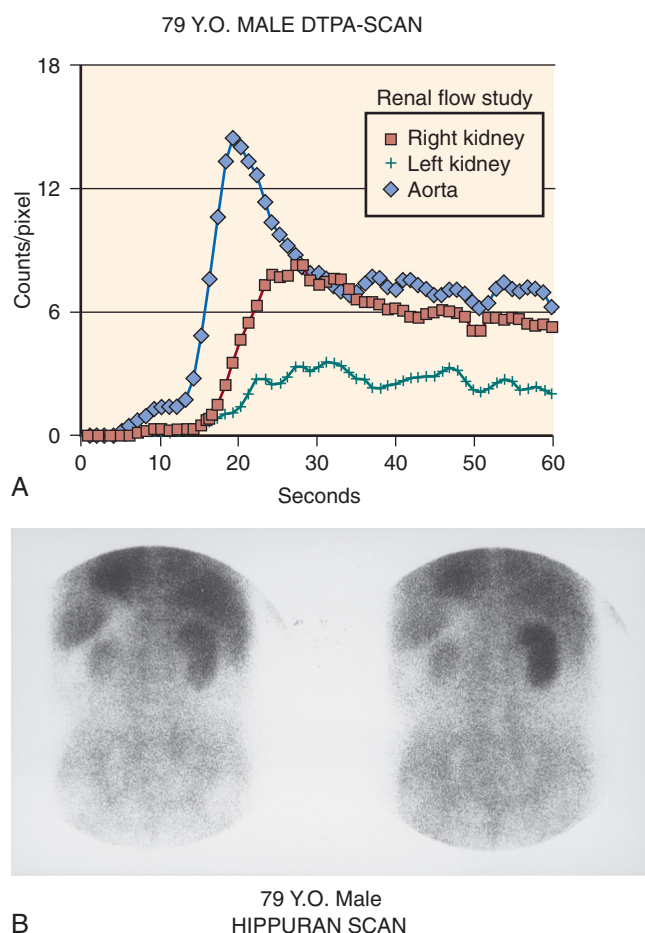


Fig. 47.22 (A and B) Isotope renography in a patient with unilateral renal artery stenosis. (A) A ^{99m}Tc -DTPA scan demonstrates delayed circulation and excretion of isotope on the *left*. (B) A hippuran scan (now replaced with ^{99m}Tc -mercaptoacetyl triglycine) provides a renogram demonstrating a small kidney with impaired renal function on the affected side. Radionuclide scans provide a comparative estimate of function from each kidney that may facilitate selection of intervention, including the potential effect of nephrectomy. *DTPA*, Diethylenetriamine pentaacetic acid.

Renographic imaging studies are less sensitive and specific for RVD in the presence of renal insufficiency (usually defined as creatinine level >2.0 mg/dL). These performance characteristics also deteriorate for patients who cannot be prepared carefully (i.e., withdrawal of diuretics and ACE inhibitors for 4–14 days before the study).¹⁷⁸ Renography, however, only provides functional information and no direct anatomic information—the location of renal artery disease, the number of renal arteries or associated aortic, and/or ostial disease (Fig. 47.22). Prospective studies of RVD from the Netherlands did observe changes in the renogram during follow-up but did not find captopril renography predictive of angiographic findings or outcomes of revascularization.¹³⁵ A prospective study of 74 patients undergoing both renography and Doppler ultrasound evaluation before renal revascularization has reported a limited predictive value of scintigraphy (sensitivity, 58%; specificity, 57%) regarding blood pressure outcomes.¹⁸¹

Under carefully controlled conditions, some authors have argued that changes in renographic appearance correlate with

changes in blood pressure to be expected after revascularization. Changes in split renal function indicate that stenotic kidneys regain GFR after revascularization, sometimes with a decrement in contralateral GFR, thereby maintaining overall kidney function unchanged.^{127,182} Currently, radionuclide studies have a role in identifying the relative contributions of each kidney to overall GFR—often as part of the evaluation for residual renal function after nephrectomy.¹⁸³

Invasive Imaging

Intraarterial angiography currently remains the gold standard for the definition of vascular anatomy and stenotic lesions in the kidney, often performed at the time of a planned intervention, such as endovascular angioplasty and/or stenting. The role of including angiography of the renal arteries during imaging of other vascular beds, such as so-called drive-by angiography during coronary artery imaging, is unclear. Studies have confirmed that the prevalence of renal artery lesions exceeding 50% lumen occlusion in patients with hypertension and coronary artery disease is high, usually from 18% to 24%.¹⁴⁸ Some individuals (7% to 10%) will have high-grade stenoses above 70% occlusion, and some will be bilateral. Noting that an arterial puncture and catheterization of the aorta and coronary vessels produce some risk, the added risk from including aortography of the renal vessels may be small. Follow-up studies of individuals with incidental renal artery lesions have suggested that the presence of these lesions does provide additive predictive risk for mortality.^{90,184} Screening angiography has been less commonly performed since the publication of prospective RCTs suggesting limited benefit of revascularization in stable patients with atherosclerotic RVD (see later).^{148,185} Hence, endovascular procedures for such lesions should be confined to those with strong indications for renal revascularization, as even ardent advocates of catheter-based intervention have suggested.

Contrast toxicity remains an issue with conventional iodinated agents given intraarterially.¹⁸⁶ Intravascular ultrasound procedures have been undertaken using papaverine to evaluate flow reserve beyond stenotic lesions.¹⁸⁷ Various reports have indicated that either a reduction in or preserved flow reserve may identify a poststenotic kidney that may benefit from successful revascularization.^{187,188} Previous studies of pressure gradients measured across stenotic lesions failed to predict the clinical response to renal revascularization. Measurements using currently available low-profile wire probes do, however, indicate a relationship between pressure gradients and activation of the renin-angiotensin system.¹⁴ Outcomes of patients with translesional pressure gradients measured after vasodilation have suggested that measurement of the hyperemic systolic gradient above 21 mm Hg (defined after vasodilation with papaverine) most accurately predicts high-grade stenosis (average, 78% by intravascular ultrasound) and a beneficial response of blood pressure after stenting.¹⁸⁹ The latter observation, and increasing reliability of technical measurements, underscore the value of measuring gradients to establish a hemodynamic role for vascular lesions of marginal severity.

Physiologic and Functional Studies of the Renin-Angiotensin System

Efforts have long been made to use activation of the RAS as a marker of underlying renovascular hypertension.^{161,190}

Although these studies are promising when performed in patients with known renovascular hypertension, they are less reliable when applied to wider populations. Plasma renin activity is sensitive to changes of sodium intake, volume status, renal function, and many medications. The sensitivity and specificity of such maneuvers are heavily dependent on the a priori probability of renovascular hypertension. In practice, the major utility of these studies depends on their negative predictive value, specifically the certainty with which one can exclude significant RVD if the test is negative. Because the negative predictive value rarely exceeds 60% to 70%, these tests offer limited value in clinical decision making.¹⁹¹

Measurement of renal vein renin levels previously had been widely used in planning surgical revascularization for hypertension. These measurements are obtained by sampling renal vein and inferior vena cava blood individually. The level obtained from the vena cava is taken as comparable to the arterial levels in each kidney and allows estimation of the contribution of each kidney to total circulating levels of plasma renin activity. Lateralization is defined usually as a ratio exceeding 1.5 between the renin activity of the stenotic kidney and nonstenotic kidney.¹⁹² Some authors have proposed a detailed examination, not only of the relative ratio between kidneys, but also the degree of suppression of renin release from the nonstenotic or contralateral kidney.¹⁹² In general, the greater the degree of lateralization, the more probable that blood pressure will benefit from surgical or other revascularization.

Results from many studies have supported the observation that large differences between kidneys identify high-grade RAS.¹⁹³ These findings have been supported by studies of renal vein measurements before consideration of nephrectomy for refractory hypertension and advanced RVD.¹⁸³ As with many tests of hormonal activation, study conditions are crucial. A number of measures to enhance renin release and magnify differences between kidneys have been proposed, including sodium depletion with diuretic administration, hydralazine, tilt table stimulation, or captopril.¹⁹⁴ Strong and colleagues have demonstrated that nonlateralization can be changed to strongly lateralizing measurements by the administration of diuretics between sequential studies.¹⁹⁵ A review of more than 50 studies of renal vein renin measurements has indicated that when lateralization could be demonstrated, clinical benefit regarding blood pressure control could be expected in more than 90% of cases.¹⁹⁴ Failure to demonstrate lateralization, however, was still associated with significant benefit in more than 50% of cases.¹³⁰

More recent series have reached similar conclusions, indicating that overall sensitivity of renal vein renin measurements is no better than 65% and that the positive predictive value is 18.5%.¹⁹⁶ For these and other reasons, renal vein assays are now performed less commonly. A major additional factor is that the goals of renal revascularization have shifted substantially and are often directed toward preservation of renal function, rather than for blood pressure control alone. In cases for which it is important to establish the degree of pressor effect of a specific kidney or site, such pre-nephrectomy measurement of renal vein renin levels can provide strong supportive evidence.¹⁸³

Studies of Individual Renal Function. Serum creatinine levels, iothalamate clearance, and other estimates of the total GFR

are measures of total combined renal excretory function and do not address changes within each kidney. A large body of literature has addressed the potential for individual split renal function studies to establish the functional importance of each kidney in RVD.¹⁹⁷

Split renal function studies classically used separate ureteral catheters to allow individual urine collection for the measurement of separate GFRs, renal blood flow, sodium excretion, concentrating ability, and the response to blockade of Ang II. These studies have demonstrated that hemodynamic effects of renal artery lesions can produce functional changes, such as avid sodium retention, even before major changes in blood flow occur.¹⁹⁸ They emphasize that autoregulation of blood flow and GFR can occur over a wide range of pressures in humans and may be affected in both stenotic and contralateral kidneys by the effects of Ang II.^{199,200} These studies require urinary tract instrumentation and provide only indirect information regarding the probability of benefit from revascularization. They are now rarely performed.

Separate renal functional measurements can be obtained less invasively with radionuclide techniques. These methods use a variety of radioisotopes (e.g., ⁹⁹Tc-MAG3 or DTPA) to estimate fractional blood flow and filtration to each kidney. Prior administration of captopril magnifies differences between kidneys, primarily by delaying excretion of the filtered isotope due to removal of the efferent arteriolar effects of Ang II.²⁰¹ Some authors have advocated such measurements to follow progressive renal artery disease and its effect on unilateral kidney function as a guide to consider revascularization. Serial measurements of individual renal function allow more precise identification of progressive ischemic injury to the affected kidney in unilateral renal artery disease than can be determined from the overall GFR.²⁰² The poststenotic kidney is susceptible to gradual loss of filtration that is accompanied by a rise in blood flow and filtration in the contralateral kidney, both of which can partially reverse after restoring blood flow.¹²⁷ Single-kidney GFR measurements using radionuclide scans accurately reflect changes in three-dimensional volume parameters measured by MRI.¹⁷³ These authors argued that demonstrating preserved parenchymal volume with a disproportionate reduction in single-kidney GFR supports the concept of so-called hibernating kidney parenchyma and might provide a predictive parameter for the recovery of kidney function after revascularization.¹⁷³

MANAGEMENT OF RENAL ARTERY STENOSIS AND ISCHEMIC NEPHROPATHY

OVERVIEW

The last decade has fostered a major shift from promoting renal revascularization for a cure of renovascular hypertension to primary reliance on medical antihypertensive therapy for atherosclerotic RVD. This pendulum swing may be excessive in some cases, however, eventually requiring nephrologists to confront advanced renal failure that might have been prevented. Considering the array of potential interventions and the complexity of these patients, clinicians need to formulate a clear set of therapeutic goals for each patient. Because all therapeutic measures—ranging from medical therapy alone to endovascular or surgical revascularization

Table 47.4 Summary of Observational Series in Renal Revascularization**T1 Outcomes of Renal Artery Angioplasty, ±Stent Placement; Hypertension**

	Cured ^a	Improved	No Change
14 series, 678 patients, 98% technical success	Weighted mean, 17%; range, 3%–68%	Weighted mean, 47%; range, 5%–61%	Weighted mean, 36%; range, 0%–61%
Renovascular hypertension, 472 patients,	12%	73%	15%

T1 Renal Artery Stents: Effect on Renal Function^b in Azotemic Patients

	“Improved”	“Stabilized”	Worse
14 series; reporting “impaired renal function”; 496 patients	Weighted mean, 30%; range, 10%–41%	Weighted mean, 42%; range, 32%–71%	Weighted mean, 29%; range, 19%–34%
Ischemic nephropathy; 469 patients	41%	37%; no change	22%

^aCured, blood pressures at normal levels, no medication; improved, blood pressure lower by at least 10 mm Hg, fewer medications; no change, same blood pressure and/or same medications

^bRenal function—variably defined as improved, persistent fall in creatinine, variable from 0.3–1.0 mg/dL or more; worse, variable, but persistent increase in creatinine 0.3–1.0 mg/dL or more.

Modified from Textor SC. Renovascular hypertension and ischemic nephropathy. In: Skorecki K, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's: The kidney*. 10th ed. Philadelphia: Elsevier; 2016:1567–1609.

with adjunctive therapy—carry both benefits and risks, the clinician's task is to weigh the role of each of these within the context of the individual's comorbidity.

In most cases, long-term management of the RVD patient requires integrated pharmacologic management of blood pressure and cardiovascular risk combined with optimal timing of renal revascularization for those with high-risk manifestations and/or disease progression. The objective of this section is to provide a framework in which to plan a balanced approach to the patient with unilateral or bilateral RAS. It should be emphasized that consideration of renal artery disease takes place in the context of managing other cardiovascular risk factors, cessation of tobacco use, reduction of cholesterol levels, and treatment of diabetes and obesity.

MEDICAL THERAPY OF RENOVASCULAR DISEASE

The overall goals of therapy are summarized in [Box 47.3](#). Foremost among these is the objective identified from multiple reports from the Joint National Committee of the National High Blood Pressure Education Program (NHBPEP): “The goal of treating patients with hypertension is to prevent morbidity and mortality associated with high blood pressure.”²⁰³ This may include the effort to simplify or potentially eliminate long-term antihypertensive drug therapy, particularly in younger individuals with FMD. A further goal is to preserve kidney function and prevent loss of kidney function related to impaired renal blood flow. In some cases, renal revascularization is undertaken to allow improved management of salt and water balance in the process of managing patients with CHF. This may allow safer use of diuretic agents, ACE inhibitor, and ARB classes of medication in patients with critical renal artery lesions affecting the entire renal mass. Because prospective clinical trial data have been limited to relatively low-risk patients included for randomization, each patient must be considered individually.

Several prospective RCTs have been undertaken since 2005. Although each has limitations, results of medical therapy

over periods ranging from 2 to 5 years have been equivalent to those observed with medical therapy plus renal revascularization, with relatively few treatment crossovers ([Tables 47.4–47.6](#)).¹³⁹ Taken together, these studies underscore the importance of optimizing medical antihypertensive therapy as part of the decision-making process. If blood pressure can be well controlled with a tolerable regimen, and kidney function remains stable, it is difficult to justify moving forward with costly and potentially hazardous imaging and/or vascular intervention procedures. As a practical matter, these measures must be considered within the entire context of patient management over time.

RENAL ARTERY STENOSIS

UNILATERAL VERSUS BILATERAL RENAL ARTERY STENOSIS

“Bilateral” in this context refers to the circumstances when the entire functional renal mass is affected by vascular occlusion. This may be caused by bilateral stenoses or stenosis to a solitary functioning kidney. Not only are the putative mechanisms related to blood pressure and volume control different in the presence of a nonstenosed, functioning contralateral kidney with unilateral disease (as described earlier), but the potential hazards of intervention and/or medical therapy differ. Patient survival is reduced in patients with bilateral disease or stenosis to a solitary functioning kidney. Progressive arterial disease in this group also poses the most immediate hazard of declining renal function. Patient survival depends on the extent of vascular involvement,¹³¹ regardless of whether renal revascularization is undertaken.

UNILATERAL RENAL ARTERY STENOSIS

Most patients with atherosclerotic renal artery disease have preexisting hypertension.²⁰⁴ As a result, many are exposed to antihypertensive drug therapy before identification of the lesion and may be well controlled with only moderate

Table 47.5 Prospective Randomized Trials of Medical Therapy Versus Angioplasty for Hypertension^a

Source	Inclusion, Blood Pressure Measurement	Blood Pressure Outcome (mm Hg)	Renal Outcome (mm Hg)	Comments
SNRAS, Webster et al, 1998 ¹²⁹ ; <i>n</i> = 55 (unilateral = 27); <i>n</i> = 135 eligible; RAS > 50%	DBP ≥95 mm Hg, two drugs; exclusion: CVA, MI within 3 mo; creatinine >500 μmol/L; BP, random-zero device; no ACE inhibitors allowed	Unilateral—PTRA, 173/95 Med Rx, 161/88 Bilateral—PTRA, 152/83; Med Rx, 171/91; <i>P</i> < .01	Creatinine (μmol/L), bilateral PTRA, 188 Med Rx, 157 Unilateral PTRA, 144 Med Rx, 168	“...unable to demonstrate any benefit in respect of renal function or event free survival” (F/U 40 mos).
EMMA: Plouin et al, 1998 ²⁴³ ; <i>n</i> = 49 (unilateral RAS only); RAS >75% or >60%, lateralizing study	Age <75 years; normal contralateral kidney; exclusion: malignant HTN CVA, CHF, MI within 6 mo; BP, automated sphygmomanometer, ABPM at 6 mo	PTRA, 140/81 Med Rx, 141/84 No. drugs (DDD)—PTRA, 1.0 Med Rx, 1.78, <i>P</i> < .01 Crossover to PTRA: 7/26 (27%)	Creatinine Clearance mL/min: (6 mos) PTRA: 77 Med Rx: 74 Renal artery occlusion: PTRA: 0 Med Rx: 0	“BP levels and the proportion of patients given antihypertensive treatment were similar one year after randomization in the control and angioplasty groups, confirming that the BP-lowering effect of angioplasty in the short and medium terms is limited in atherosclerotic RAS.”
DRASTIC: Van Jaarsveld, et al. 2000 ¹³⁵ ; <i>n</i> = 106 ASO RAS >50%	Resistant: two drugs; DBP >95 mm Hg or creatinine rise with ACEI; exclusion: creatinine ≥2.3 mg/dL; solitary kidney, total occlusion—kidney <8 cm; BP, automated oscillometric	BP outcomes at 3 mo—PTRA, 169/89 Med Rx, 163/88 At 12 mo—PTRA, 152/84 Med Rx, 162/88 No. of drugs, 1.9 vs. 2.4, <i>P</i> < .01	Creatinine Cl (3 mos) mL/min: PTRA: 70 Med Rx: 59 (<i>P</i> = .03) Abnormal renograms PTRA: 36% Med Rx: 70% (<i>P</i> = .002) Renal artery occlusion PTRA 0 Med Rx 8	“In the treatment of patients with hypertension and renal artery stenosis, angioplasty has little advantage over antihypertensive drug therapy.”

^aIn patients with atherosclerotic renal artery stenosis.

ABPM, Overnight ambulatory reading; ACEI, angiotensin-converting enzyme inhibitor; ASO, atherosclerotic occlusive disease; BP, blood pressure; CHF, congestive heart failure; CVA, cerebrovascular accident; DBP, diastolic blood pressure; DDD, defined daily doses; DRASTIC, Dutch Renal Artery Stenosis Intervention Cooperative; EMMA, Essai Multicentrique Medicaments vs Angioplastie; F/U, follow-up; HTN, hypertension; Med Rx, medical therapy; MI, myocardial infarction; *n*, number of patients; PTRA, percutaneous renal artery angioplasty; RAS, renal artery stenosis; SNRAS, Scottish and Newcastle Renal Artery Stenosis Collaborative Group; Tx, therapy.

medication use.⁸⁸ As noted, such patients commonly come to clinical attention when recognizable clinical progression occurs. Occasionally, clinical decision making is strongly influenced by concerns about the hazards of medical therapy and failing to achieve restored blood flow soon enough. Examination of the results of medical therapy alone is important before evaluating the role of vascular reconstruction or dilation.

Since the introduction of agents blocking the renin-angiotensin system have been introduced, most patients (86%–92%) with unilateral renal artery disease can achieve blood pressure levels lower than 140/90 mm Hg with medical regimens based on these medications. Treatment trials have confirmed that target blood pressure levels can regularly be achieved.^{98,121} Given the widespread use of these agents for patients with many forms of cardiovascular disease, it is likely that subcritical cases of RVD are treated without being identified.

The crux of clinical debates centers around whether the treatment of RAS with antihypertensive drug therapy poses a long-term hazard to kidney function. Early studies with

experimental “clip” hypertension have described renal fibrosis and scarring in the stenotic kidney in animals treated with ACE inhibitors. It is well recognized that removal of the efferent arteriolar effects of Ang II leads to loss of glomerular filtration in a kidney with reduced renal perfusion (Fig. 47.23). Experimental studies in two-kidney, one-clip rats have indicated that the loss of kidney function is sometimes irreversible, although survival is improved with the use of ACE inhibitors compared with minoxidil-treated animals.²⁰⁵ The unique role of ACE inhibitors and ARBs must be understood in this regard. Any drug capable of reducing systemic arterial pressure has the potential to lower renal pressures beyond a critical stenosis. As a result, successful antihypertensive therapy in RVD has the theoretic result of reducing blood flow to the poststenotic kidney sufficient to induce vascular thrombosis.

The unique feature of renin-angiotensin system blockade is the specific reduction of angiotensin-dependent efferent arteriolar resistance sufficient to lower transcapillary filtration pressures. This can occur despite preserved blood flow to the glomerulus (see Fig. 47.23 B).^{206,207} This property is central

Table 47.6 Prospective Randomized Clinical Trials for Renal Function and/or Cardiovascular Outcomes Comparing Medical Therapy With Renal Artery Stenting

Trial	N	Population	Inclusion Criteria	Exclusion Criteria	Outcomes
STAR (2009) ¹²⁰ : 10 centers, f/up 2 years	Med Tx, 76; PTRA, 64	Patients with impaired renal function, ostial ARVD detected by various imaging studies and stable blood pressure (BP) on statin and aspirin	ARVD >50%; creatinine clearance <80 mL/min/1.73 m ² ; Controlled blood pressure 1 month before inclusion	Kidney <8 cm, renal artery diameter <4 mm eCrCl <15 mL/min/1.73 m ² DM with proteinuria (>3 g/day), malignant hypertension	No difference in GFR decline (primary endpoint $\geq 20\%$ change in clearance), but many did not undergo PTRA ARVD <50% on angiography Serious complication in the PTRA group Study was underpowered
ASTRAL (2009) ¹²¹ : 57 centers, f/up 5 years	Med Tx, 403; PTRA, 403	Patients with uncontrolled or refractory hypertension or unexplained renal dysfunction with unilateral or bilateral ARVD on statin and aspirin	ARVD—substantial disease suitable for endovascular Rx and patient's physician uncertainty of clinical benefit from revascularization	High likelihood of PTRA in <6 months Without ARVD, previous ARVD, PTRA, FMD	No difference in BP, renal function, mortality, CV events (primary endpoint: 20% reduction of in the mean slope of the reciprocal of serum creatinine level) Substantial risk in PTRA group
CORAL (2013) ⁹⁸ : 109 centers, f/up 5 years	Med Tx, 480; PTRA, 467	Hypertension, two or more antihypertensives or CKD stage ≥ 3 with ARVD, with unilateral or bilateral disease on statin	SBP >155 mm Hg, at least two drugs; ARVD >60%; subsequent changes included no longer requiring systolic hypertension if eGFR was <60 mL/min	FMD Creatinine >4.0 mg/dL Kidney length <7 cm and use of more than one stent	No difference of death from CV or renal causes modest improvement of SBP in stented group Total, 26 complications (5.5%)

ARVD, Atherosclerotic renovascular disease; CKD, chronic kidney disease; CV, cardiovascular; DM, diabetes mellitus; eCrCl, estimated creatinine clearance; FMD, fibromuscular dysplasia; f/up, follow-up; GFR, glomerular filtration rate; Med Tx, medical therapy; N, number of patients; PTRA, percutaneous renal artery angioplasty; RVD, renovascular disease; SBP, systolic blood pressure; Tx, therapy.

Modified from Herrmann SM, Saad A, Textor SC. Management of atherosclerotic renovascular disease after Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL). *Nephrol Dial Transplant*. 2015;30(3):366–375.

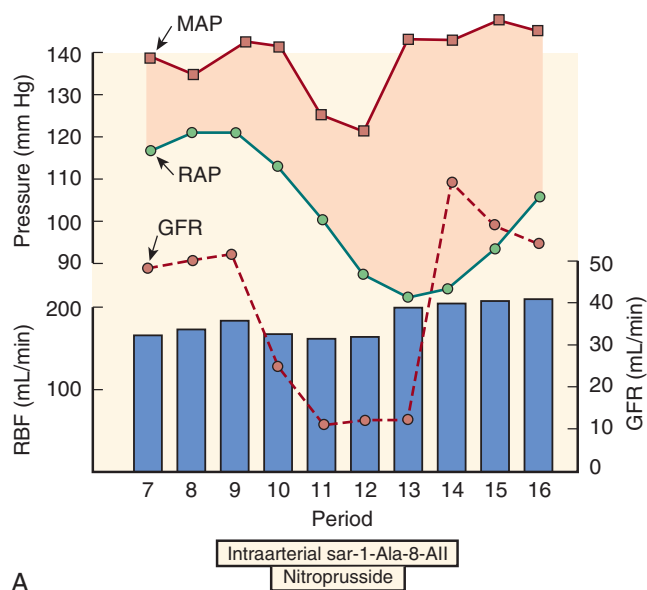
to the benefits of this class of agent in hyperfiltration states thought to accelerate renal damage in other settings, such as diabetes. In the presence of RVD, the fall in glomerular filtration beyond a stenotic lesion can be observed despite relatively preserved plasma flows.²⁰⁷ The fall in the GFR heralds an approaching degree of critical vascular compromise before blood flow itself is reduced.²⁰⁶ Studies in a porcine renovascular model suggested that angiotensin receptor blockade may have microvascular protective effects in the kidney as compared with drug therapy without renin-angiotensin system blockade.²⁰⁸ Earlier studies in renovascular hypertensive animals confirmed that despite a reduction in filtration, renal structural integrity can be preserved and recovered after removal of the clip and/or removal of the ACE inhibitor.²⁰⁹ Hence, it is unlikely that ACE inhibitors or ARBs themselves pose a unique hazard beyond that attributable to reduction in renal blood flow.

It is important to recognize that the contralateral kidney usually supports total glomerular filtration, despite reduced filtration in the stenotic kidney. Changes in overall GFR may be undetectable, in part due to increased filtration in the contralateral kidney.¹²⁷ This fact may be used in several ways. Some authors have argued in favor of using “split” renal

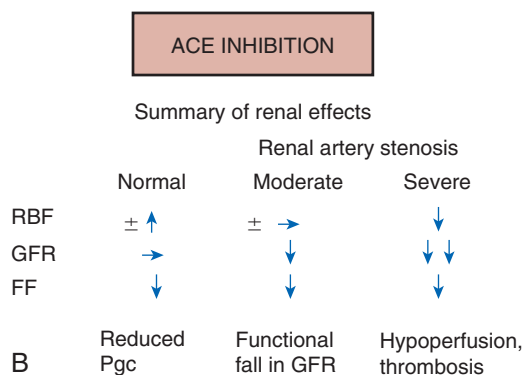
function measurements, such as radionuclide renal scans, to detect loss of individual kidney function as a means of timing revascularization.²⁰² Depending on the circumstances, loss of one kidney might be acceptable if one can ensure the patient that the remaining kidney has adequate function and blood supply. The fall in the GFR from a loss of one kidney may represent a loss of the GFR analogous to that of nephrectomy for malignancy or donating a kidney for renal transplantation. In such cases, the long-term hazard to the remaining kidney is small, although not negligible.^{210–212} As the age and comorbidity of the population at risk rise, the loss of one kidney may pose an acceptable additional hazard if the overall glomerular filtration is adequate. Subsequent follow-up to monitor the condition of the remaining kidney is essential.

The experience with ACE inhibition in trials of CHF is reassuring in this regard. Thousands of patients with marginal arterial pressures and clinical heart failure have been treated with a variety of ACE inhibitors and ARBs.²⁰⁶ These patients are at high risk for undetected renal artery lesions as part of the atherosclerotic burden associated with coronary disease. Although a minor change in the creatinine level is observed in 8% to 10% of these individuals, a rise sufficient to lead

RENAL HEMODYNAMICS AND FUNCTION DURING SYSTEMIC PRESSURE REDUCTION AND ANGIOTENSIN BLOCKADE IN A DOG WITH RENAL ARTERY CONSTRICTION



A



B

Fig. 47.23 (A) The glomerular filtration rate (GFR) falls beyond a renal artery stenotic lesion during blockade of angiotensin II (produced by intraarterial infusion of sar-1-al-8-angiotensin II) and pressure reduction induced by sodium nitroprusside. The fall in GFR occurs despite preserved renal blood flow (RBF), measured by electromagnetic flow probe. These observations illustrate the specific role of angiotensin II in maintaining GFR in the poststenotic kidney at reduced perfusion pressures. (B) Summary of the effects of angiotensin-converting enzyme (ACE) inhibition and/or angiotensin receptor blockade on RBF, GFR, and filtration fraction (FF) in normal, moderate, and severe levels of renal artery stenosis. As compared with other antihypertensive agents, ACE inhibitors and angiotensin receptor blockers (ARBs) lead to a fall in GFR and filtration fraction due to removal of the efferent arteriolar effects of angiotensin II. When stenosis is sufficiently severe that pressure reduction compromises renal blood flow, the potential for complete occlusion is present, as with other effective antihypertensive agents as well. MAP, Mean arterial pressure; RAP, renal artery pressure. (From Textor SC. Renal failure related to ACE inhibitors. *Semin Nephrol*. 1997;17:67–76.)

to withdrawal of these agents under trial monitoring conditions occurs in only 1% to 2%. Data from patients with high cardiovascular disease risk treated with ramipril included patients with creatinine levels up to 2.3 mg/dL. Those with creatinine between 1.4 and 2.3 mg/dL were at higher risk for cardiovascular mortality and had a major survival benefit

from ACE inhibition.¹⁴¹ Close follow-up of kidney function has indicated that withdrawal of ACE inhibition due to deterioration of renal function is less than 5% and no greater than placebo.¹⁴¹ All participants in the CORAL trial were treated with an ARB, with rare discontinuation.⁹⁸ Registry data from the United Kingdom have further supported the tolerability of renin-angiotensin system blockade in more than 90% of cases of atherosclerotic RVD, including more than 75% of patients with bilateral disease.¹⁰³ Importantly, there appeared to be a mortality benefit in patients treated with these agents.²¹³

PROGRESSIVE RENAL ARTERY STENOSIS IN MEDICALLY TREATED PATIENTS

The potential for vascular occlusive disease to progress further is central to long-term management of patients with RVD. It may be argued that failure to revascularize the kidneys exposes the individual to the hazard of undetected progressive occlusion, potentially leading to irreversible loss of renal function. Because the treatment trials have been relatively short (2–4 years), this issue remains an important consideration for the long-term treatment of atherosclerotic RVD. A firm understanding of the evidence regarding progressive atherosclerotic disease of the kidney is important for planning endovascular and surgical revascularization.

Atherosclerosis is a variably progressive disorder. Those involved in the management of disorders of the carotid, coronary, aortic, and peripheral vasculatures all recognize the potential for progression, which occurs at widely different individual rates. Medical therapy of all of these should incorporate measures aimed at intensive risk factor reduction. Treatment of these risk factors reduces mortality rates from cardiovascular disease.²¹⁴

An important clinical question is whether progressive renal artery occlusive disease affects the progression of renovascular hypertension. Moderate anatomic progression alone does not reliably predict functional changes in terms of deteriorating blood pressure control or renal function. Results from prospective Doppler ultrasound studies have indicated that a decrement in measured renal size by 1 cm (renal atrophy) developed in 5.5% of those with normal initial vessels and 20.8% of those with baseline stenosis greater than 60% during a follow-up interval of 33 months.²¹⁵ Changes in the serum creatinine level were infrequent but did occur in a subset of patients, particularly those with bilateral RAS. These findings are in general agreement with early studies during medical therapy of renovascular hypertension, in which 35% had a detectable fall in measured renal length, but only 8 of 41 (19%) had a significant rise in the creatinine level during a follow-up of 33 months.¹³⁶ Follow-up of the medical treatment arms during short-term studies have failed to show major changes in kidney function, although occasional loss of renal perfusion was observed by radionuclide scanning.¹³⁵

Another important question is whether the management of RAS without revascularization leads to clinical progression, either in terms of refractory hypertension or advancing renal insufficiency. Follow-up of patients with incidentally identified RAS is helpful in this regard. A review of peripheral aortograms has identified 69 patients with high-grade renal arterial stenoses (>70%) followed without revascularization for more than 6 months.²¹⁶ Their long-term follow-up identified

generally satisfactory blood pressure control, although some required more intensive antihypertensive therapy during an average of 36 months of follow-up. Of these, 4 eventually underwent renal revascularization for refractory hypertension and/or renal dysfunction, and 5 developed ESKD, of which only one case was thought to be related to RAS directly. Overall, the serum creatinine level rose from 1.4 to 2.0 mg/dL. These data indicate that many of these patients can be managed without revascularization for years, and that clinical progression leading to urgent revascularization develops in 10% to 14% of them. Expansion of this data set to 160 individuals has allowed comparison of different antihypertensive regimens.¹³⁰ The rates of progression were not specifically related to the introduction of ACE inhibitors, although the level of blood pressure control improved in later years with all medications.¹³⁰

These observations have been supported by a report of 126 patients with incidental RAS compared with 397 patients matched for age.¹³⁷ The measured serum creatinine level was higher, and the calculated GFR (estimated using the Cockcroft-Gault equation) was lower in patients with RAS followed for 8 to 10 years. However, no patient progressed to ESKD. These observations are consistent with the results of prospective trials of medical versus surgical intervention, which started in the 1980s and extended into the 1990s.²¹⁷ Remarkably similar data are evident from other RCTs, in which both the medical therapy and revascularization arms progressed to a renal endpoint in 16% to 22% of cases in the CORAL and ASTRAL trials (see Table 47.6).^{98,121} No differences in patient survival or renal function were identified. Taken together, these studies suggest that the rates of progression of RVD are moderate and occur at widely varying rates. Many of these patients can be managed without revascularization for years, although some will inevitably progress. These observations underscore the importance of close follow-up for those identified with RVD.

Although these reports are informative, they leave many questions unanswered:

- How often does suboptimal blood pressure control of renovascular hypertension accelerate cardiovascular morbidity and mortality?
- Does one lose the opportunity to reverse hypertension effectively by delaying renal revascularization?

These issues require further prospective studies. It is clear that progressive disease develops in a subset of high-risk patients who present with pulmonary edema, accelerating hypertension, and rapidly falling GFR. In such patients, optimal stability of kidney function and blood pressure control over the long term can be achieved by successful surgical or endovascular restoration of the renal blood supply.^{108,218}

ENDOVASCULAR RENAL ANGIOPLASTY AND STENTING

The ability to restore renal perfusion in high-risk patients with renovascular hypertension and ischemic nephropathy using endovascular methods represents a major advance in regard to treatment of this disorder. Restoration of blood flow to the kidney beyond a stenotic lesion intuitively should provide a means to improve renovascular hypertension and

halt progressive vascular occlusive injury. In the 1990s, a major shift from surgical reconstruction ensued, toward preferential application of endovascular procedures.⁹ The total volume of renal revascularization procedures registered for the US Medicare population older than 65 years rose 62%, from 13,380 to 21,600 between 1996 and 2000.²¹⁹ This change reflected an increase in endovascular procedures by 2.4-fold, whereas surgical renovascular procedures fell by 45%. The trend continued, with an estimate of 35,000 endovascular procedures in 2005.^{219,220} Since the publication of the ASTRAL trial, this trend has reversed dramatically, leading to few revascularization procedures in the United Kingdom.

Revascularizing the kidney has benefits and risks. With older patients developing RAS in the context of preexisting hypertension, the likelihood of a total cure for hypertension is low, particularly in atherosclerotic disease. Although complications are not common, they can be catastrophic, including atheroembolic disease and aortic dissection. Knowing when to pursue renal revascularization is central to the dilemma of managing RVD.

ANGIOPLASTY FOR FIBROMUSCULAR DISEASE

Most lesions of medial fibroplasia are located at a distance away from the renal artery ostium. Many of these have multiple webs within the vessel, which can be successfully traversed and opened by balloon angioplasty. Stenting is rarely required unless vessel dissection occurs. Experience in the 1980s indicated more than 94% technical success rates.²²¹ Some of these lesions ($\approx 10\%$) develop restenosis, for which repeat procedures have been used.⁹¹ Clinical benefit regarding blood pressure control has been reported in observational outcome studies in 65% to 75% of patients, although the rates of cure are less secure.⁸³ Cure of hypertension, defined as sustained blood pressure levels less than 140/90 mm Hg with no antihypertensive medications, may be obtained in 35% to 50% of patients.²²² Predictors of cure—normal arterial pressures without medication at 6 months and beyond after angioplasty—include lower systolic blood pressures, younger age, and shorter duration of hypertension.^{222,223}

A large majority of patients with FMD are female, at younger ages than those with atherosclerotic disease.⁸³ In general, such patients have less aortic disease and are at a lower risk for major complications of angioplasty. Because the risk for major procedural complications is low, most clinicians favor early intervention for hypertensive patients with FMD with the hope of reduced antihypertensive medication requirements after successful angioplasty.

ANGIOPLASTY AND STENTING FOR ATHEROSCLEROTIC RENAL ARTERY STENOSIS

After the introduction of percutaneous transluminal renal angioplasty (PTR), it was soon evident that ostial lesions commonly failed to respond, in part because of extensive recoil of the plaque, which extended into the main portion of the aorta.²²⁴ These lesions develop restenosis rapidly, even after early technical success. Endovascular stents were introduced for ostial lesions in the late 1980s and early 1990s.²²⁵

The technical advantage of stents is indisputable. An example of successful renal artery stenting is shown in Fig. 47.24. Prospective comparisons between angioplasty alone versus angioplasty with stents has shown that intermediate

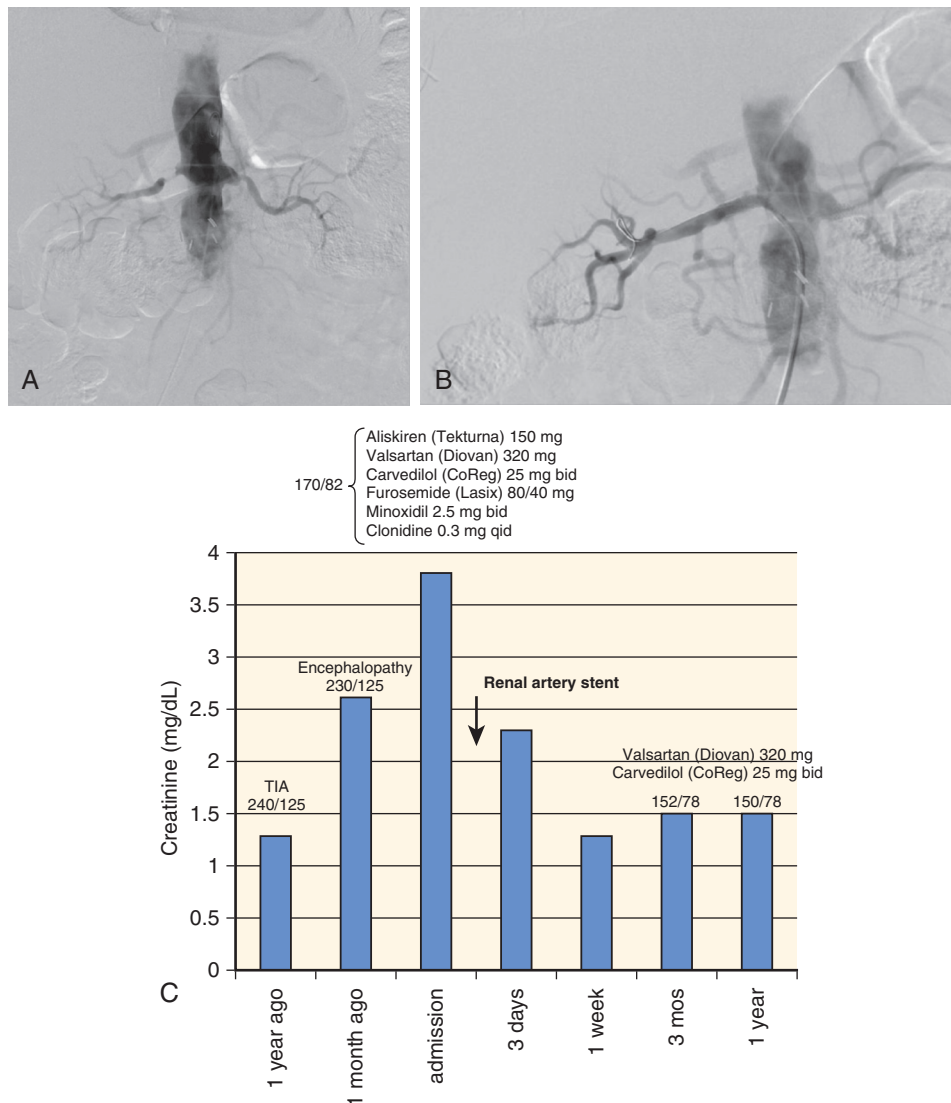


Fig. 47.24 (A) Renal aortogram illustrating high-grade, bilateral renal arterial lesions in a 73-year-old male who developed accelerated hypertension a few months before. (B) Aortogram after placement of endovascular stents, illustrating excellent vessel patency and early technical success. This was followed by resolution of his hypertension. (C) Serum creatinine levels, blood pressure levels, and antihypertensive drug regimen in an individual with high-grade bilateral renal artery stenosis. Rapidly developing hypertension and loss of glomerular filtration rate were associated with transient neurologic deficits. Successful renal artery stenting led to sustained reduction in serum creatinine, improved blood pressure levels, and reduced medication requirements. Such cases reflect high-risk subsets that were not well represented in the prospective randomized controlled trials. (Modified from Textor SC. Attending rounds: a patient with accelerated hypertension and an atrophic kidney. *Clin J Am Soc Nephrol*. 2014;9(6):1117–1123.)

(6–12 months) vessel patency is 29% and 75%, respectively.²²⁴ Restenosis fell from 48% to 14% in stented patients. As technical success continues to improve, many reports have suggested nearly 100% technical success in early vessel patency, although rates of restenosis continue to reach 14% to 25%.^{9,148,226} The introduction of endovascular stents has expanded renal revascularization, in part because of improved technical patency possible with ostial atherosclerotic lesions and applicability in older patients with substantial comorbidity.

Regarding outcomes of renal stenting, a systematic analysis under the direction of the Agency for Health Care Research and Quality has found that the “strength of evidence” through 2016 regarding the benefits and risks of stent revascularization

is low, in part because of major differences between subjects included in prospective RCTs and those reported in observational series.²²⁷ These outcomes are commonly considered in terms of the following: (1) blood pressure control; and (2) preservation or salvage of renal function in ischemic nephropathy. Observational cohort blood pressure studies after stenting face the same limitations as those observed with angioplasty alone. Results over 1- to 4-year follow-ups are summarized from representative series in [Tables 47.4 to 47.6](#) and reviewed in detail elsewhere.^{228,229} Typically reported falls in blood pressure in observational cohorts are in the range of 25 to 30 mm Hg systolic, the best predictor of the initial systolic blood pressure.²³⁰ Some authors have reported 42% improvement in blood pressure with fewer medications

needed, although cures were rare, and renal function was unchanged.²³¹ Careful attention to the degree of residual patency in 210 patients with stents reported more than 91% patency at 1 year and 79% patency at 5 years.¹⁵³ Blood pressures were cured or improved in more than 80%. In some cases, episodes of angina and recurrent CHF subsided.^{104,232} As noted in the trials summarized later, prospective RCTs have been less impressive regarding the benefits of stenting^{98,120,121} (Fig. 47.25).

Outcomes regarding the recovery of renal function after endovascular revascularization are summarized in Tables 47.4 to 47.6. In general, average changes in renal function for atherosclerotic RAS, as reflected by serum creatinine levels, have been small.¹⁰⁹ Remarkably, the changes in renal function in azotemic patients after surgical reconstruction are similar.^{233,234} The average group changes in kidney function can be misleading. Careful evaluation of the literature indicates that three distinctly different clinical outcomes are routinely observed. In some cases ($\approx 27\%$), revascularization produces clinical improvements in kidney function.¹⁰⁹ For this group, in one reported series, the mean serum creatinine level fell from a mean value of 4.5 mg/dL to an average of 2.2 mg/dL. There can be no doubt that such patients benefit from the procedure and avoid the major morbidity (and probably mortality) associated with advanced renal failure. The bulk of patients ($\approx 52\%$) have little measurable change in renal function, however.¹⁰⁹ Whether such patients benefit clinically depends on the actual likelihood of progressive renal injury if the stenotic lesion were managed without revascularization, as discussed previously. Those without much

risk of progression gain little, other than potential reversal of hypertrophy in the unaffected kidney.

The most significant concern, however, is the group of patients whose renal function deteriorates further after a revascularization procedure. In most reports, this ranges from 19% to 25%.^{109,235} In some cases, this represents atheroembolic disease or a variety of complications, including vessel dissection with thrombosis.²³⁶ Hence, nearly 20% of azotemic patients face progression of renal insufficiency and the potential of requiring renal replacement therapy.^{234,237,238} Predictors of both mortality and ESKD include the level of renal function before intervention and the levels of proteinuria.¹²² Importantly, patients with minimal proteinuria enrolled in CORAL had improved mortality outcomes.³ Deterioration of renal function after stenting sometimes develops rapidly and may reflect atheroembolic injury, which may be nearly universal after any vascular intervention,²³⁹ and acceleration of oxidative stress, producing interstitial inflammation and fibrosis.^{240,241} Whether improved techniques, including the application of distal protection devices for endovascular catheters, will reduce these complications is not yet certain.

Several studies have suggested that the progression of renal failure attributed to ischemic nephropathy can be reduced by endovascular procedures.^{108,235,242} Harden and coworkers presented reciprocal creatinine plots in 23 (of 32 patients), suggesting that the slope of loss of the GFR could be favorably changed after renal artery stenting.²³⁵ Renal function improved or stabilized in 69%, indicating that in 31% it worsened, consistent with results from other series. These reports and a guideline document from the American Heart Association

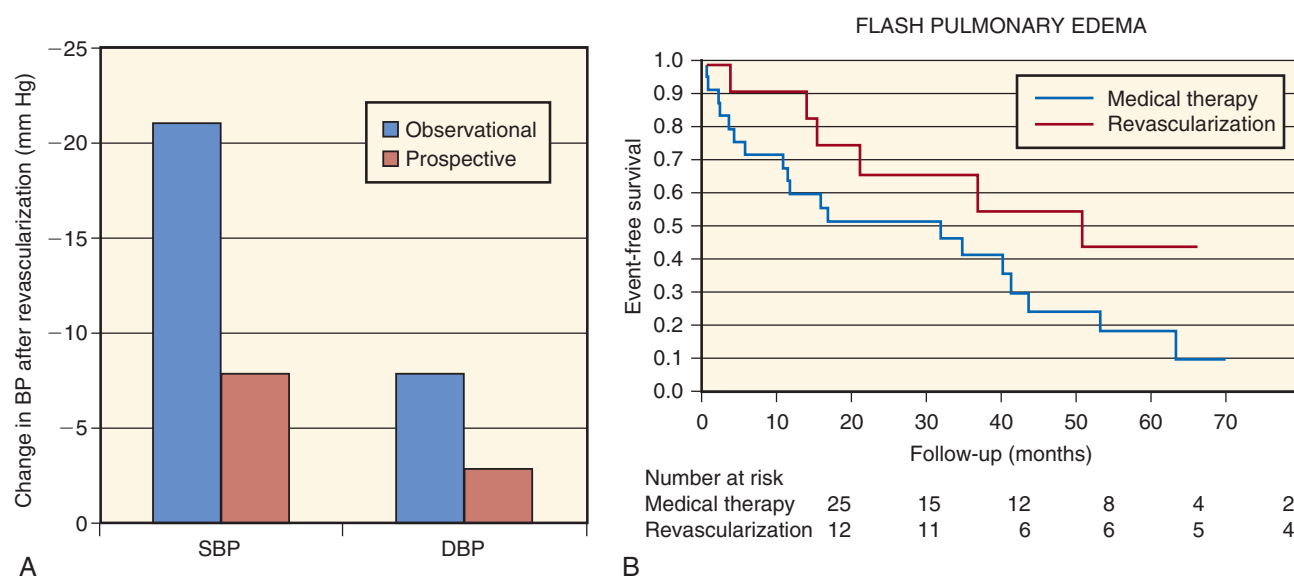


Fig. 47.25. (A) Comparison of blood pressure (BP) changes reported after renal revascularization from a large observational registry (>1000 patients) and from a meta-analysis of three prospective randomized trials (210 patients). The large difference between these BP effects is reflected in variable enthusiasm for intervention between clinicians. Initial results from prospective trials (ASTRAL and CORAL; see text) identified only minor differences in achieved BP levels. Although results from observational series may overstate treatment benefits, results from prospective trials likely underestimate changes, in part due to limitations in patient recruitment and crossover between treatment arms, ranging from 27% to 44% in early series (see text). (B) Follow-up data from a registry in the United Kingdom, most of whom were not considered candidates for ASTRAL. This report identified a major mortality risk for patients with renal artery stenosis and episodes of pulmonary edema that was reduced for patients submitted for renal revascularization. A similar mortality benefit was identified for patients with rapidly progressive renal failure (see text). DBP, Diastolic BP; SBP, systolic BP. (From Ritchie J, Green D, Chrysochou C, et al. High-risk clinical presentations in atherosclerotic renovascular disease: prognosis and response to renal artery revascularization. *Am J Kidney Dis.* 2014;63(2):186–197.)

have promoted the use of breakpoint analysis to analyze and report the results of revascular procedures.¹⁵¹ Caution must be applied regarding the application of breakpoints using reciprocal creatinine plots in this disorder, however. Vascular disease does not affect both kidneys symmetrically, nor is it likely to follow a constant course of progression, in contrast to diabetic nephropathy, for example. As a result, a gradual loss of renal function with subsequent stabilization can be observed equally with unilateral disease leading to total occlusion, as well as successful revascularization.

Perhaps the most convincing group data in this regard derives from serial renal functional measurement in 33 patients with high-grade (>70%) stenosis affecting the entire renal mass (bilateral disease or stenosis in a solitary functioning kidney), with creatinine levels between 1.5 and 4.0 mg/dL.²⁴² Follow-up over a mean of 20 months has shown that the slope of GFR loss converted from negative (-0.0079 dL/mg/month) to positive (0.0043 dL/mg/month). These studies agree with other observations that long-term survival is reduced in bilateral disease and that the potential for renal dysfunction and accelerated cardiovascular disease risk is highest in such patients.

PROSPECTIVE TREATMENT TRIALS

MEDICAL THERAPY COMPARED WITH ANGIOPLASTY PLUS STENTS

In regard to renovascular hypertension, as noted previously, many patients with unilateral RAS are managed without restoration of blood flow for a long period, sometimes indefinitely. The judgment about endovascular intervention in a specific case is based on the anticipated outcome, as summarized later. There are now several prospective randomized trials comparing medical therapy with revascularization using PTRAs plus stents on which to base conclusions. Familiarity with the available trials and their limitations is important. The major features of these trials are summarized in [Tables 47.5 and 47.6](#).

Three small trials in renovascular hypertension from the 1990s have addressed the relative value of endovascular repair, specifically PTRAs as compared with medical therapy for atherosclerotic RAS.^{129,135,243} To the credit of these investigators, care was taken to standardize blood pressure measurement before and after endovascular repair and to select antihypertensive regimens carefully. All these trials have limitations, but they are instructive.

Webster and colleagues randomized 55 patients with atherosclerotic RAS to medical therapy or PTRAs.¹²⁹ Follow-up blood pressures were obtained using a random-zero sphygmomanometer after a run-in period. The run-in period produced considerable reduction in blood pressures in all patients. Those with unilateral disease showed no difference between medical therapy and PTRAs¹²⁹ after 6 months. There was greater blood pressure benefit after PTRAs in those with bilateral RAS. The authors concluded that they were “unable to demonstrate any benefit with respect to renal function or event free survival” during follow-up, up to 40 months.

Plouin and colleagues randomized 49 patients with unilateral atherosclerotic RAS more than 75% or more than 60% with lateralizing functional studies.²⁴³ Blood pressure measurements were based on overnight ambulatory readings (ABPM), which are thought to yield more reproducible data

and to be relatively free from placebo or office effects. Seven of 26 subjects assigned to medical therapy eventually crossed over to the PTRAs group (27%) for refractory hypertension. There were six procedural complications in the PTRAs group, including branch dissection and segmental infarction. Final blood pressures were not different between groups, but fewer medications were required in the PTRAs group. Taken together, this trial suggested that PTRAs produced more complications in the short term, was useful in some medical treatment failures, and required slightly fewer medications after 6 months. Medical therapy for this study specifically excluded renin-angiotensin system blockers.

A third prospective trial included 106 patients enrolled in the DRASTIC study.¹³⁵ These patients were selected for resistance to therapy, including two drugs, and were required to have serum creatinine values below 2.3 mg/dL. Blood pressures were evaluated using automated oscillometric devices at 3 and 12 months after entry. Patients were evaluated on an intention to treat basis. Blood pressures did not differ between groups overall at 3 or 12 months, although the PTRAs group was taking fewer medications (2.1 ± 1.3 vs. 3.2 ± 1.5 defined daily doses; $P < .001$). The authors concluded that in the treatment of hypertension and RAS, “angioplasty has little advantage over antihypertensive drug therapy.”¹³⁵ Many critics emphasize, however, that 22 of 50 patients (44%) assigned to medical therapy were considered treatment failures and referred for PTRAs after 3 months. There were eight cases of total arterial occlusion in the medical group, as compared with none in the angioplasty group.

Many clinicians interpret these data to support an important role for PTRAs in the management of patients with refractory hypertension and RAS. Regardless, the results of these trials indicate that benefits of endovascular procedures, even in the short term, are moderate compared with effective antihypertensive therapy. Patients failing to respond to medical therapy often improve after revascularization. Some authors have combined these prospective studies into meta-analyses, indicating that taken together, renal revascularization produces modest but definite reductions in blood pressures, averaging $-7/-3$ mm Hg.^{227,244,245}

STENTS FOR PROGRESSION OF RENAL INSUFFICIENCY AND CARDIOVASCULAR OUTCOMES

Several prospective RCTs have sought to compare medical therapy with endovascular stenting targeted primarily to the progression of chronic kidney disease (CKD) and/or adverse cardiovascular (CV) outcomes, such as stroke and coronary disease events. Among these, the largest are the US-based CORAL trial and the UK-based ASTRAL trial (see [Table 47.6](#)).^{98,246} Although these have been published and add important information, they have substantial limitations.

CORAL was designed to test whether renal artery stenting, when added to optimal (standardized) medical therapy, including blockade of the renin-angiotensin system, improves CV outcomes for individuals with atherosclerotic RAS.⁹⁸ The authors listed more than 110 participating centers, with a total of 947 participants enrolled over a 5-year period to stenting plus medical therapy (467 patients) or to medical therapy alone (480 patients). Average entry estimated GFR (eGFR) values were 58 mL/min/1.73 m². This trial attempted to standardize the evaluation of stenosis by requiring transluminal gradient measurement and review by an angiographic

core laboratory. After a mean follow-up of 43 months, no differences were apparent for any or all of the composite endpoints—death from CV or renal causes, myocardial infarction, stroke, hospitalization for CHF, progressive renal insufficiency, or the need for renal replacement therapy—between the stent and medical therapy alone groups (35.1% and 35.8%, respectively; hazard ratio, 0.94; 95% confidence interval, 0.76–1.17; $P = .58$).⁹⁸ Systolic blood pressure was slightly lower in the revascularization arm (−2.3 mm Hg). Post hoc analysis of 413 participants in CORAL, with a urine albumin-to-creatinine ratio below the median (22.5 mg/g), showed reduced cardiovascular event rates and improved mortality compared with those with higher entry levels of proteinuria.³ Importantly, procedural complications per vessel treated were only 5.2% in the stent group (Box 47.4). CORAL has several important limitations, which are discussed elsewhere and listed later.^{1,115}

The authors acknowledged that recruitment for CORAL was difficult and took longer than anticipated. Most centers enrolled fewer than 10 subjects over the 4- to 5-year period. Several criteria for enrollment and intervention changed during the trial. The original intention had been to include patients with severe RAS and systolic blood pressure above

155 mm Hg while receiving two or more antihypertensive medications. “Severe RAS” was defined as more than 80% stenosis in isolation or 60% to 80% with a gradient of at least 20 mm Hg. As the protocols evolved, patients could be enrolled with or without hypertension if their eGFR was less than 60 mL/min/1.73 m². Requirements for translational gradients were dropped, and patients could be enrolled using duplex ultrasound, MRA, or CTA criteria. Ultimately, the average level of stenosis (67%) measured in the core laboratory was lower than estimates by the investigators on site (73%).

As with other RCTs, it is likely that many of these lesions were below the threshold of 75% to 80% usually required to produce a reduction in blood flow in experimental studies. Of these patients, 25% had reached goal blood pressures before entry. Specific high-risk groups, including those with CHF within 30 days, were excluded. Based on Medicare claims data indicating more than 20,000 renal artery stent procedures annually (estimates from 2000), it is clear that screening and enrollment for CORAL included only a small fraction of treatment candidates, for whom no contemporaneous US registry comparison is available. Based on difficult enrollment and relaxed criteria, it appears that the CORAL cohort represented a relatively low-risk atherosclerotic population. This is supported by well-preserved levels of eGFR at baseline, easily controlled blood pressures, and exclusion of recent CHF and low 5-year CV mortality. Remarkably, fewer than 5% of enrolled patients died of cardiovascular causes during CORAL, substantially less than the registry population reported by Ritchie and associates,¹⁰⁸ and less even than the ASTRAL cohort,¹²¹ which specifically excluded patients that clinicians thought would definitely benefit from revascularization.

Results from the Angioplasty and Stenting for Renal Atherosclerotic Lesions (ASTRAL) trial from the United Kingdom were published for 806 subjects followed for several years.²⁴⁶ The mean serum creatinine level was above 2.0 mg/dL, and the severity of vascular occlusion estimated by a variety of imaging methods was above 70%. Patients were considered eligible for the trial if clinicians were uncertain about optimal management. No differences were apparent regarding changes in kidney function, blood pressure, hospitalizations, mortality, or episodes of circulatory congestion.¹²¹ Rates of progression to a renal endpoint were 16% to 22% in both medically treated and revascularized groups. A smaller trial, Stent Placement in patients with Atherosclerotic Renal Artery Stenosis (STAR), enrolled 140 subjects and aimed to evaluate progression of CKD based on initial creatinine clearance of less than 80 mL/min and atherosclerotic stenosis of over 50%. No differences in change in creatinine clearance were detected at 2 years, although some substantial complications occurred in the stent-treated group.¹²⁰ This study highlights the drawbacks of including trivial vascular lesions in a treatment trial. Only 46 of 64 patients assigned to stent therapy underwent stenting, predominantly because lesions were often not hemodynamically significant.

These modest benefits present a striking contrast between the current era and a few decades ago. Reports from the 1970s underscore the fact that some patients experienced recurrent episodes of malignant phase hypertension, with encephalopathy, fluid retention, and progressive renal insufficiency. Since then, malignant hypertension has become

Box 47.4 Complications After Percutaneous Transluminal Renal Angioplasty and Stenting of the Renal Arteries

Minor—most frequently reported)

- Groin hematoma
- Puncture site trauma

Major—reported in 71/799 treated arteries (9%)²⁴⁹

- Hemorrhage requiring transfusion
- Femoral artery pseudoaneurysm needing repair
- Brachial artery traumatic injury needing repair
- Renal artery perforation leading to surgical intervention
- Stent thrombosis, surgical or antithrombotic intervention
- Distal renal artery embolus
- Iliac artery dissection
- Segmental renal infarction
- Cholesterol embolism: renal
- Peripheral atheroemboli
- Aortic dissection¹⁶⁷
- Restenosis, 16% (range, 0%–39%)
- Deterioration of renal function: 26% (range, 0%–45%)

Combined complication rate over 24 months—stenting after failed PTRA, 24%²²²

Mortality attributed to procedure, 0.5%

Procedure-related complications—51/379 patients in 10 series, 13.5%

Procedural complications reported from CORAL angiographic core laboratory (2014)¹⁵⁹

- Dissection 11/495 (2.2%)
- Branch occlusion 6/495 (1.2%)
- Distal embolization 6/495 (1.2%)

Single events—wire perforation, vessel rupture, pseudoaneurysm

Total events; 26/495 (5.2%)

PTRA, Percutaneous transluminal renal angioplasty.

Table 47.7 Clinical Scenarios in Which Treatment of Significant Renal Artery Stenosis May Be Considered^a

Type of Care	Examples
Appropriate care	• Cardiac disturbance syndromes (flash pulmonary edema or acute coronary syndrome [ACS]) with severe hypertension • Resistant hypertension (HTN; uncontrolled with failure of maximally tolerated doses of at least three antihypertensive agents, one of which is a diuretic, or intolerance to medications) • Ischemic nephropathy with chronic kidney disease (CKD), with eGFR <45 mL/min/1.73 m² and global renal ischemia (unilateral significant RAS with a solitary kidney or bilateral significant RAS) without other explanation
May be appropriate care	• Unilateral RAS with CKD (eGFR <45 mL/min/1.73 m²) • Unilateral RAS with prior episodes of congestive heart failure (stage C) • Anatomically challenging or high-risk lesion (early bifurcation, small vessel, severe concentric calcification, severe aortic atheroma or mural thrombus)
Rarely appropriate care	• Unilateral, solitary, or bilateral RAS with controlled BP and normal renal function. • Unilateral, solitary, or bilateral RAS with kidney size <7 cm in pole to pole length • Unilateral, solitary, or bilateral RAS with chronic end-stage kidney disease on hemodialysis >3 months • Unilateral, solitary, or bilateral renal artery chronic total occlusion

^aSignificant RAS is an angiographically moderate lesion (50%–70%) with physiologic confirmation of severity or >70% stenosis.

BP, Blood pressure; eGFR, estimated glomerular filtration rate.

Modified from Parikh SA, Shishebor MH, Gray BH, et al. SCAI expert consensus statement for renal artery stenting appropriate use. *Catheter Cardiovasc Intervent.* 2014;84:1163–1171.

less prevalent in most Western countries. The introduction of ACE inhibitors and calcium channel blockers in the 1980s led to reductions in the number of people with severe hypertension and improved the medical management of patients with high-renin states, including renovascular hypertension. Results from the prospective trials of angioplasty are less favorable now than those from retrospective series. A registry report from 2002 of over 1000 successfully stented patients has suggested that average blood pressure levels fall during follow-up, $-21/-10$ mm Hg¹⁴³ (see Fig. 47.23). The differences between prospective trials and registry data may reflect outcome reporting bias. Alternatively, enrollment in prospective trials itself may reflect recruitment bias in favor of more stable patients in less urgent clinical need of restoring renal circulation. Hence, the randomized trials almost certainly underestimate the benefits of renal revascularization for patients at the greatest risk of accelerated hypertension and/or renal failure.^{185,227} Taken together, however, results of recent trials underscore the efficacy of medical therapy for many patients and weaken the argument for revascularization for moderate atherosclerotic RAS. Recent consensus recommendations regarding appropriate use of renal revascularization from the Society for Cardiovascular Angiography and Interventions are summarized in Table 47.7.¹⁸⁵

COMPLICATIONS OF RENAL ARTERY ANGIOPLASTY AND STENTING

Atherosclerotic plaque is commonly composed of multiple layers, with calcified, fibrotic, and inflammatory components. The physical expansion of such lesions applies considerable force on the wall and may lead to cracking and release of particulate debris into the bloodstream. Effective balloon angioplasty and stenting require techniques to limit damage to blood vessels during the procedure. A review of 10 published series with 416 stented vessels has indicated that

significant complications arise in 13% of cases, not counting those that lead to the need for dialysis.⁹ Complications are listed in Box 47.4; these include hematomas and retroperitoneal bleeding requiring transfusion. On average, renal function deteriorated in 26% of subjects in these series, and 50% (7 of 14) of those with preprocedure creatinine levels above 400 μ mol (4.5 mg/dL) progressed to ESKD requiring dialysis.²³⁷ Follow-up data regarding both mortality and ESKD for recent stented cohorts tend to support these observations.¹²² Most complications are minor, including local hematomas and false aneurysms at the insertion site. Occasional severe complications develop, including aortic dissection,²³⁶ stent migration, and vessel occlusion with thrombosis.⁹ Local renal dissections can be managed by the judicious application of additional stents. Mortality related directly to this procedure is low, but has been reported in 0.5% to 1.5% of patients.^{237,247}

Restenosis remains a significant clinical limitation. Rates vary widely, from 13% to 30%, usually developing within the first 6 to 12 months.^{226,248,249} Most more recent series have reported 13% to 16% restenosis, sometimes leading to repeat procedures.²²⁶

RENAL DENERVATION

Activation of sympathoadrenergic pathways has long been recognized as a major regulatory pathway for blood pressure. Studies of the renal nerves identify afferent and efferent signaling between the kidney and brain.^{250,251} These pathways affect numerous mechanisms, including renin release, sodium transport, and peripheral sympathetic nerve activity. Clinical studies in patients with resistant hypertension have indicated that systemic blood pressures fall after bilateral renal denervation using radiofrequency or ultrasonic energy.^{252,253} Initial studies suggested that reductions in blood pressure may exceed 34/14 mm Hg after 36 months.²⁵⁴ Follow-up studies in wider ranges of patients using sham controls had smaller

decrements in pressure, in the range of 14 mm Hg systolic as compared with 11 mm Hg in controls.²⁵⁵ Additional studies in less resistant subjects have observed decreases of 6 to 8 mm Hg (systolic) as compared with controls.^{253,256} The durability of these changes is not yet known, and most patients continue to require antihypertensive drug therapy. Experimental studies in RVD rats have indicated that denervation may reduce blood pressure and vascular remodeling. Importantly, human studies using endovascular denervation have excluded patients with RVD to this point, specifically to minimize safety concerns regarding renal arterial injury.

SURGICAL TREATMENT OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

Early experience with vascular disease of the kidney was based entirely on surgical intervention, either nephrectomy or vascular reconstruction, with the objective of “surgical curability.”⁶ For that reason, much of the original data regarding split renal function measurement was geared toward identifying functionally significant lesions as a guide for patient selection for a major surgical procedure. Surgical intervention is less commonly performed today and is usually reserved for complex vascular reconstruction and/or failed endovascular procedures. Generally, age and comorbidities in patients with atherosclerotic disease favor endovascular procedures, when feasible.

Methods of surgical intervention have changed over the decades. A review in 1982 emphasized the role for ablative techniques, including partial nephrectomy.²⁵⁷ The use of ablative operative means was guided by the difficulty of controlling blood pressure during this period. Such interventions have become less common since the introduction of effective medication regimens. In some cases, nephrectomy of a totally infarcted kidney provides major improvement in blood pressure control at low operative risk. The introduction of laparoscopic surgical techniques makes nephrectomy technically easier in some patients for whom vascular reconstruction is not an option. These series reflect widely variable methods of determining blood pressure benefit, as discussed later.^{183,257}

Surgical series from the 1960s and early 1970s indicated that cure of hypertension occurred in only 30% to 40% of subjects, despite attempts at preselection.^{6,258} Survival of groups chosen for surgery appeared to be better than those chosen for medical management. This likely reflected the high preoperative risks identified in those for whom surgery was excluded. The cooperative study of RVD in the 1960s and 1970s examined many of the clinical characteristics of renovascular hypertension.³⁵ These studies identified some of the limitations and hazards of surgical intervention and reported mortality rates of 6.8%, even in excellent institutions. The mean age in this series was 50.5 years. Definitions of operative mortality included events as late as 375 days after the procedure and may have overestimated the hazard. Had the authors considered only deaths within the first week, for example, the immediate perioperative mortality would be 1.7%.

The subsequent development of improved techniques for patient selection, including screening for coronary and carotid disease, for renal artery bypass and endarterectomy, and for combined aortic and renal artery repair have altered the

Box 47.5 Surgical Procedures Applied to Reconstruction of the Renal Artery and/or Reversal of Renovascular Hypertension

Ablative surgery: removal of a “pressor” kidney

- Nephrectomy: direct or laparoscopic
- Partial nephrectomy

Renal artery reconstruction (requires aortic approach)

- Renal endarterectomy
- Transaortic endarterectomy
- Resection and reanastomosis: suitable for focal lesions
- Aortorenal bypass graft

Extraanatomic procedures (may avoid direct manipulation of the aorta)—require adequate alternate circulation without stenosis at celiac origin

- Splenorenal bypass graft
- Hepatorenal bypass graft
- Gastroduodenal, superior mesenteric, iliac to renal bypass grafts
- Autotransplantation with exvivo reconstruction

Modified from Bower TC, Oderich GS. *Surgical Revascularization*. London: Springer; 2014; and Libertino JA, Zinman L. Surgery for renovascular hypertension. In: Breslin DL, Swinton NW, Libertino JA, Zinman L, eds. *Renovascular hypertension*. Baltimore: Williams and Wilkins; 1982:166–212.

practice of vascular surgery.^{6,259} Some options developed for renal artery reconstruction are listed in Box 47.5. Most methods focus on reconstruction of the vascular supply for preservation of the nephron mass. Transaortic endarterectomy can effectively restore circulation to both kidneys. It requires aortic cross-clamping and may be undertaken as part of a combined procedure with aortic replacement. Identification and treatment of carotid and coronary disease have led to reductions in surgical morbidity and mortality. By addressing associated CV risk before surgery, early surgical mortality falls below 2% for patients without other major diseases.²⁶⁰

Surgical reconstruction of the renal blood supply usually requires access to the aorta. A variety of alternative surgical procedures have been designed to avoid manipulation of the badly diseased aorta or when previous surgical procedures make access difficult. These include extraanatomic repair of the renal artery using hepatorenal or splenorenal conduits that avoid manipulation of a badly diseased aorta.²⁶¹ An example of contemporary surgical reconstruction to protect renal function is illustrated in Fig. 47.26. Success with extrarenal conduits depends on the integrity of the alternative blood supply. Hence, careful preoperative assessment of stenotic orifices of the celiac axis is undertaken before using the hepatic or splenic arteries. Results of these procedures have been good, both in short- and long-term follow-up studies.²⁶² Analysis of 222 patients treated over 10 years earlier has shown that these procedures were performed with 2.2% mortality, low rates of restenosis (7.3%), and good long-term survival.²⁶² The predictors of late mortality were age older than 60 years, coronary disease, and previous vascular surgery.

The durability of surgical vascular reconstruction is well established. Follow-up studies after 5 and 10 years for all forms of renal artery bypass procedures indicate excellent

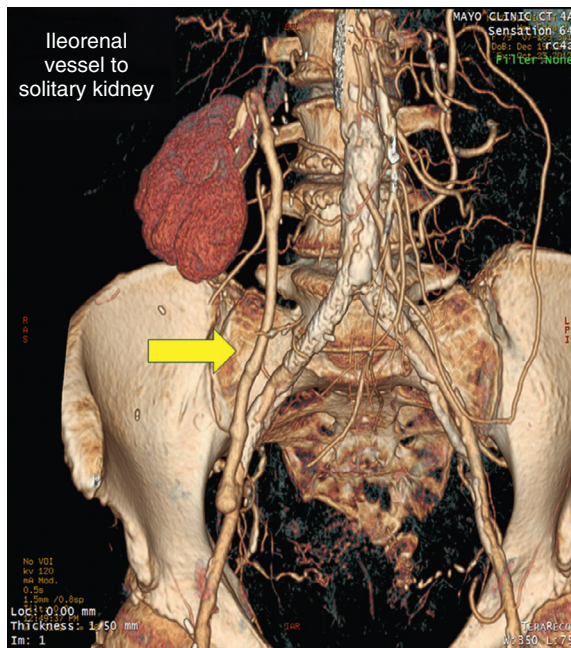


Fig. 47.26 Computed tomography angiogram illustrating complex surgical repair of aortic and vascular perfusion to a solitary functioning kidney. There is a vascular graft from the right common iliac artery to the right renal artery (arrow). Although surgical reconstruction of the renal vessels is less commonly performed since the introduction of endovascular techniques, it provides an essential alternative for complex occlusive disease and failed endovascular stenting in selected patients.

long-term patency (>90%), both for renal artery procedures alone and when combined with aortic reconstruction.²⁶³ Results of surgery have been good, despite increasing age of subjects in the reported series. Patient selection has been important in all these series. Although long-term outcome data are established for surgery, limited information is available for endovascular stent procedures, which are more prone to restenosis and technical failure. Some studies have compared endovascular intervention (PTRA without stents) and surgical repair. A single study of nonstent, unilateral atherosclerotic disease, in which patients were randomly assigned to surgery or PTRA, has shown that although surgical success rates were higher and PTRA was needed on a repeat basis in several cases, the 2-year patency rates were 90% for PTRA and 97% for surgery.²⁶⁴ A prospective comparison of endovascular stents compared with open surgical renal revascularization has suggested that patency over 4 years is better with open surgical repair, but that overall the outcome of the two procedures does not differ.²⁶⁵

In many institutions, surgical reconstruction of the renal arteries is now most often undertaken as part of aortic surgery.²⁶⁶ In one series, patients with impaired renal function (creatinine >2.0 mg/dL) underwent simultaneous aortic and renal procedures in 75% of cases.²³³ Studies have indicated that combining renal revascularization with aortic repair does not increase the risk of the aortic operation. As with endovascular techniques, the results regarding changes in renal function include improvement in 22% to 26%, no change in 46% to 52% (some consider this as stabilization), and progressive deterioration in 18% to 22% (Fig. 47.27).

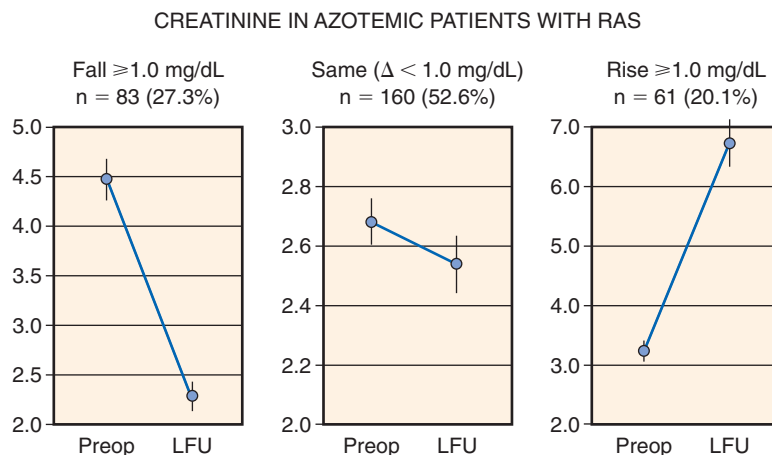


Fig. 47.27 Renal functional outcomes after surgical renal revascularization in 304 azotemic patients (creatinine >2.0 mg/dL) with atherosclerotic renal artery stenosis. On average, the mean serum creatinine level does not change during a follow-up period exceeding 36 months, as observed with most reported series. Group mean values obscure major differences in clinical outcomes, as shown here. Some patients experience major clinical benefit (defined as a fall in serum creatinine of >1.0 mg/dL; left panel). The largest group has minor changes (<1.0 mg/dL), which might be considered “stabilization” of renal function. The degree of benefit in these patients depends on whether renal function has been deteriorating before intervention. The data for the group in the right panel emphasize the failure to observe consistent overall improvement in function because 18% to 22% of patients developed worsening renal function. The exact causes of this deterioration are not clear, although atherosclerotic disease is responsible for a portion. This potential hazard of revascularization must be considered when offering these procedures. *LFU*, Latest follow-up; *RAS*, renin angiotensin system (Modified from Textor SC, Wilcox CS: Renal artery stenosis: a common, treatable cause of renal failure? *Annu Rev Med.* 2001;52:421–442.)

Table 47.8 Selected Series of Surgical Renal Artery Revascularization, 2000–2005

Source	No. of Patients	No. of Patients (%)		Renal Function Response (%)		Hypertension Response (%)		Perioperative Outcome (%)	
		Bilateral Repair	Preoperative RI	Improved	Unchanged	Cured	Improved or stable	Death	Morbidity
Hansen	232	64	100	58	35	11	76	7.3	30
Paty ^a	414	NR	4	26	68	NR		5.5	11.4
Cherr	500	59	49	43	47	73	12	4.6	16
Marone	96	27	100	42	41	NR		4.1	NR
Mozes	198	65	57	28	67	2	59	2.5	19

^aPatients operated for hypertension or renal salvage. Improvement was noted in 26%, 68% remained stable, and 6% worsened. Specific renal function decline occurred in 3%.

NR, No response; RI, resistive index.

Modified from Bower TC, Oderich GS. *Surgical revascularization*. London: Springer; 2014.

Using intraoperative color flow Doppler ultrasound allows for the immediate correction of suboptimal results and improves long-term patency.²⁶⁷ Results from several surgical series are summarized in Table 47.8. Despite good results, the number of open operations for renal artery revascularization continues to decline. A review of the national inpatient sample indicates relatively high mortality rates ($\approx 10\%$) overall, leading the authors to support lower-risk endovascular methods, where possible, or referral to high-volume surgical centers.²⁶⁸ For experienced centers using current techniques, operative risk is below 4% in good risk candidates.^{269,270} Risk factors for higher risk include advanced age, elevated creatinine level (>2.7 – 3.0 mg/dL), and associated aortic or other vascular disease.

Studies in patients with bilateral renal artery lesions or vascular occlusion to the entire renal mass have indicated that restoration of blood flow can lead to the preservation of renal function in some cases.²⁷¹ Usually this has been undertaken when evidence for preserved blood supply, sometimes from capsular vessels, is evident by renography. Occasionally, revascularization can lead to functional recovery sufficient to eliminate the need for dialysis.

PREDICTORS OF LIKELY BENEFIT REGARDING RENAL REVASCLARIZATION

Identification of patients most likely to improve blood pressure and/or renal function after renal revascularization remains elusive. As noted previously, functional tests of renin release, such as measurement of renal vein renin levels, have not performed universally well. Many of these studies are most useful when positive, for example, the likelihood of benefit improves with more evident lateralization, but have relatively poor negative predictive value—that is when such studies are negative, outcomes of vessel repair may still be beneficial. Clinically, recent progression of hypertension, deterioration of renal function, and/or pulmonary edema remain among the most consistent predictors of improved blood pressure after intervention.

Clinical Relevance

Predictors

Progressive hypertension, worsening renal function, and pulmonary edema are the most consistent predictors of improved blood pressure after revascularization.

Predicting favorable renal functional outcomes is also difficult. Surgical and endovascular procedures are least likely to benefit those with advanced renal insufficiency, usually characterized by a serum creatinine level above 3.0 mg/dL. Nonetheless, occasional patients with recent progression to advanced renal dysfunction can recover their GFR, with durable improvement over many years (see Fig. 47.27). Small kidneys, as identified by length less than 8 cm, are less likely to recover function, particularly when little function can be identified on radionuclide renography.²⁷² Reports of the renal resistance index measured by Doppler ultrasound in 5950 patients have indicated that identification of lower resistance was a favorable marker for improvement in GFR and blood pressure, whereas an elevated resistance index was an independent marker of poor outcomes¹⁵⁹ (see Fig. 47.20). None of these associations are absolute, and recent studies have identified favorable outcomes in some patients with adverse predictors.²⁴⁷ As noted, outcomes for patients with a low urinary albumin-to-creatinine ratio are distinctly better than for those with higher levels.^{3,122} Some authors have suggested that detecting abnormalities in fractional flow reserve measured by translesional flows and gradients after dilation with papaverine may predict benefits of revascularization.^{189,273} Recent deterioration of kidney function or hypertension portends a more likely improvement with revascularization.

FUTURE RESEARCH IN RENOVASCULAR DISEASE

How best to identify and recover filtration in poststenotic kidneys remains a challenging area of investigation. Measurement of renal vein cytokines have indicated that poststenotic kidneys exhibit abundant proinflammatory signaling, even

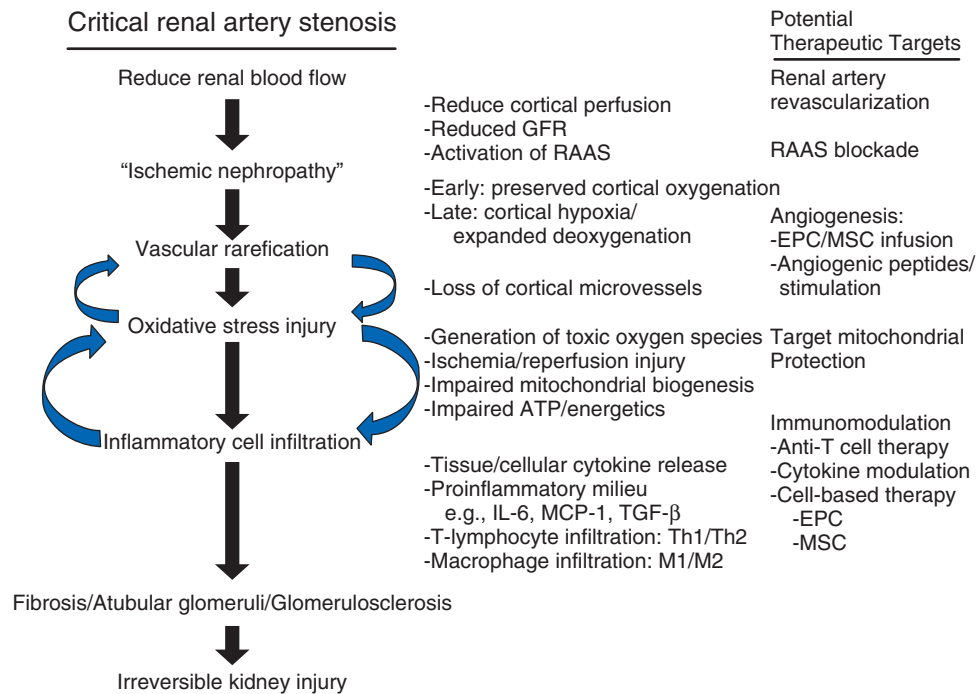


Fig. 47.28 Injury pathways and targets in renovascular disease. The working diagram highlights recent experimental studies delineating specific pathways of oxidative stress injury and inflammatory injury pathways in the poststenotic kidney. *Right panel*, Specific therapeutic targets are identified that may alleviate these injury pathways, over and above simply restoring blood flow. Many of these pathways likely overlap in many forms of kidney injury. EPC, Endothelial progenitor cells; GFR, glomerular filtration rate; IL-6, interleukin 6; MCP, monocyte chemoattractant protein; RAAS, renin-angiotensin-aldosterone system; TGF- β , transforming growth factor beta. (From Textor SC, Lerman LO. Paradigm shifts in atherosclerotic renovascular disease: where are we now? *J Am Soc Nephrol*. 2015;26:2074–2080.)

with relatively preserved oxygenation.²⁴¹ Studies of human kidneys undergoing temporary arterial occlusion have indicated that mitochondrial swelling and disruption develop rapidly.²⁷⁴ These data support experimental models showing increased oxidative stress and mitochondrial dysfunction under conditions of chronically reduced renal blood flow.^{275,276} Protection of mitochondrial structure and function using infused agents that stabilize cardiolipin and inhibit opening of the mitochondrial transition pore appears to facilitate microvascular recovery and kidney regeneration after renal revascularization.⁵¹ Preliminary studies in human subjects have supported this approach.²⁷⁷

Some studies have emphasized the role of inflammatory pathways and accumulation of activated T cells and macrophages with vessels and tissue of the atherosclerotic kidney.³² Experimental models using infused endothelial progenitor cells⁵⁶ and mesenchymal stem cells²⁷⁸ respond to revascularization with improved microvascular regeneration. Initial studies in human atherosclerotic RAS using autologous mesenchymal stem cells delivered into the renal artery have suggested that cortical blood flow and tissue oxygenation increase over a 3-month period.²⁷⁹

Recognition of the role and interactions among these pathways offers the possibility for targeted therapy in patients with occlusive RVD. Some of these are highlighted in Fig. 47.28.

SUMMARY

RVD is common, particularly in older subjects with other atherosclerotic disease. It can have multiple clinical conse-

quences, ranging from asymptomatic incidentally discovered disease to accelerated hypertension and progressive renal failure. With improved imaging and increasing patient age, significant renal artery disease is being detected more frequently. The clinician must evaluate the impact of the renal artery disease in the individual patient and the potential risk-benefit ratio for renal revascularization. Many patients with fibromuscular dysplasia and other disorders benefit from restoring renal artery patency in terms of blood pressure control. An algorithm to guide the treatment and reevaluation of patients with atherosclerotic RAS is presented in Fig. 47.29. Application of this strategy relies heavily on considering comorbid risks and the evolution of blood pressure control and kidney function over time. The management of cardiovascular risk and hypertension are the primary objectives of medical therapy.

For most patients, the realistic goals of renal revascularization are to reduce blood pressure medication requirements and stabilize renal function over time. Patients with bilateral disease or stenosis in a solitary functioning kidney may have a lower risk of circulatory congestion (flash pulmonary edema or its equivalent) and lower risk for advancing renal failure after revascularization of the kidney. It is essential to appreciate the risks inherent in surgical or endovascular manipulation of the diseased aorta. These include atheroembolic complications and potential deterioration of renal function related to the procedure itself (estimated at 20% for patients with preexisting kidney dysfunction). Hence, the decision to undertake these procedures should include consideration of whether the potential gain warrants such risks. In many cases, improved blood pressure and recovery of renal function

MANAGEMENT OF RENOVASCULAR HYPERTENSION AND ISCHEMIC NEPHROPATHY

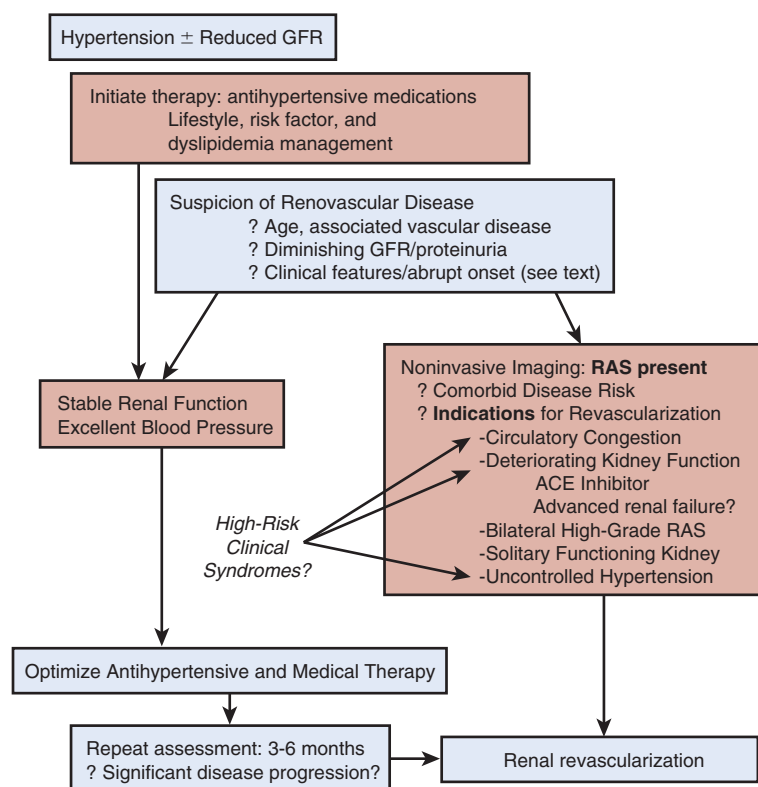


Fig. 47.29 Algorithm summarizing a management scheme for patients with renovascular hypertension and/or ischemic nephropathy. Optimizing antihypertensive and medical therapy for comorbid conditions, including dyslipidemia and smoking, is paramount to reducing cardiovascular morbidity and mortality in atherosclerotic disease. Decisions regarding the timing of renal revascularization procedures depend on the clinical manifestations (see text) and whether blood pressures and kidney function remain stable. *ACE*, Angiotensin-converting enzyme; *GFR*, glomerular filtration rate; *PTRA*, percutaneous renal angioplasty; *RAS*, renal artery stenosis.

justify the costs and hazards. Long-term follow-up of blood pressure and renal function is important in all cases, whether revascularization is performed or not, particularly because of the potential for disease progression, restenosis, and/or recurrent disease. Optimal selection and timing for medical management and revascularization depend largely on the comorbid conditions for each patient.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. A 73-year-old man is identified as having a 70% stenosis of his right renal artery during CT angiography for abdominal pain. The left was normal. He has two-vessel coronary disease, for which stents were placed a month earlier. His cardiologist recommends that he undergo renal artery stenting after six months of clopidogrel therapy.

He has hypertension for 15 years and now feels well. His current medications include an ACE inhibitor, a beta-blocker, an alpha-blocker, a diuretic, clopidogrel, and aspirin.

BP: 145/74 mm Hg P: 65

Chest Clear; No edema

Labs: CBC normal. Serum creatinine 1.4 mg/dL (eGFR: 49 mL/min/1.73 m²), sodium 140 mEq/L, potassium 4.6 mEq/L

Which of the following most accurately describes the likely results of renal artery stenting for this individual?

- Lower antihypertensive drug requirements
- Improved kidney function
- Reduced angina
- Regression of left ventricular hypertrophy
- Lower mortality

Answer: a

Rationale: Several studies demonstrate that antihypertensive medications requirements are lower, but usually not withdrawn after revascularization, making (a) most likely. Other potential benefits including changes in kidney function rarely occur, especially with unilateral disease (making (b) incorrect). Most importantly, no mortality benefits have been observed (e). (c) and (d) sometimes are benefits with effective BP control, but are not directly related to stenting. These points have been underscored in several trials including ASTRAL, STAR and CORAL.²⁸⁰⁻²⁸²

2. A 69-year-old woman returns after a recent hospitalization with congestive heart failure (ejection fraction 40%) and a rise in creatinine from 1.2 to 2.6 mg/dL. An ultrasound examination identifies a small right kidney (7.4 cm) and normal sized left (10.8 cm). Doppler ultrasound demonstrates no flow to the right kidney. The left has a peak systolic velocity of 425 cm/sec and estimated more than 70% stenosis.

Blood pressure was 180/80 mm Hg initially but fell to 130 / 80 after treatment with ramipril, amlodipine and furosemide. Serum creatinine 2.7 mg/dL, sodium 139 mEq/L, potassium 4.9 mEq/L.

Which of the following is most likely to stabilize kidney function and fluid status for this patient?

- Laparoscopic nephrectomy of the right kidney
- Withdrawal of ramipril
- Addition of spironolactone
- Transition to an angiotensin receptor blocker
- Endovascular stent placement

Answer: e

Rationale: This individual has both impaired cardiac pump function and reduced GFR with a solitary functioning kidney. In this situation, hospitalization for recurrent CHF can be reduced by renal revascularization,^{283,284} making (e) the best choice. (a) would have no effect on kidney function. One

might withdraw ramipril (b) but this would impair cardiac function and allow higher BP, making it less correct. No benefits in this situation to a transition to ARB (same effect as doing nothing) or spironolactone (c).

3. Which of the following is most probable for an 81-year-old man with recently identified, high-grade renal arterial stenosis (serum creatinine 1.5 mg/dL) and a small abdominal aortic aneurysm (3.2 cm) over the next two years?

- Cerebrovascular event (stroke, TIA)
- Congestive cardiac failure
- End-stage kidney disease
- Aneurysm rupture
- Development of angina

Answer: e

Rationale: Patients with atherosclerotic renal artery stenosis are at highest risk for developing new cardiovascular symptoms. New angina and coronary symptoms (e) are most common, as directly reflected from the chapter Figure 2 from the Medicare Claims review of Kalra, et al.²⁸⁵ End-stage disease (c) and aneurysm rupture (d) are distinctly less probable, while congestive heart failure (b) and CNS symptoms (a) are intermediate.

4. A 65-year-old woman is referred for CKD Stage 3. Two years ago she developed a rise in blood pressure to 170/95 mmHg during a routine examination. Previous BP levels always had been below 140/90. Current medications include losartan, amlodipine and indapamide.

BP: 134/70 mmHg P: 74 bpm trace edema. BMI: 27 kg/m²

Serum creatinine 1.5 mg/dL (CKD EPI eGFR: 36 mL/min/1.73 m²)

Doppler ultrasound identified high velocities (310 cm/sec) to the left kidney (9.0 cm) and normal velocities (85 cm/sec) to the right kidney (10.8 cm).

Which of the following MOST accurately describes the state of the left kidney?

- Despite reduced blood flow, cortical oxygenation is preserved
- Progressive medullary hypoxia is inevitable
- Reduced GFR magnifies medullary oxygen deprivation
- Angiotensin receptor blockade limits renin release
- Restoring renal blood flow will reverse cortical ischemia

Answer: a

Rationale: Experimental and human studies indicate that moderate RVD is associated with a loss of blood flow and volume, but well preserved cortical oxygenation, in part because the cortex receives a large excess of oxygenated blood, making (a) correct.²⁸⁶ Biopsy information indicates that medullary hypoxia can be stable, in part because reduced GFR lowers oxygen consumption related to solute transport, making (b) and (c) incorrect.²⁸⁷ Blockade of the renin-angiotensin system tends to stimulate renin release (opposite of [d]), and when inflammatory pathways become activated, restoring blood flow alone fails to abrogate that process, making (e) incorrect.

5. You are asked to evaluate resistant hypertension and azotemia in a 56-year-old woman with atherosclerotic renovascular disease. Serum creatinine had been 1.7 mg/dL during therapy with losartan, which had lowered BP to 130/70 mm Hg. She had undergone renal artery angioplasty a year previously, after which her angiotensin receptor blocker had been withdrawn.

Current medications: carvedilol, amlodipine and furosemide.

BP 180/98 mmHg. Well-appearing. Audible epigastric bruit.

Urinalysis: normal. Serum creatinine 1.4 mg/dL (CKD EPI eGFR: 42 mL/min/1.73 m²), potassium 3.4 mEq/L

CT angiography demonstrates two normal sized kidneys with residual stenosis in both renal arteries.

Which of the following MOST accurately describes the role of renin-angiotensin blockade in this patient?

- a. ACE inhibition should be avoided due to bilateral RVD
- b. A direct renin inhibitor (aliskiren) would be the preferred agent

c. Administration of losartan would likely lower BP

d. An angiotensin receptor blocker should be started only after repeat angioplasty

Answer: c

The vast majority of patients (92%) with renovascular disease (either atherosclerotic or fibromuscular in origin) tolerate blockade of the renin-angiotensin system, including bilateral disease (78%).²⁸⁸ The finding of recurrent hypertension and lower S[K] is consistent with activation of the renin-angiotensin system and likely would respond to blockade, making (c) the preferred answer. It is incorrect to infer that these agents should be avoided, as long-term outcomes and mortality appear to be improved with RAAS blockade,²⁸⁹ making (a) incorrect. The role of aliskiren is not precisely defined but has no advantage. If azotemia worsens, repeat revascularization may be considered, but should not be a necessary first step to therapy, making (d) less correct.